Introduction to Algebra I (revisit)

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Contents

1. Groups I **3**
1.1. Permutations 3
1.2. Groups 12
1.3. Subgroups and Lagrange's Theorem 25
1.4. Residue classes 33
1.5. The additive group $\mathbb{Z}_n$ 34
1.6. Multiplication and the group of units 35
1.7. Group Actions 37
1.8. Homomorphisms 52
1.9. Quotient Groups 59

1. Groups I

Lecture 1: Permutations, Groups, Lagrange's Theorem

§1.1. Permutations

Rotman §2.2–§2.4

In high school mathematics the words *permutation* and *arrangement* are used interchangeably. We draw a distinction between them.

Definition 1.1.1 (Permutation, arrangement)**.** A **permutation** of a set X is a bijection $\alpha\colon X \to X$. If X is finite with $|X| = n$, then an **arrangement** of X is a list $x_1, x_2, \ldots, x_n$ of all the elements of X with no repetitions.

Given an arrangement $x_1, \ldots, x_n$, define $f\colon \{1, \ldots, n\} \to X$ by $f(i) = x_i$; thus the list displays the values of f. That there are no repetitions says f is injective, and that every $x \in X$ occurs on the list says f is surjective. So an arrangement of X is the same thing as a bijection $f\colon \{1, \ldots, n\} \to X$.

Example 1.1.2 (Arrangements of a three-element set)**.** There are six arrangements of $X = \{a, b, c\}$:

$$abc; \quad acb; \quad bac; \quad bca; \quad cab; \quad cba.$$

In general there are exactly $n!$ arrangements of an n-element set.

If $X = \{1, 2, \ldots, n\}$, a permutation $\alpha\colon X \to X$ gives the list $\alpha(1) = i_1, \ldots, \alpha(n) = i_n$, which we record in **two-rowed notation**:

$$\alpha = \begin{pmatrix} 1 & 2 & \cdots & j & \cdots & n \\ \alpha(1) & \alpha(2) & \cdots & \alpha(j) & \cdots & \alpha(n) \end{pmatrix}.$$

> **Intuition .** Informally, arrangements (lists) and permutations (bijections) are two ways of describing the same thing. The advantage of viewing a permutation as a *bijection* is that bijections can be *composed*, and the composite of two permutations is again a permutation. Lists can only be counted.

Definition 1.1.3 (Symmetric group)**.** The family of all permutations of a set X, denoted by S_X, is called the **symmetric group** on X. When $X = \{1, 2, \ldots, n\}$ we write S_n and call it the **symmetric group on n letters**.

Example 1.1.4 (Composition in S_3 is not commutative)**.** If

$$\alpha = \begin{pmatrix} 1 & 2 & 3 \\ 2 & 3 & 1 \end{pmatrix} \quad \text{and} \quad \beta = \begin{pmatrix} 1 & 2 & 3 \\ 2 & 1 & 3 \end{pmatrix},$$

then

$$\alpha \circ \beta = \begin{pmatrix} 1 & 2 & 3 \\ 3 & 2 & 1 \end{pmatrix} \quad \text{and} \quad \beta \circ \alpha = \begin{pmatrix} 1 & 2 & 3 \\ 1 & 3 & 2 \end{pmatrix},$$

so $\alpha \circ \beta \neq \beta \circ \alpha$: for instance $\alpha \circ \beta\colon 1 \mapsto \alpha(\beta(1)) = \alpha(2) = 3$, whereas $\beta \circ \alpha\colon 1 \mapsto 2 \mapsto 1$.

Some permutations do commute, however; the reader may check that $\left(\begin{smallmatrix} 1 & 2 & 3 & 4 \\ 2 & 1 & 3 & 4 \end{smallmatrix}\right)$ and $\left(\begin{smallmatrix} 1 & 2 & 3 & 4 \\ 1 & 2 & 4 & 3 \end{smallmatrix}\right)$ commute.

Note 1.1.5 (Order of multiplication)**.** Some authors put "functions on the right", writing $(i)\alpha$ instead of $\alpha(i)$. Their $\alpha \circ \beta$ is our $\beta \circ \alpha$: the notational switch causes a switch in the order of multiplication.

Proposition 1.1.6 (Cancellation in S_X)
If $\gamma, \alpha, \beta \in S_X$ and $\gamma\alpha = \gamma\beta$, then $\alpha = \beta$. Similarly $\alpha\gamma = \beta\gamma$ implies $\alpha = \beta$.

Proof. Compose with γ^{-1} and use associativity of composition:

$$\alpha = 1_X \circ \alpha = (\gamma^{-1} \circ \gamma) \circ \alpha = \gamma^{-1} \circ (\gamma \circ \alpha) = \gamma^{-1} \circ (\gamma \circ \beta) = (\gamma^{-1} \circ \gamma) \circ \beta = 1_X \circ \beta = \beta.$$

A similar argument works on the other side. ∎

Aside from being cumbersome, the two-rowed notation hides the answers to elementary questions: is the square of a permutation the identity? what is the smallest $m > 0$ with $\alpha^m = 1$? can one factor a permutation into simpler pieces? The special permutations below remedy this defect.

Notation . From now on we write $\beta\alpha$ instead of $\beta \circ \alpha$, and (1) instead of 1_X.

§1.1.i. Cycles

Definition 1.1.7 (Fixes, moves)**.** If $\alpha \in S_n$ and $i \in \{1, \dots, n\}$, then α **fixes** i if $\alpha(i) = i$, and α **moves** i if $\alpha(i) \neq i$.

Definition 1.1.8 (Cycle)**.** Let $i_1, i_2, \dots, i_r$ be distinct integers in $\{1, \dots, n\}$. If $\alpha \in S_n$ fixes the other integers (if any) and

$$\alpha(i_1) = i_2, \quad \alpha(i_2) = i_3, \quad \dots, \quad \alpha(i_{r-1}) = i_r, \quad \alpha(i_r) = i_1,$$

then α is called an **r-cycle**, or a **cycle of length r**, and we write

$$\alpha = (i_1\ i_2\ \dots\ i_r).$$

A 2-cycle is called a **transposition**. A 1-cycle is the identity, so all 1-cycles are equal: $(i) = (1)$ for every i.

Example 1.1.9 (Cycle notation)**.** The two-rowed notation does not help us recognize that $\alpha = \left(\begin{smallmatrix}1&2&3&4&5\\4&3&1&5&2\end{smallmatrix}\right)$ is a 5-cycle, but in fact $\alpha(1) = 4$, $\alpha(4) = 5$, $\alpha(5) = 2$, $\alpha(2) = 3$, $\alpha(3) = 1$; that is, $\alpha = (1\ 4\ 5\ 2\ 3)$. Likewise

$$\begin{pmatrix}1&2&3&4\\2&3&4&1\end{pmatrix} = (1\ 2\ 3\ 4), \qquad \begin{pmatrix}1&2&3&4&5\\5&1&4&2&3\end{pmatrix} = (1\ 5\ 3\ 4\ 2),$$
$$\begin{pmatrix}1&2&3&4&5\\2&3&1&4&5\end{pmatrix} = (1\ 2\ 3).$$

On the other hand $\beta = \left(\begin{smallmatrix}1&2&3&4\\2&1&4&3\end{smallmatrix}\right)$ is *not* a cycle; in fact $\beta = (1\ 2)(3\ 4)$.

The term *cycle* comes from the Greek word for circle: one pictures the cycle $(i_1\ i_2\ \dots\ i_r)$ as a clockwise rotation of a circle (Figure 1.1). Any i_j may be taken as the starting point, so an r-cycle has r different cycle notations:

$$(i_1\ i_2\ \dots\ i_r) = (i_2\ i_3\ \dots\ i_r\ i_1) = \cdots = (i_r\ i_1\ i_2\ \dots\ i_{r-1}).$$

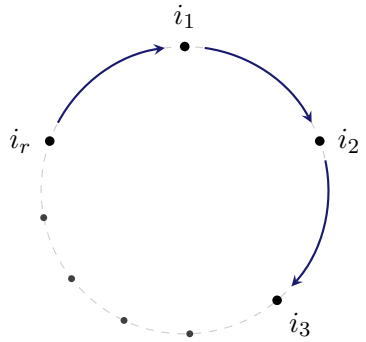


Figure 1.1.: A cycle as a (clockwise) rotation.

Example 1.1.10 (The factorization algorithm)**.** Take

$$\alpha = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 & 7 & 8 & 9 \\ 6 & 4 & 7 & 2 & 5 & 1 & 8 & 9 & 3 \end{pmatrix}.$$

Begin by writing "(1". Now $\alpha\colon 1 \mapsto 6$, so write "(1 6". Next $\alpha\colon 6 \mapsto 1$, so the parentheses close: α begins "(1 6)". The smallest number not yet used is 2, and $2 \mapsto 4$, $4 \mapsto 2$, giving "(1 6)(2 4)". The smallest remaining number is 3, and $3 \mapsto 7 \mapsto 8 \mapsto 9 \mapsto 3$, giving the 4-cycle (3 7 8 9). Finally $\alpha(5) = 5$, and so

$$\alpha = (1\ 6)(2\ 4)(3\ 7\ 8\ 9)(5).$$

To verify a claim like this one evaluates both sides. For instance, writing $\beta = (1\ 6)$, $\gamma = (2\ 4)$, $\delta = (3\ 7\ 8\ 9)$,

$$\beta\gamma\delta(1) = \beta(\gamma(\delta(1))) = \beta(\gamma(1)) = \beta(1) = 6 = \alpha(1),$$

since δ and γ fix 1. (Proposition 1.1.16 gives a more satisfactory proof that the algorithm works.)

Example 1.1.11 (Multiplying with the algorithm)**.** Factorizations into cycles are convenient for *multiplying* permutations: one simply displays the partial outputs. In S_5 let

$$\sigma = (1\ 2)(1\ 3\ 4\ 2\ 5)(2\ 5\ 1\ 3).$$

Then $\sigma\colon 1 \mapsto 3 \mapsto 4 \mapsto 4$, so σ begins "(1 4". Next $4 \mapsto 4 \mapsto 2 \mapsto 1$, so σ begins "(1 4)". Then $2 \mapsto 5 \mapsto 1 \mapsto 2$, so σ fixes 2; and $3 \mapsto 2 \mapsto 5 \mapsto 5$, $5 \mapsto 1 \mapsto 3 \mapsto 3$. Therefore $\sigma = (1\ 4)(2)(3\ 5)$.

Definition 1.1.12 (Disjoint permutations)**.** Two permutations $\alpha, \beta \in S_n$ are **disjoint** if every i moved by one is fixed by the other: if $\alpha(i) \neq i$ then $\beta(i) = i$, and if $\beta(j) \neq j$ then $\alpha(j) = j$. A family $\beta_1, \ldots, \beta_t$ is disjoint if each pair of them is disjoint.

Observation . If $\alpha = (i_1\ \ldots\ i_r)$ and $\beta = (j_1\ \ldots\ j_s)$, then any $k \in \{i_1, \ldots, i_r\} \cap \{j_1, \ldots, j_s\}$ is moved by both. Hence two cycles are disjoint if and only if the underlying sets $\{i_1, \ldots, i_r\}$ and $\{j_1, \ldots, j_s\}$ are disjoint sets. When α and β are disjoint there are exactly three possibilities for a number i: it is moved by α, it is moved by β, or it is fixed by both.

Lemma 1.1.13 (Disjoint permutations commute)

Disjoint permutations $\alpha, \beta \in S_n$ commute.

Proof. It suffices to prove $\alpha\beta(i) = \beta\alpha(i)$ for $1 \leq i \leq n$. If β moves i, say $\beta(i) = j \neq i$, then β also moves j (otherwise $\beta(j) = j$ and $\beta(i) = j$ contradict injectivity of β); since α and β are disjoint, $\alpha(i) = i$ and $\alpha(j) = j$, so $\beta\alpha(i) = \beta(i) = j = \alpha(j) = \alpha\beta(i)$. A similar argument applies if α moves i. The last possibility is that neither moves i, and then $\alpha\beta(i) = i = \beta\alpha(i)$. ∎

Remark 1.1.14. The converse fails: permutations that are *not* disjoint may still commute. The reader may check that (1 2 3)(4 5) and (1 3 2)(6 7) commute; even simpler, $\alpha\alpha^2 = \alpha^2\alpha$.

Lemma 1.1.15

Let $X = \{1, \dots, n\}$, let $\alpha \in S_X = S_n$, and given $i_1 \in X$ define $i_{j+1} = \alpha(i_j)$ for $j \geq 1$. Write $Y = \{i_j : j \geq 1\}$ and let Y' be the complement of Y in X.

(i) If α moves i_1, then there is $r > 1$ with $i_1, \dots, i_r$ all distinct and $i_{r+1} = \alpha(i_r) = i_1$.

(ii) $\alpha(Y) = Y$ and $\alpha(Y') = Y'$.

Proof. (i) Since X is finite, there is a smallest $r > 1$ with $i_1, \dots, i_r$ all distinct but $i_{r+1} = \alpha(i_r) \in \{i_1, \dots, i_r\}$; say $\alpha(i_r) = i_j$ with $1 \leq j \leq r$. If $j > 1$, then $\alpha(i_r) = i_j = \alpha(i_{j-1})$, and injectivity of α gives $i_r = i_{j-1}$, contradicting distinctness. Therefore $\alpha(i_r) = i_1$.

(ii) Clearly $\alpha(Y) \subseteq Y$, since $i_j \in Y$ implies $\alpha(i_j) = i_{j+1} \in Y$. If $k \in Y'$ and $\alpha(k) \in Y$, then $\alpha(k) = i_j = \alpha(i_{j-1})$ for some j (by part (i) this holds even for $i_j = i_1$); injectivity gives $k = i_{j-1} \in Y$, contradicting $Y \cap Y' = \varnothing$. Hence $\alpha(Y') \subseteq Y'$.

These inclusions are equalities: $\alpha(X) = \alpha(Y) \cup \alpha(Y')$ is a disjoint union because α is injective, and $|\alpha(Y)| \leq |Y|$, $|\alpha(Y')| \leq |Y'|$. If either inequality were strict, then $|\alpha(X)| < |X|$, contradicting surjectivity of α. ∎

Proposition 1.1.16 (Factorization into disjoint cycles)

Every permutation $\alpha \in S_n$ is either a cycle or a product of disjoint cycles.

Proof. Induct on the number $k \geq 0$ of points moved by α. The base step $k = 0$ is true, for then α is the identity, which is a 1-cycle.

If $k > 0$, let i_1 be a point moved by α. As in Lemma 1.1.15, define $Y = \{i_1, \dots, i_r\}$ with the i_j distinct, $\alpha(i_j) = i_{j+1}$ for $j < r$, and $\alpha(i_r) = i_1$. Let $\sigma = (i_1\ i_2\ \dots\ i_r)$, so that σ fixes each point of Y'. If $r = n$, then $\alpha = \sigma$. If $r < n$, then $\alpha(Y') = Y'$ by the lemma. Put $\alpha' = \alpha\sigma^{-1}$; we claim α' and σ are disjoint. If σ moves i, then $i = i_j \in Y$, and $\alpha'(i_j) = \alpha\sigma^{-1}(i_j) = \alpha(i_{j-1}) = i_j$, so α' fixes i_j. Conversely, if α' moves k, then $k \notin Y$, so $k \in Y'$; but σ fixes every element of Y'. Hence $\alpha = \alpha'\sigma$ is a factorization into disjoint permutations. The number of points moved by α' is $k - r < k$, so the inductive hypothesis gives $\alpha' = \beta_1 \cdots \beta_t$ with $\beta_1, \dots, \beta_t$ disjoint cycles. Therefore $\alpha = \beta_1 \cdots \beta_t\sigma$ is a product of disjoint cycles. ∎

Definition 1.1.17 (Complete factorization)**.** A **complete factorization** of a permutation α is a factorization of α into disjoint cycles that contains one 1-cycle (i) for every i fixed by α.

The algorithm of Example 1.1.10 always yields a complete factorization. In a complete factorization $\alpha = \beta_1 \cdots \beta_t$, every symbol i between 1 and n occurs in exactly one β_λ.

Example 1.1.18. If $\alpha = \left(\begin{smallmatrix} 1 & 2 & 3 & 4 & 5 \\ 1 & 3 & 4 & 2 & 5 \end{smallmatrix}\right)$, the algorithm gives the complete factorization $\alpha = (1)(2\ 3\ 4)(5)$. By contrast the factorizations

$$\alpha = (2\ 3\ 4) = (1)(2\ 3\ 4) = (2\ 3\ 4)(5)$$

are *not* complete.

Write $\beta = (i_0\ i_1\ \dots\ i_{r-1})$. Then $i_1 = \beta(i_0)$, $i_2 = \beta(i_1) = \beta^2(i_0)$, and in general

$$i_k = \beta^k(i_0) \qquad (k \leq r - 1). \tag{1.1}$$

Since $\beta(i_{r-1}) = i_0$, equation (1.1) holds for all k if the subscripts j in i_j are read mod r.

Lemma 1.1.19 (Powers of a cycle)

Let $\beta = (i_0\ i_1\ \dots\ i_{r-1})$ be an r-cycle. Then $\beta^r = (1)$, and r is the smallest positive integer k with $\beta^k = (1)$.

Proof. Reading subscripts mod r, equation (1.1) gives $\beta^k(i_j) = i_{j+k}$ for all j and all $k \geq 1$. Taking $k = r$ yields $\beta^r(i_j) = i_j$ for every j; since β fixes every other symbol, so does β^r, and hence $\beta^r = (1)$. If $0 < k < r$, then $\beta^k(i_0) = i_k \neq i_0$, because $i_0, \dots, i_{r-1}$ are distinct; therefore $\beta^k \neq (1)$. ■

Lemma 1.1.20 (i) Let $\alpha = \beta\delta$ be a factorization into disjoint permutations. If β moves i, then $\alpha^k(i) = \beta^k(i)$ for all $k \geq 1$.

(ii) If β and γ are cycles both of which move $i = i_0$, and if $\beta^k(i) = \gamma^k(i)$ for all $k \geq 1$, then $\beta = \gamma$.

Remark 1.1.21. The hypothesis in (ii) does *not* assume that β and γ have the same length; that is part of the conclusion.

Proof. (i) Since β moves i, disjointness implies δ fixes i, and hence so does every power of δ. Now β and δ commute by Lemma 1.1.13, so $(\beta\delta)^k(i) = \beta^k(\delta^k(i)) = \beta^k(i)$.

(ii) By (1.1), if $\beta = (i_0\ i_1\ \dots\ i_{r-1})$, then $i_k = \beta^k(i_0)$, and if $\gamma = (i_0\ j_1\ \dots\ j_{s-1})$, then $j_k = \gamma^k(i_0)$. We may assume $r \leq s$, so that $i_1 = j_1, \dots, i_{r-1} = j_{r-1}$. Since $j_r = \gamma^r(i_0) = \beta^r(i_0) = i_0$, it follows that $s = r$ and $j_k = i_k$ for all k. Therefore $\beta = \gamma$. ■

Theorem 1.1.22 (Uniqueness of the complete factorization)

Let $\alpha \in S_n$ and let $\alpha = \beta_1 \cdots \beta_t$ be a complete factorization into disjoint cycles. This factorization is unique except for the order in which the cycles occur.

Intuition . This is the analog, for permutations, of the fundamental theorem of arithmetic: cycles play the role of primes.

Proof. Let $\alpha = \gamma_1 \cdots \gamma_s$ be a second complete factorization into disjoint cycles. Every complete factorization has exactly one 1-cycle for each i fixed by α, so it suffices to prove, by induction on ℓ, the larger of t and s, that the cycles of length > 1 are uniquely determined by α.

The base step $\ell = 1$ is the hypothesis $\beta_1 = \alpha = \gamma_1$.

For the inductive step, if β_t moves $i = i_0$, then $\beta_t^k(i_0) = \alpha^k(i_0)$ for all $k \geq 1$, by Lemma 1.1.20(i). Some γ_j must also move i_0; since disjoint cycles commute, we may re-index so that γ_s moves i_0, and then $\gamma_s^k(i_0) = \alpha^k(i_0)$ for all k. Now Lemma 1.1.20(ii) gives $\beta_t = \gamma_s$, so that cancellation (Proposition 1.1.6) yields

$$\beta_1 \cdots \beta_{t-1} = \gamma_1 \cdots \gamma_{s-1}.$$

By the inductive hypothesis $s = t$ and the γ's may be re-indexed so that $\gamma_\lambda = \beta_\lambda$ for all λ. ■

Proposition 1.1.23 (Inverses) (i) $(i_1\ i_2\ \dots\ i_r)^{-1} = (i_r\ i_{r-1}\ \dots\ i_1)$.

(ii) If $\gamma \in S_n$ and $\gamma = \beta_1 \cdots \beta_k$, then $\gamma^{-1} = \beta_k^{-1} \cdots \beta_1^{-1}$ (note that the order of the factors is reversed).

Intuition . In the picture of a cycle as a clockwise rotation of a circle (Figure 1.1), the inverse is just the counterclockwise rotation.

Proof. (i) The composite $(i_1\ i_2\ \dots\ i_r)(i_r\ i_{r-1}\ \dots\ i_1)$ fixes every integer between 1 and n other than $i_1, \dots, i_r$; it sends $i_1 \mapsto i_r \mapsto i_1$, and for $j \geq 2$ it sends $i_j \mapsto i_{j-1} \mapsto i_j$. Thus every integer is fixed, so the composite is (1). A similar argument treats the other order.

(ii) Induct on $k \geq 2$. For $k = 2$,

$$(\beta_1\beta_2)(\beta_2^{-1}\beta_1^{-1}) = \beta_1(\beta_2\beta_2^{-1})\beta_1^{-1} = \beta_1\beta_1^{-1} = (1),$$

and similarly $(\beta_2^{-1}\beta_1^{-1})(\beta_1\beta_2) = (1)$. For the inductive step, let $\delta = \beta_1 \cdots \beta_k$; then $\beta_1 \cdots \beta_k\beta_{k+1} = \delta\beta_{k+1}$, and so

$$(\beta_1 \cdots \beta_{k+1})^{-1} = (\delta\beta_{k+1})^{-1} = \beta_{k+1}^{-1}\delta^{-1} = \beta_{k+1}^{-1}(\beta_1 \cdots \beta_k)^{-1} = \beta_{k+1}^{-1}\beta_k^{-1} \cdots \beta_1^{-1}. \qquad \blacksquare$$

Thus $(1\ 2\ 3\ 4)^{-1} = (4\ 3\ 2\ 1) = (1\ 4\ 3\ 2)$, and $(1\ 2)^{-1} = (2\ 1) = (1\ 2)$: every transposition is its own inverse.

Example 1.1.24. Proposition 1.1.23 applies in particular when the factors are disjoint cycles, in which case reversing the order is unnecessary because disjoint cycles commute. Thus if $\alpha = \left(\begin{smallmatrix}1&2&3&4&5&6&7&8&9\\6&4&7&2&5&1&8&9&3\end{smallmatrix}\right) = (1\ 6)(2\ 4)(3\ 7\ 8\ 9)(5)$, then

$$\alpha^{-1} = (5)(9\ 8\ 7\ 3)(4\ 2)(6\ 1) = (1\ 6)(2\ 4)(3\ 9\ 8\ 7).$$

§1.1.ii. Cycle Structure and Conjugation

Definition 1.1.25 (Cycle structure)**.** Two permutations $\alpha, \beta \in S_n$ have the same **cycle structure** if their complete factorizations have the same number of r-cycles for every $r \geq 1$.

There are $\frac{1}{r}\big[n(n-1)\cdots(n-r+1)\big]$ r-cycles in S_n (Rotman, Exercise 2.21). This formula counts permutations of a given cycle structure, provided one is careful when several cycles have the same length. For example, the number of permutations in S_4 of cycle structure $(a\ b)(c\ d)$ is

$$\tfrac{1}{2}\Big[\tfrac{1}{2}(4 \times 3)\Big] \cdot \Big[\tfrac{1}{2}(2 \times 1)\Big] = 3,$$

the extra factor $\frac{1}{2}$ occurring so that we do not count $(a\ b)(c\ d) = (c\ d)(a\ b)$ twice. Similarly, the number of permutations in S_n of the form $(a\ b)(c\ d)(e\ f)$ is

$$\frac{1}{3!\,2^3}\big[n(n-1)(n-2)(n-3)(n-4)(n-5)\big].$$

Cycle structure	Number
(1)	1
(1 2)	6
(1 2 3)	8
(1 2 3 4)	6
(1 2)(3 4)	3
	24

(a) Permutations in S_4.

Cycle structure	Number
(1)	1
(1 2)	10
(1 2 3)	20
(1 2 3 4)	30
(1 2 3 4 5)	24
(1 2)(3 4 5)	20
(1 2)(3 4)	15
	120

(b) Permutations in S_5.

Table 1.1.: Counting permutations by cycle structure.

Lemma 1.1.26

Let $\alpha, \gamma \in S_n$. For all i, if $\gamma\colon i \mapsto j$, then $\alpha\gamma\alpha^{-1}\colon \alpha(i) \mapsto \alpha(j)$.

Proof. $\alpha\gamma\alpha^{-1}(\alpha(i)) = \alpha\gamma(i) = \alpha(j)$. ∎

Proposition 1.1.27

If $\gamma, \alpha \in S_n$, then $\alpha\gamma\alpha^{-1}$ has the same cycle structure as γ. In more detail, if the complete factorization of γ is

$$\gamma = \beta_1\beta_2\cdots(i\ j\ \ldots)\cdots\beta_t,$$

then $\alpha\gamma\alpha^{-1}$ is the permutation obtained from γ by applying α to every symbol occurring in the cycles of γ.

Remark 1.1.28. For example, if $\gamma = (1\ 3)(2\ 4\ 7)(5)(6)$ and $\alpha = (2\ 5\ 6)(1\ 4\ 3)$, then

$$\alpha\gamma\alpha^{-1} = (\alpha 1\ \ \alpha 3)(\alpha 2\ \ \alpha 4\ \ \alpha 7)(\alpha 5)(\alpha 6) = (4\ 1)(5\ 3\ 7)(6)(2).$$

Proof. Let σ denote the permutation obtained from γ by applying α to all the symbols. If γ fixes i, then Lemma 1.1.26 shows that $\alpha\gamma\alpha^{-1}$ fixes $\alpha(i)$, as does σ. Assume that γ moves i, say $\gamma(i) = j$, so that one of the cycles of γ is $(i\ j\ \ldots)$; by definition, one of the cycles of σ is $(\alpha(i)\ \alpha(j)\ \ldots)$, that is, $\sigma\colon \alpha(i) \mapsto \alpha(j)$. But Lemma 1.1.26 says $\alpha\gamma\alpha^{-1}\colon \alpha(i) \mapsto \alpha(j)$, so σ and $\alpha\gamma\alpha^{-1}$ agree on every number of the form $\alpha(i)$. Since α is surjective, every $k \in X$ has this form, and so $\sigma = \alpha\gamma\alpha^{-1}$. ∎

Proposition 1.1.29

If $\gamma, \gamma' \in S_n$, then γ and γ' have the same cycle structure if and only if there exists $\alpha \in S_n$ with $\gamma' = \alpha\gamma\alpha^{-1}$.

Proof. Sufficiency is Proposition 1.1.27.

Conversely, assume γ and γ' have the same cycle structure; that is, $\gamma = \beta_1\cdots\beta_t$ and $\gamma' = \sigma_1\cdots\sigma_t$ are complete factorizations with β_λ and σ_λ of the same length for all $\lambda \le t$. Write $\beta_\lambda = (i_1^\lambda, \ldots, i_{r(\lambda)}^\lambda)$ and $\sigma_\lambda = (j_1^\lambda, \ldots, j_{r(\lambda)}^\lambda)$, and define

$$\alpha(i_1^\lambda) = j_1^\lambda, \quad \alpha(i_2^\lambda) = j_2^\lambda, \quad \ldots, \quad \alpha(i_{r(\lambda)}^\lambda) = j_{r(\lambda)}^\lambda$$

for all λ. Since $\beta_1\cdots\beta_t$ is a *complete* factorization, every $i \in X = \{1, \ldots, n\}$ occurs in exactly one β_λ; hence $\alpha(i)$ is defined for every $i \in X$ and $\alpha\colon X \to X$ is a single-valued function. Since every $j \in X$ occurs in some σ_λ, α is surjective, hence a bijection, and so $\alpha \in S_n$. By Proposition 1.1.27, the λth cycle of $\alpha\gamma\alpha^{-1}$ is $\big(\alpha(i_1^\lambda)\ \ldots\ \alpha(i_{r(\lambda)}^\lambda)\big) = \sigma_\lambda$. Therefore $\alpha\gamma\alpha^{-1} = \gamma'$. ∎

Example 1.1.30. If $\gamma = (1\ 2\ 3)(4\ 5)(6)$ and $\gamma' = (2\ 5\ 6)(3\ 1)(4)$, then $\gamma' = \alpha\gamma\alpha^{-1}$ where

$$\alpha = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 \\ 2 & 5 & 6 & 3 & 1 & 4 \end{pmatrix} = (1\ 2\ 5)(3\ 6\ 4).$$

Note that there are other choices for α as well.

§1.1.iii. Parity

Proposition 1.1.31

If $n \geq 2$, then every $\alpha \in S_n$ is a product of transpositions.

Proof. Of course $(1) = (1\ 2)(1\ 2)$ is such a product, as is every transposition: $(i\ j) = (i\ j)(1\ 2)(1\ 2)$. By Proposition 1.1.16 it suffices to factor an r-cycle β. If $r = 1$, then $\beta = (1\ 2)(1\ 2)$. If $r \geq 2$, then

$$\beta = (1\ 2\ \ldots\ r) = (1\ r)(1\ r-1)\cdots(1\ 3)(1\ 2).$$

One checks this by evaluating each side: the left side sends $1 \mapsto 2$, and each of $(1\ r), (1\ r-1), \ldots, (1\ 3)$ fixes 2, so the right side also sends $1 \mapsto 2$; and so on. ∎

Remark 1.1.32 (Transposition factorizations are far from unique)**.** Such a factorization is not as nice as the factorization into disjoint cycles. The transpositions occurring need not commute: $(1\ 2\ 3) = (1\ 3)(1\ 2) \neq (1\ 2)(1\ 3)$; and neither the factors nor their number is determined. For example, in S_4,

$$\begin{aligned}(1\ 2\ 3) &= (1\ 3)(1\ 2)\\ &= (2\ 3)(1\ 3)\\ &= (1\ 3)(4\ 2)(1\ 2)(1\ 4)\\ &= (1\ 3)(4\ 2)(1\ 2)(1\ 4)(2\ 3)(2\ 3).\end{aligned}$$

The only invariant that survives is the *parity* of the number of factors, and proving this is the goal of the rest of the section.

Example 1.1.33 (The 15-puzzle)**.** The **15-puzzle** consists of a starting position, which is a 4×4 array of the numbers 1 through 15 together with a symbol # (a blank), and **simple moves**. Consider the starting position

3	15	4	12
10	11	1	8
2	5	13	9
6	7	14	#

A simple move interchanges the blank with an adjacent symbol; here there are two beginning moves, interchanging # with 14 or with 9. One *wins* if, after a sequence of simple moves, the starting position is transformed into the standard array $1, 2, \ldots, 15, \#$.

To analyze the game, note that the given array is really a permutation $\alpha \in S_{16}$ of $\{1, 2, \ldots, 15, \#\}$: if the *squares* are labelled 1 through $15, \#$, then $\alpha(i)$ is the symbol occupying the ith square. The position above is

$$\alpha = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 & 7 & 8 & 9 & 10 & 11 & 12 & 13 & 14 & 15 & \# \\ 3 & 15 & 4 & 12 & 10 & 11 & 1 & 8 & 2 & 5 & 13 & 9 & 6 & 7 & 14 & \# \end{pmatrix}.$$

Each simple move is a transposition τ that moves #, and performing it from the position β yields the position $\tau\beta$. Hence to win one needs special transpositions $\tau_1, \ldots, \tau_m$ with

$$\tau_m \cdots \tau_2 \tau_1 \alpha = (1).$$

Some starting positions can be won and others cannot, as we shall see in Example 1.1.41.

Lemma 1.1.34

If $k, \ell \geq 0$ and the letters a, b, c_i, d_j are all distinct, then

$$(a\ b)(a\ c_1 \dots c_k\ b\ d_1 \dots d_\ell) = (a\ c_1 \dots c_k)(b\ d_1 \dots d_\ell)$$

and

$$(a\ b)(a\ c_1 \dots c_k)(b\ d_1 \dots d_\ell) = (a\ c_1 \dots c_k\ b\ d_1 \dots d_\ell).$$

Proof. The left side of the first equation sends

$$a \mapsto c_1 \mapsto c_1; \quad c_i \mapsto c_{i+1} \mapsto c_{i+1}\ (i < k); \quad c_k \mapsto b \mapsto a;$$

$$b \mapsto d_1 \mapsto d_1; \quad d_j \mapsto d_{j+1} \mapsto d_{j+1}\ (j < \ell); \quad d_\ell \mapsto a \mapsto b.$$

Evaluating the right side gives the same values on a, b, and all c_i, d_j, and each side fixes every other number, so the two sides are equal.

For the second equation, reverse the first and multiply both sides on the left by $(a\ b)$, using $(a\ b)(a\ b) = (1)$. ■

An illustration: $(1\ 2)(1\ 3\ 4\ 2\ 5\ 6\ 7) = (1\ 3\ 4)(2\ 5\ 6\ 7)$.

Definition 1.1.35 (Signum)**.** If $\alpha \in S_n$ and $\alpha = \beta_1 \cdots \beta_t$ is a *complete* factorization into disjoint cycles, then the **signum** of α is

$$\operatorname{sgn}(\alpha) = (-1)^{n-t}.$$

Theorem 1.1.22 shows that sgn is a well-defined function, for t is determined by α. If ε is a 1-cycle, then $t = n$ and $\operatorname{sgn}(\varepsilon) = 1$. If τ is a transposition, then it moves two numbers and fixes the other $n-2$, so $t = 1 + (n-2) = n-1$ and $\operatorname{sgn}(\tau) = (-1)^{n-(n-1)} = -1$.

Lemma 1.1.36

If $\alpha, \tau \in S_n$ with τ a transposition, then $\operatorname{sgn}(\tau\alpha) = -\operatorname{sgn}(\alpha)$.

Proof. Let $\alpha = \beta_1 \cdots \beta_t$ be a complete factorization and let $\tau = (a\ b)$.

If a and b occur in the same β, say β_1, then $\beta_1 = (a\ c_1 \dots c_k\ b\ d_1 \dots d_\ell)$ with $k, \ell \geq 0$, and Lemma 1.1.34 gives $\tau\beta_1 = (a\ c_1 \dots c_k)(b\ d_1 \dots d_\ell)$. This exhibits a complete factorization of $\tau\alpha = (\tau\beta_1)\beta_2 \cdots \beta_t$ with $t+1$ cycles, for β_1 has split in two. Therefore $\operatorname{sgn}(\tau\alpha) = (-1)^{n-(t+1)} = -\operatorname{sgn}(\alpha)$.

Otherwise a and b occur in different cycles, say $\beta_1 = (a\ c_1 \dots c_k)$ and $\beta_2 = (b\ d_1 \dots d_\ell)$. Then $\tau\alpha = (\tau\beta_1\beta_2)\beta_3 \cdots \beta_t$ and Lemma 1.1.34 gives $\tau\beta_1\beta_2 = (a\ c_1 \dots c_k\ b\ d_1 \dots d_\ell)$. So $\tau\alpha$ has a complete factorization with $t-1$ cycles, and $\operatorname{sgn}(\tau\alpha) = (-1)^{n-(t-1)} = -\operatorname{sgn}(\alpha)$. ■

Theorem 1.1.37 (sgn is multiplicative)

For all $\alpha, \beta \in S_n$,

$$\operatorname{sgn}(\alpha\beta) = \operatorname{sgn}(\alpha)\operatorname{sgn}(\beta).$$

Proof. By Proposition 1.1.31 we may write $\alpha = \tau_1 \cdots \tau_m$ as a product of transpositions; induct on m. The base step $m = 1$ is Lemma 1.1.36. If $m > 1$, the inductive hypothesis applies to

$\tau_2 \cdots \tau_m$, and

$$\begin{aligned}
\operatorname{sgn}(\alpha\beta) &= \operatorname{sgn}(\tau_1 \cdots \tau_m \beta) \\
&= -\operatorname{sgn}(\tau_2 \cdots \tau_m \beta) && \text{(Lemma 1.1.36)} \\
&= -\operatorname{sgn}(\tau_2 \cdots \tau_m)\operatorname{sgn}(\beta) && \text{(induction)} \\
&= \operatorname{sgn}(\tau_1 \cdots \tau_m)\operatorname{sgn}(\beta) && \text{(Lemma 1.1.36)} \\
&= \operatorname{sgn}(\alpha)\operatorname{sgn}(\beta). && \blacksquare
\end{aligned}$$

It follows by induction on $k \geq 2$ that $\operatorname{sgn}(\alpha_1\alpha_2 \cdots \alpha_k) = \operatorname{sgn}(\alpha_1)\operatorname{sgn}(\alpha_2) \cdots \operatorname{sgn}(\alpha_k)$.

Definition 1.1.38 (Even, odd, parity)**.** A permutation $\alpha \in S_n$ is **even** if $\operatorname{sgn}(\alpha) = 1$, and **odd** if $\operatorname{sgn}(\alpha) = -1$. We say α and β have the same **parity** if both are even or both are odd.

Theorem 1.1.39 (i) Let $\alpha \in S_n$. If α is even, then α is a product of an even number of transpositions; if α is odd, then α is a product of an odd number of transpositions.

(ii) If $\alpha = \tau_1 \cdots \tau_q = \tau'_1 \cdots \tau'_p$ are two factorizations into transpositions, then q and p have the same parity.

Proof. (i) If $\alpha = \tau_1 \cdots \tau_q$, then Theorem 1.1.37 gives $\operatorname{sgn}(\alpha) = \operatorname{sgn}(\tau_1) \cdots \operatorname{sgn}(\tau_q) = (-1)^q$, since every transposition is odd. Hence $\operatorname{sgn}(\alpha) = 1$ forces q even, and $\operatorname{sgn}(\alpha) = -1$ forces q odd.

(ii) If there were two factorizations, one into an odd and one into an even number of transpositions, then $\operatorname{sgn}(\alpha)$ would have two different values. ∎

Corollary 1.1.40

Let $\alpha, \beta \in S_n$. If α and β have the same parity, then $\alpha\beta$ is even; if they have distinct parity, then $\alpha\beta$ is odd.

Proof. If $\operatorname{sgn}(\alpha) = (-1)^q$ and $\operatorname{sgn}(\beta) = (-1)^p$, then Theorem 1.1.37 gives $\operatorname{sgn}(\alpha\beta) = (-1)^{q+p}$. ∎

Example 1.1.41 (The 15-puzzle, resolved)**.** One can show that if $\alpha \in S_{16}$ is the starting position of the 15-puzzle, then the game can be won if and only if α is an *even* permutation fixing #. One direction is clear. The blank # starts in position 16; each simple move takes # up, down, left, or right, so the total number of moves is $m = u + d + l + r$. If # is to return home, then $u = d$ and $l = r$, so $m = 2u + 2r$ is even. That is, if $\tau_m \cdots \tau_1 \alpha = (1)$, then m is even; hence $\alpha = \tau_1 \cdots \tau_m$ (because $\tau^{-1} = \tau$ for every transposition), and so α is even.

For the starting position of Example 1.1.33, the cycle notation is

$$\alpha = (1\ 3\ 4\ 12\ 9\ 2\ 15\ 14\ 7)(5\ 10)(6\ 11\ 13)(8)(\#),$$

so $t = 5$ and $\operatorname{sgn}(\alpha) = (-1)^{16-5} = -1$. Therefore α is odd, and the game starting with α *cannot* be won.

§1.2. Groups

Generalizations of the quadratic formula for cubics and quartics were discovered in the early 1500s. Over the next three centuries many tried to find analogous formulas for higher degree, but in 1824 N. H. Abel (1802–1829) proved that no such formula exists for the general polynomial of degree 5. In 1831 E. Galois (1811–1832) solved the problem completely, determining precisely

which polynomials admit such a formula; his fundamental idea was the invention of the notion of *group*. Since Galois's time groups have arisen throughout mathematics, for they are also the way to describe *symmetry*.

Intuition . The essence of a "product" is that two things are combined to form a third thing of the same kind: multiplication, addition and subtraction combine two numbers to give a number; composition combines two permutations to give a permutation.

Definition 1.2.1 (Binary operation)**.** A **(binary) operation** on a set G is a function $*\colon G \times G \to G$.

In more detail, an operation assigns an element $*(x, y) \in G$ to each *ordered* pair (x, y) of elements of G; it is more natural to write $x * y$. Thus composition of functions is the operation $(f, g) \mapsto g \circ f$, while multiplication, addition and subtraction are $(x, y) \mapsto xy$, $(x, y) \mapsto x + y$ and $(x, y) \mapsto x - y$. Composition and subtraction show why we want *ordered* pairs, for $x * y$ and $y * x$ may differ.

Note 1.2.2 (Law of substitution)**.** An operation is, as any function, single-valued. Stated explicitly, this is called the **law of substitution**: if $x = x'$ and $y = y'$, then $x * y = x' * y'$.

Definition 1.2.3 (Group)**.** A **group** is a set G equipped with an operation $*$ and a special element $e \in G$, called the **identity**, such that

(i) the **associative law** holds: $a * (b * c) = (a * b) * c$ for all $a, b, c \in G$;

(ii) $e * a = a$ for all $a \in G$;

(iii) for every $a \in G$ there is $a' \in G$ with $a' * a = e$.

Remark 1.2.4. Only a *left* identity and *left* inverses are assumed. The two-sided statements $a * e = a$ and $a * a' = e$ are theorems, proved in Proposition 1.2.8.

The set S_X of all permutations of a set X, with composition as operation and 1_X as identity, is a group: the symmetric group on X.

Intuition (Why abstraction pays)**.** We are now at the precise point where algebra becomes *abstract* algebra. In contrast to the concrete group S_n, we will prove general results about groups without specifying either their elements or their operation. Theorems then apply to *many* different groups at once, and it is often simpler to work abstractly even when dealing with one particular concrete group (see Example 1.2.19).

Definition 1.2.5 (Abelian group)**.** A group G is **abelian** if it satisfies the **commutative law**: $x * y = y * x$ for all $x, y \in G$.

The name honours Abel, who proved in 1827 a theorem equivalent to the existence of a formula for the roots of a polynomial whose Galois group is commutative. The groups S_n for $n \geq 3$ are *not* abelian, because $(1\ 2)(1\ 3) = (1\ 3\ 2)$ while $(1\ 3)(1\ 2) = (1\ 2\ 3)$.

§1.2.i. First Consequences of the Axioms

Associativity says $a*(b*c) = (a*b)*c$, so it is unambiguous to write $a*b*c$. Not all operations are associative: if $c \neq 0$, then $a - (b - c) \neq (a - b) - c$. The next lemma shows that some associativity carries over to products of four factors.

Lemma 1.2.6

If $*$ is an associative operation on a set G, then

$$(a*b)*(c*d) = \big[a*(b*c)\big]*d$$

for all $a, b, c, d \in G$.

Proof. Write $g = a*b$. Then $(a*b)*(c*d) = g*(c*d) = (g*c)*d = \big[(a*b)*c\big]*d = \big[a*(b*c)\big]*d$. ■

Lemma 1.2.7

If G is a group and $a \in G$ satisfies $a*a = a$, then $a = e$.

Proof. There is $a' \in G$ with $a'*a = e$. Multiplying both sides of $a*a = a$ on the left by a' gives $a'*(a*a) = a'*a = e$; but $a'*(a*a) = (a'*a)*a = e*a = a$. Hence $a = e$. ■

Proposition 1.2.8

Let G be a group with operation $*$ and identity e.

(i) $a*a' = e$ for all $a \in G$.

(ii) $a*e = a$ for all $a \in G$.

(iii) If $e' \in G$ satisfies $e'*a = a$ for all $a \in G$, then $e' = e$; that is, the identity is unique.

(iv) Let $a \in G$. If $b \in G$ satisfies $b*a = e$, then $b = a'$; that is, inverses are unique.

Proof. (i) By Lemma 1.2.6,

$$(a*a')*(a*a') = \big[a*(a'*a)\big]*a' = (a*e)*a' = a*(e*a') = a*a',$$

and Lemma 1.2.7 gives $a*a' = e$.
(ii) Using part (i), $a*e = a*(a'*a) = (a*a')*a = e*a = a$.
(iii) If $e'*a = a$ for all $a \in G$, then in particular $e'*e' = e'$, so $e' = e$ by Lemma 1.2.7.
(iv) By part (i), $a*a' = e$. Hence $b = b*e = b*(a*a') = (b*a)*a' = e*a' = a'$. ■

Definition 1.2.9 (Inverse)**.** If G is a group and $a \in G$, then the unique element $a' \in G$ with $a'*a = e$ is called the **inverse** of a, denoted by a^{-1}.

Lemma 1.2.10

Let G be a group.

(i) The **cancellation laws** hold: if $a, b, x \in G$ and either $x*a = x*b$ or $a*x = b*x$, then $a = b$.

(ii) $(a^{-1})^{-1} = a$ for all $a \in G$.

(iii) If $a, b \in G$, then $(a*b)^{-1} = b^{-1}*a^{-1}$. More generally, for all $n \geq 2$,

$$(a_1*a_2*\cdots*a_n)^{-1} = a_n^{-1}*\cdots*a_2^{-1}*a_1^{-1}.$$

Proof. (i)

$$a = e * a = (x^{-1} * x) * a = x^{-1} * (x * a) = x^{-1} * (x * b) = (x^{-1} * x) * b = e * b = b.$$

A similar proof, using $x * x^{-1} = e$, works when x is on the right.

(ii) By Proposition 1.2.8(i), $a * a^{-1} = e$. Uniqueness of inverses, Proposition 1.2.8(iv), says that $(a^{-1})^{-1}$ is the unique $x \in G$ with $x * a^{-1} = e$; therefore $(a^{-1})^{-1} = a$.

(iii) By Lemma 1.2.6,

$$(a * b) * (b^{-1} * a^{-1}) = \big[a * (b * b^{-1})\big] * a^{-1} = (a * e) * a^{-1} = a * a^{-1} = e,$$

so $(a * b)^{-1} = b^{-1} * a^{-1}$ by Proposition 1.2.8(iv). The general statement follows by induction on n. ■

Notation . From now on we usually write ab for $a * b$ and 1 for the identity. When a group is abelian we often use *additive* notation instead: an **additive group** is a set G with an operation $+$ and an element $0 \in G$ such that $a + (b + c) = (a + b) + c$, $0 + a = a$, and for every a there is $-a \in G$ with $(-a) + a = 0$. The inverse of a is written $-a$ rather than a^{-1}.

§1.2.ii. Examples of Groups

Example 1.2.11 (A first list of groups). (i) S_X, the set of all permutations of a set X, is a group under composition; in particular S_n is a group.

(ii) $\mathbb{Z}$ is an additive abelian group with $a * b = a + b$, identity $e = 0$, and inverse $-n$ of n. Similarly $\mathbb{Q}$, $\mathbb{R}$ and $\mathbb{C}$ are additive abelian groups.

(iii) $\mathbb{Q}^\times$, the set of all nonzero rationals, is an abelian group under ordinary multiplication, with identity 1 and inverse $1/r$ of r. Similarly $\mathbb{R}^\times$ is a multiplicative abelian group. Note that $\mathbb{Z}^\times$ is *not* a group, for no element other than ± 1 has a multiplicative inverse in $\mathbb{Z}^\times$.

(iv) The nonzero complex numbers $\mathbb{C}^\times$ form an abelian group under multiplication. For inverses, if $z = a + ib$ with $a, b \in \mathbb{R}$, let $\overline{z} = a - ib$ be its complex conjugate; then $z\overline{z} = a^2 + b^2$, so $z \neq 0$ if and only if $z\overline{z} \neq 0$, and for $z \neq 0$,

$$z^{-1} = \overline{z}/z\overline{z} = (a/z\overline{z}) - (b/z\overline{z})i.$$

(v) The **circle group** is $S^1 = \{z \in \mathbb{C} : |z| = 1\}$ under multiplication of complex numbers; the identity is 1, and the inverse of a number of modulus 1 is its complex conjugate. Although S^1 is abelian we still write it multiplicatively.

(vi) For a positive integer n, let $\Gamma_n = \{\zeta^k : 0 \leq k < n\}$ be the set of all nth roots of unity, where $\zeta = e^{2\pi i/n} = \cos(2\pi/n) + i\sin(2\pi/n)$. De Moivre's theorem shows Γ_n is an abelian group under multiplication, the inverse of a root of unity being its complex conjugate.

(vii) The plane $\mathbb{R} \times \mathbb{R}$ is an additive abelian group under vector addition: $(x, y) + (x', y') = (x + x', y + y')$. The identity is the origin $O = (0, 0)$ and the inverse of $\mathbf{v} = (x, y)$ is $-\mathbf{v} = (-x, -y)$.

(viii) The **parity group** $\mathbb{P}$ has two elements, the words "even" and "odd", with

$$\text{even} + \text{even} = \text{even} = \text{odd} + \text{odd}, \qquad \text{even} + \text{odd} = \text{odd} = \text{odd} + \text{even}.$$

(ix) The **Boolean group** $\mathcal{B}(X)$ of a set X (named after the logician G. Boole, 1815–1864) is the family of all subsets of X with addition given by symmetric difference $A + B = (A - B) \cup (B - A)$. Symmetric difference is commutative, the identity is $\varnothing$, and every element is its own inverse, since $A + A = \varnothing$. Thus $\mathcal{B}(X)$ is abelian.

Example 1.2.12 (Matrix groups and affine maps)**.** (i) A $(2 \times 2$ real) matrix is $A = \left(\begin{smallmatrix} a & c \\ b & d \end{smallmatrix}\right)$ with $a, b, c, d \in \mathbb{R}$, and the product is

$$AB = \begin{pmatrix} a & c \\ b & d \end{pmatrix} \begin{pmatrix} w & y \\ x & z \end{pmatrix} = \begin{pmatrix} aw + cx & ay + cz \\ bw + dx & by + dz \end{pmatrix};$$

each entry of AB is a dot product of a row of A with a column of B. The **determinant** of A is $\det(A) = ad - bc$, and A is **nonsingular** if $\det(A) \neq 0$. Since $\det(AB) = \det(A)\det(B)$, a product of nonsingular matrices is nonsingular. The set $GL(2, \mathbb{R})$ of all nonsingular matrices, under matrix multiplication, is a *nonabelian* group, the 2×2 real **general linear group**: the identity is $I = \left(\begin{smallmatrix} 1 & 0 \\ 0 & 1 \end{smallmatrix}\right)$ and

$$A^{-1} = \begin{pmatrix} d/\Delta & -c/\Delta \\ -b/\Delta & a/\Delta \end{pmatrix}, \qquad \Delta = ad - bc = \det(A).$$

(ii) Allowing entries in $\mathbb{Q}$ or $\mathbb{C}$ gives $GL(2, \mathbb{Q})$ and $GL(2, \mathbb{C})$; one may even allow entries in $\mathbb{Z}$, in which case $GL(2, \mathbb{Z})$ is defined to be the set of such matrices with determinant ± 1 (so that A^{-1} also has integer entries). More generally, all nonsingular $n \times n$ matrices form a group $GL(n, k)$.

(iii) All **special orthogonal** matrices, that is, all matrices $A = \left(\begin{smallmatrix} \cos\alpha & -\sin\alpha \\ \sin\alpha & \cos\alpha \end{smallmatrix}\right)$, form the 2×2 **special orthogonal group** $SO(2, \mathbb{R})$. Indeed

$$\begin{pmatrix} \cos\alpha & -\sin\alpha \\ \sin\alpha & \cos\alpha \end{pmatrix} \begin{pmatrix} \cos\beta & -\sin\beta \\ \sin\beta & \cos\beta \end{pmatrix} = \begin{pmatrix} \cos(\alpha+\beta) & -\sin(\alpha+\beta) \\ \sin(\alpha+\beta) & \cos(\alpha+\beta) \end{pmatrix}$$

by the addition theorems for sine and cosine; this calculation also shows that $SO(2, \mathbb{R})$ is abelian. The identity matrix is special orthogonal, and so is the inverse of a special orthogonal matrix. We shall see that $SO(2, \mathbb{R})$ is a disguised version of the circle group S^1, and that it consists of all rotations of the plane about the origin.

(iv) The **affine group** $\mathrm{Aff}(1, \mathbb{R})$ consists of all **affine maps** $f_{a,b} \colon \mathbb{R} \to \mathbb{R}$, $f_{a,b}(x) = ax + b$, with a, b fixed reals and $a \neq 0$. It is a group under composition: if $f_{c,d}(x) = cx + d$, then

$$f_{a,b} f_{c,d}(x) = a(cx + d) + b = acx + (ad + b) = f_{ac,\, ad+b}(x),$$

and $ac \neq 0$; the identity is $1_{\mathbb{R}} = f_{1,0}$, and the inverse of $f_{a,b}$ is $f_{a^{-1},\, -a^{-1}b}$. The composition is reminiscent of matrix multiplication:

$$\begin{pmatrix} a & b \\ 0 & 1 \end{pmatrix} \begin{pmatrix} c & d \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} ac & ad + b \\ 0 & 1 \end{pmatrix}.$$

§1.2.iii. Generalized Associativity and Powers

Definition 1.2.13 (n-expression, ultimate product)**.** An **n-expression** is an n-tuple $(a_1, \ldots, a_n)$ in the product of n copies of G. It yields elements of G as follows: choose two adjacent entries of the tuple, multiply them, and obtain an $(n-1)$-expression; repeat until a 2-expression (W, X) is reached, and then form $WX \in G$. Such an element WX is called an **ultimate product** derived from the expression. The expression **needs no parentheses** if all ultimate products derived from it are equal.

For example, from (a, b, c, d) one can reach $[(ab)c]d$, $(ab)(cd)$, $a[(bc)d]$, and so on. To say that an operation is associative is exactly to say that the two ultimate products from every 3-expression agree; it is not obvious that all ultimate products from a longer expression agree.

Theorem 1.2.14 (Generalized associativity)
If $n \geq 3$, then every n-expression $(a_1, \ldots, a_n)$ in a group G needs no parentheses.

Remark 1.2.15. Neither the identity element nor inverses are used in the proof; the hypothesis can be weakened to G being a **semigroup**, that is, a nonempty set with an associative binary operation.

Proof. Induct on n (second form). The base step $n = 3$ is associativity. For the inductive step, consider two 2-expressions obtained from $(a_1, \ldots, a_n)$:

$$(W, X) = (a_1 \cdots a_i,\ a_{i+1} \cdots a_n) \quad \text{and} \quad (Y, Z) = (a_1 \cdots a_j,\ a_{j+1} \cdots a_n).$$

We must prove $WX = YZ$. By induction each of W, X, Y, Z is the one and only ultimate product of an m-expression with $m < n$. Without loss of generality $i \leq j$. If $i = j$, then $W = Y$ and $X = Z$, so $WX = YZ$.

Assume $i < j$. Let A be the ultimate product of $(a_1, \ldots, a_i)$, let B be that of $(a_{i+1}, \ldots, a_j)$, and let C be that of $(a_{j+1}, \ldots, a_n)$; these are unambiguous by the inductive hypothesis. Then $W = A$, $Z = C$, $X = BC$ and $Y = AB$, so $WX = A(BC)$ and $YZ = (AB)C$. The base step $n = 3$ gives $WX = YZ$. ∎

Definition 1.2.16 (Powers)**.** If G is a group and $a \in G$, define a^n for $n \geq 1$ inductively by $a^1 = a$ and $a^{n+1} = aa^n$; define $a^0 = 1$ and, for $n > 0$, $a^{-n} = (a^{-1})^n$.

One shows $(a^{-1})^n = (a^n)^{-1}$; this is a special case of Lemma 1.2.10(iii). There is a hidden complication: we defined $a^3 = aa^2 = a(aa)$, but $(aa)a = a^2a$ is another reasonable contender. Associativity makes them equal, and generalized associativity shows that all powers are unambiguously defined.

Corollary 1.2.17
If G is a group, $a \in G$, and $m, n \geq 1$, then

$$a^{m+n} = a^m a^n \quad \text{and} \quad (a^m)^n = a^{mn}.$$

Proof. Both a^{m+n} and $a^m a^n$ are ultimate products of the expression having $m + n$ factors each equal to a; both $(a^m)^n$ and a^{mn} arise from the expression having mn factors each equal to a. Apply Theorem 1.2.14. ∎

In particular any two powers of the same element commute: $a^m a^n = a^{m+n} = a^{n+m} = a^n a^m$.

Proposition 1.2.18 (Laws of exponents)
Let G be a group, $a, b \in G$, and let m, n be (not necessarily positive) integers.

(i) If a and b commute, then $(ab)^n = a^n b^n$.

(ii) $(a^n)^m = a^{mn}$.

(iii) $a^m a^n = a^{m+n}$.

Proof. Exercises for the reader. ∎

Notation . If the operation is $+$, it is more natural to write na for $a + a + \cdots + a$. In additive notation Proposition 1.2.18 reads

$$n(a+b) = na + nb, \qquad m(na) = (mn)a, \qquad ma + na = (m+n)a.$$

§1.2.iv. The Order of an Element

Example 1.2.19 (Shuffling a deck)**.** Suppose a deck of cards is shuffled, changing the order from $1, 2, 3, 4, \ldots, 52$ to $2, 1, 4, 3, \ldots, 52, 51$. Shuffling again the same way restores the original order. In fact for *any* permutation α of the 52 cards, repeating α sufficiently often restores the deck. One proof uses our knowledge of permutations: write $\alpha = \beta_1 \cdots \beta_t$ with β_i a disjoint r_i-cycle. Then $\beta_i^{r_i} = (1)$ by Lemma 1.1.19, so $\beta_i^k = (1)$ for $k = r_1 \cdots r_t$; since disjoint cycles commute, Proposition 1.2.18(i) gives

$$\alpha^k = (\beta_1 \cdots \beta_t)^k = \beta_1^k \cdots \beta_t^k = (1).$$

Here is a more general result with a *simpler* proof (abstract algebra can be easier than algebra): if G is a finite group and $a \in G$, then $a^k = 1$ for some $k \geq 1$. Indeed, consider $1, a, a^2, \ldots$. Since G is finite there must be a repetition, say $a^m = a^n$ with $m > n$; then $1 = a^m a^{-n} = a^{m-n}$.

Definition 1.2.20 (Order of an element)**.** Let G be a group and $a \in G$. If $a^k = 1$ for some $k \geq 1$, then the smallest such exponent $k \geq 1$ is called the **order** of a; if no such power exists, a is said to have **infinite order**.

Every element of a finite group has finite order, by Example 1.2.19. In any group the identity is the only element of order 1; an element has order 2 if and only if it is not the identity and equals its own inverse. The matrix $A = \left(\begin{smallmatrix} 1 & 1 \\ 0 & 1 \end{smallmatrix}\right) \in GL(2, \mathbb{R})$ has infinite order, since $A^k = \left(\begin{smallmatrix} 1 & k \\ 0 & 1 \end{smallmatrix}\right) \neq I$ for all $k \geq 1$.

Lemma 1.2.21

Let G be a group and let $a \in G$ have finite order k. If $a^n = 1$, then $k \mid n$. In fact $\{n \in \mathbb{Z} : a^n = 1\}$ is the set of all multiples of k.

Proof. Let $I = \{n \in \mathbb{Z} : a^n = 1\}$. Then $0 \in I$ because $a^0 = 1$; if $n, m \in I$, then $a^{n-m} = a^n a^{-m} = 1$, so $n - m \in I$; and if $n \in I$, $q \in \mathbb{Z}$, then $a^{qn} = (a^n)^q = 1$, so $qn \in I$. Therefore I consists of all the multiples of its smallest positive element, which by definition is the order k of a. Hence $a^n = 1$ implies $k \mid n$. ∎

Proposition 1.2.22 (Order of a permutation)

Let $\alpha \in S_n$.

(i) If α is an r-cycle, then α has order r.

(ii) If $\alpha = \beta_1 \cdots \beta_t$ is a product of disjoint r_i-cycles β_i, then α has order $m = \operatorname{lcm}(r_1, \ldots, r_t)$.

(iii) If p is a prime, then α has order p if and only if it is a p-cycle or a product of disjoint p-cycles.

Proof. (i) This is Lemma 1.1.19.

(ii) Each β_i has order r_i by (i). Suppose $\alpha^M = (1)$. Since the β_i commute, $(1) = \alpha^M = \beta_1^M \cdots \beta_t^M$. The β_i are disjoint, hence so are their powers, and a product of disjoint permutations can be the identity only if every factor is; thus $\beta_i^M = (1)$ for each i, and then Lemma 1.2.21 gives $r_i \mid M$ for all i, so M is a common multiple of $r_1, \ldots, r_t$. Conversely, if $m = \operatorname{lcm}(r_1, \ldots, r_t)$, then $\alpha^m = (1)$. Hence α has order m.

(iii) Write α as a product of disjoint cycles and use (ii). ■

For example, a permutation in S_n has order 2 if and only if it is either a transposition or a product of disjoint transpositions. We can now augment Table 1.1.

Cycle structure	Number	Order	Parity
(1)	1	1	Even
(1 2)	10	2	Odd
(1 2 3)	20	3	Even
(1 2 3 4)	30	4	Odd
(1 2 3 4 5)	24	5	Even
(1 2)(3 4 5)	20	6	Odd
(1 2)(3 4)	15	2	Even
	120		

Table 1.2.: Permutations in S_5.

§1.2.v. Symmetry

What do we mean when we say that an isosceles triangle $\triangle$ is symmetric? Put $\triangle = \triangle ABC$ with base AB on the x-axis and the y-axis the perpendicular bisector of AB. Close your eyes; let $\triangle$ be reflected in the y-axis (interchanging A and B); open your eyes. You cannot tell that $\triangle$ has been reflected: $\triangle$ is symmetric about the y-axis. Had $\triangle$ been reflected in the x-axis it would be obvious. Reflection is a special kind of *isometry*.

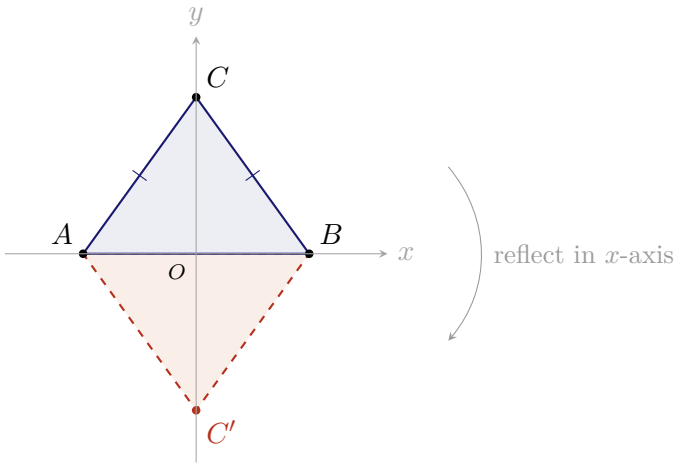


Figure 1.2.: The isosceles triangle $\triangle ABC$ is symmetric about the y-axis; its reflection in the x-axis (dashed) is visibly different.

Definition 1.2.23 (Isometry)**.** An **isometry** of the plane is a function $\varphi\colon \mathbb{R}^2 \to \mathbb{R}^2$ that is *distance preserving*: for all $P = (a, b)$ and $Q = (c, d)$,

$$\|\varphi(P) - \varphi(Q)\| = \|P - Q\|, \qquad \text{where } \|P - Q\| = \sqrt{(a-c)^2 + (b-d)^2}.$$

Write $P \cdot Q = ac + bd$ for the dot product of $P = (a, b)$ and $Q = (c, d)$. It is symmetric and bilinear, and $P \cdot P = a^2 + b^2 = \|P\|^2$, where $O = (0, 0)$ is the origin and $\|P\| = \|P - O\|$. Expanding,

$$\|P - Q\|^2 = (P - Q) \cdot (P - Q) = P \cdot P - 2(P \cdot Q) + Q \cdot Q. \tag{1.2}$$

Lemma 1.2.24 (Isometries fixing O preserve dot products)
Let $\varphi\colon \mathbb{R}^2 \to \mathbb{R}^2$ be a function.

(i) If φ is an isometry with $\varphi(O) = O$, then $\varphi(P) \cdot \varphi(Q) = P \cdot Q$ for all P, Q.

(ii) Conversely, if $\varphi(P) \cdot \varphi(Q) = P \cdot Q$ for all P, Q, then φ is an isometry and $\varphi(O) = O$.

Proof. (i) Putting $Q = O$ in the definition of an isometry gives $\|\varphi(P)\| = \|\varphi(P) - \varphi(O)\| = \|P - O\| = \|P\|$, so $\varphi(P)\cdot\varphi(P) = P\cdot P$ for every P. Now expand both sides of $\|\varphi(P) - \varphi(Q)\|^2 = \|P - Q\|^2$ using (1.2):

$$\varphi(P) \cdot \varphi(P) - 2\big(\varphi(P) \cdot \varphi(Q)\big) + \varphi(Q) \cdot \varphi(Q) = P \cdot P - 2(P \cdot Q) + Q \cdot Q.$$

The first terms on the two sides are equal, and so are the last terms; cancelling them and dividing by -2 gives $\varphi(P) \cdot \varphi(Q) = P \cdot Q$.

(ii) By (1.2) and the hypothesis,

$$\begin{aligned}\|\varphi(P) - \varphi(Q)\|^2 &= \varphi(P) \cdot \varphi(P) - 2\big(\varphi(P) \cdot \varphi(Q)\big) + \varphi(Q) \cdot \varphi(Q) \\ &= P \cdot P - 2(P \cdot Q) + Q \cdot Q = \|P - Q\|^2,\end{aligned}$$

and taking nonnegative square roots shows that φ is an isometry. Moreover $\|\varphi(O)\|^2 = \varphi(O) \cdot \varphi(O) = O \cdot O = 0$, so $\varphi(O) = O$. ∎

Remark 1.2.25. The hypothesis $\varphi(O) = O$ in (i) cannot be dropped. The map $\varphi(U) = U + V$ with $V = (1, 0)$ is an isometry, since $\|(P + V) - (Q + V)\| = \|P - Q\|$, but $\varphi(O) \cdot \varphi(O) = V \cdot V = 1 \neq 0 = O \cdot O$.

Dot products measure angles. Let $P, Q \neq O$, and write $P = \|P\|(\cos\alpha, \sin\alpha)$ and $Q = \|Q\|(\cos\beta, \sin\beta)$ in polar coordinates. Then

$$P \cdot Q = \|P\|\|Q\|(\cos\alpha\cos\beta + \sin\alpha\sin\beta) = \|P\|\|Q\|\cos(\alpha - \beta).$$

The angle $\theta \in [0, \pi]$ between P and Q (that is, $\angle POQ$) differs from $\pm(\alpha - \beta)$ by a multiple of 2π; since cos is even and 2π-periodic, $\cos\theta = \cos(\alpha - \beta)$. Hence

$$P \cdot Q = \|P\|\|Q\|\cos\theta, \qquad \text{that is,} \qquad \cos\theta = \frac{P \cdot Q}{\|P\|\|Q\|}. \tag{1.3}$$

Since cos is injective on $[0, \pi]$, this determines θ. In particular, P and Q are orthogonal ($\theta = \pi/2$) if and only if $P \cdot Q = 0$.

Corollary 1.2.26 (Isometries preserve angles) (i) If φ is an isometry with $\varphi(O) = O$ and $P, Q \neq O$, then the angle between $\varphi(P)$ and $\varphi(Q)$ equals the angle between P and Q.

(ii) More generally, if φ is any isometry and P, Q, R are points with $Q, R \neq P$, then $\angle QPR = \angle\,\varphi(Q)\varphi(P)\varphi(R)$.

In particular, every isometry preserves perpendicularity.

Proof. (i) By Lemma 1.2.24, $\varphi(P)\cdot\varphi(Q) = P\cdot Q$, and, as in its proof, $\|\varphi(P)\| = \|P\| \neq 0$ and $\|\varphi(Q)\| = \|Q\| \neq 0$. If θ and θ' are the angles between P, Q and between $\varphi(P), \varphi(Q)$, then (1.3) gives

$$\cos\theta' = \frac{\varphi(P)\cdot\varphi(Q)}{\|\varphi(P)\|\|\varphi(Q)\|} = \frac{P\cdot Q}{\|P\|\|Q\|} = \cos\theta,$$

and $\theta' = \theta$ because both lie in $[0, \pi]$.

(ii) The angle $\angle QPR$ is the angle between the vectors $Q - P$ and $R - P$. Since $(Q - P) - (R - P) = Q - R$, applying (1.2) to $Q - P$ and $R - P$ gives

$$(Q - P)\cdot(R - P) = \tfrac{1}{2}\big(\|Q - P\|^2 + \|R - P\|^2 - \|Q - R\|^2\big).$$

The right side involves only distances, as do $\|Q - P\|$ and $\|R - P\|$, so none of them change when P, Q, R are replaced by $\varphi(P), \varphi(Q), \varphi(R)$. By (1.3), the two angles have the same cosine, hence are equal.

Finally, perpendicularity means meeting at an angle of $\pi/2$, which is preserved by (ii). ∎

Notation . Denote the set of all isometries of the plane by $\mathrm{Isom}(\mathbb{R}^2)$. Its subset of isometries φ with $\varphi(O) = O$ is called the **orthogonal group** of the plane, denoted $O_2(\mathbb{R})$. If $P \neq Q$ are points, let $L[P, Q]$ denote the line they determine and PQ the line segment with endpoints P and Q.

Example 1.2.27 (Rotations, reflections, translations)**.** (i) Given an angle θ, the **rotation** R_θ about the origin O is defined by $R_\theta(O) = O$, and for $P \neq O$, by rotating the segment PO through θ (counterclockwise if $\theta > 0$). One can rotate about any point of the plane.

(ii) The **reflection** ρ_ℓ in a line ℓ, called its **axis**, fixes each point of ℓ; if $P \notin \ell$, then $\rho_\ell(P) = P'$ where ℓ is the perpendicular bisector of PP'. Pretending ℓ is a mirror, P' is the mirror image of P. Now $\rho_\ell \in \mathrm{Isom}(\mathbb{R}^2)$, and if ℓ passes through the origin, then $\rho_\ell \in O_2(\mathbb{R})$.

(iii) Given a point V, **translation** by V is $\tau_V\colon \mathbb{R}^2 \to \mathbb{R}^2$, $\tau_V(U) = U + V$. A translation fixes the origin if and only if $V = O$, so the identity is the only translation that is also a rotation.

Lemma 1.2.28

If φ is an isometry of the plane, then distinct points P, Q, R are collinear if and only if $\varphi(P), \varphi(Q), \varphi(R)$ are collinear.

Proof. Suppose P, Q, R are collinear, with R between P and Q, so that $\|P - Q\| = \|P - R\| + \|R - Q\|$. If $\varphi(P), \varphi(Q), \varphi(R)$ were not collinear, they would be the vertices of a triangle, and the triangle inequality would give

$$\|\varphi(P) - \varphi(Q)\| < \|\varphi(P) - \varphi(R)\| + \|\varphi(R) - \varphi(Q)\|,$$

contradicting that φ preserves distance.

We can prove the other direction with similar method. ∎

Every isometry φ is injective: if $P \neq Q$, then $\|\varphi(P) - \varphi(Q)\| = \|P - Q\| \neq 0$. Surjectivity is less obvious.

Proposition 1.2.29

Every isometry φ of the plane fixing the origin is a linear transformation.

Proof. Let $C_d = \{Q \in \mathbb{R}^2 : \|Q - O\| = d\}$ be the circle of radius $d > 0$ centred at O. If $P \in C_d$, then $d = \|\varphi(P) - \varphi(O)\| = \|\varphi(P) - O\|$, so $\varphi(C_d) \subseteq C_d$.

Let $P \neq O$ and $r \in \mathbb{R}$. If $\|P - O\| = p$, then $\|rP - O\| = |r|p$, so $rP \in L[O, P] \cap C_{|r|p}$. Since φ preserves collinearity (Lemma 1.2.28), $\varphi(L[O, P] \cap C_{|r|p}) \subseteq L[O, \varphi(P)] \cap C_{|r|p}$, and a line meets a circle in at most two points; hence $\varphi(rP) = \pm r\varphi(P)$. To eliminate the minus sign, suppose $r > 0$. The origin lies between $-rP$ and P, so the distance from $-rP$ to P is $rp + p$, whereas the distance from rP to P is $|rp - p|$. Since $rp + p \neq |rp - p|$ and φ preserves distance, $\varphi(rP) \neq -r\varphi(P)$. A similar argument works for $r < 0$. Therefore $\varphi(rP) = r\varphi(P)$.

It remains to prove $\varphi(P + Q) = \varphi(P) + \varphi(Q)$. If O, P, Q are collinear, choose U on $L[O, P]$ at distance 1 from the origin, so $P = pU$, $Q = qU$ and $P + Q = (p + q)U$. Since φ preserves scalar multiplication,

$$\varphi(P) + \varphi(Q) = p\varphi(U) + q\varphi(U) = (p + q)\varphi(U) = \varphi(P + Q).$$

If O, P, Q are not collinear, then $P + Q$ is the point S such that O, P, Q, S are the vertices of a parallelogram (parallelogram law). Since φ preserves distance, $O = \varphi(O), \varphi(P), \varphi(Q), \varphi(S)$ are also the vertices of a parallelogram, so $\varphi(S) = \varphi(P) + \varphi(Q)$; that is, $\varphi(P+Q) = \varphi(P)+\varphi(Q)$. ■

Corollary 1.2.30

Every isometry $\varphi\colon \mathbb{R}^2 \to \mathbb{R}^2$ is a bijection, and every isometry fixing O is a nonsingular linear transformation.

Proof. First assume $\varphi(O) = O$, so φ is linear by Proposition 1.2.29. Since φ is injective, $P = \varphi(e_1)$, $Q = \varphi(e_2)$ is a basis of $\mathbb{R}^2$, where $e_1 = (1, 0)$, $e_2 = (0, 1)$. Hence $\psi\colon aP + bQ \mapsto ae_1 + be_2$ is a well-defined function inverse to φ; so φ is a bijection and is nonsingular.

Now let φ be any isometry, say $\varphi(O) = U$. Then $\tau_{-U} \circ \varphi$ fixes O, so $\tau_{-U} \circ \varphi = \theta$ is a nonsingular linear transformation, and $\varphi = \tau_U \circ \theta$ is a composite of bijections. ■

Proposition 1.2.31

Both $\mathrm{Isom}(\mathbb{R}^2)$ and $O_2(\mathbb{R})$ are groups under composition.

Proof. Clearly $1_{\mathbb{R}^2}$ is an isometry. If φ' and φ are isometries, then

$$\big\|(\varphi'\varphi)(P) - (\varphi'\varphi)(Q)\big\| = \|\varphi(P) - \varphi(Q)\| = \|P - Q\|,$$

so composition is an operation on $\mathrm{Isom}(\mathbb{R}^2)$. Each φ is a bijection by Corollary 1.2.30, and φ^{-1} is also an isometry: $\|P - Q\| = \big\|\varphi(\varphi^{-1}(P)) - \varphi(\varphi^{-1}(Q))\big\| = \big\|\varphi^{-1}(P) - \varphi^{-1}(Q)\big\|$. Composition of functions is always associative. The same proof applies to $O_2(\mathbb{R})$. ■

Corollary 1.2.32 (i) If O, P, Q are noncollinear points and φ, ψ are isometries of the plane with $\varphi(O) = \psi(O) = O$, $\varphi(P) = \psi(P)$ and $\varphi(Q) = \psi(Q)$, then $\varphi = \psi$.

(ii) More generally, if A, B, C are noncollinear points and φ, ψ are isometries of the plane agreeing on A, B and C, then $\varphi = \psi$.

Proof. (i) Since φ and ψ fix O, both are linear transformations by Proposition 1.2.29. Since O, P, Q are noncollinear, the list P, Q is linearly independent in $\mathbb{R}^2$: a nontrivial relation $aP + bQ = O$ would put P and Q on a common line through O. As $\dim(\mathbb{R}^2) = 2$, the list P, Q is a basis, and two linear transformations agreeing on a basis are equal.

(ii) Let $U = \varphi(A) = \psi(A)$, and set $\varphi' = \tau_{-U} \circ \varphi \circ \tau_A$ and $\psi' = \tau_{-U} \circ \psi \circ \tau_A$. These are composites of isometries, hence isometries, and $\varphi'(O) = \varphi(A) - U = O$; likewise $\psi'(O) = O$. Moreover

$$\varphi'(B - A) = \varphi(B) - U = \psi(B) - U = \psi'(B - A),$$

and similarly $\varphi'(C-A) = \psi'(C-A)$. The points $O, B-A, C-A$ are the translates of A, B, C by $-A$, so they are noncollinear. By (i), $\varphi' = \psi'$, and therefore $\varphi = \tau_U \circ \varphi' \circ \tau_{-A} = \tau_U \circ \psi' \circ \tau_{-A} = \psi$. ∎

Remark 1.2.33. In (i) the hypothesis $\varphi(O) = \psi(O) = O$ cannot be dropped, and in (ii) two points are not enough. If O, P, Q are noncollinear, the identity and the reflection $\rho_{L[P,Q]}$ both fix P and Q, yet they are different: the reflection moves $O \notin L[P, Q]$.

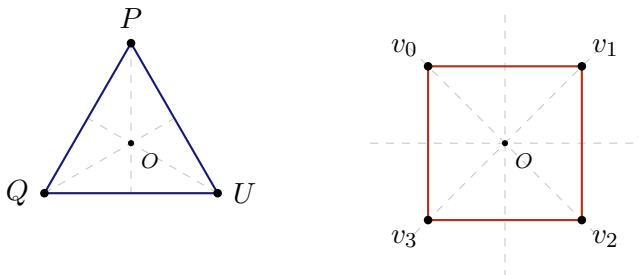
Example 1.2.34 (Triangles)**.** If $\triangle$ is a triangle with vertices P, Q, U and φ is an isometry, then $\varphi(\triangle)$ is the triangle with vertices $\varphi(P), \varphi(Q), \varphi(U)$. If moreover $\varphi(\triangle) = \triangle$, then φ permutes the vertices. Assume the centre of $\triangle$ is O.

- If $\triangle$ is *isosceles* with equal sides PQ and PU, and ρ_ℓ is the reflection with axis $\ell = L[O, P]$, then $\rho_\ell(\triangle) = \triangle$; we may describe ρ_ℓ by the transposition $(Q\ U)$. On the other hand $\rho_{\ell'}(\triangle) \neq \triangle$ for $\ell' = L[O, Q]$.
- If $\triangle$ is *equilateral*, then also $\rho_{L[O,Q]}$ and $\rho_{L[O,U]}$ carry $\triangle$ to itself (described by $(P\ U)$ and $(P\ Q)$), and so do the rotations about O by 120° and 240° (described by $(P\ Q\ U)$ and $(P\ U\ Q)$).
- If $\triangle$ is *scalene*, then $\varphi(\triangle) = \triangle$ implies $\varphi = 1$.

So an equilateral triangle is "more symmetric" than an isosceles triangle, which in turn is "more symmetric" than a scalene one.

Definition 1.2.35 (Symmetry group)**.** The **symmetry group** $\Sigma(\Omega)$ of a figure Ω in the plane is the set of all isometries φ of the plane with $\varphi(\Omega) = \Omega$. The elements of $\Sigma(\Omega)$ are called **symmetries** of Ω.

It is straightforward to see that $\Sigma(\Omega)$ is always a group.



$\Sigma(\pi_3) = D_6$, $|\Sigma(\pi_3)| = 6$ $\quad$ $\Sigma(\pi_4) = D_8$, $|\Sigma(\pi_4)| = 8$

Figure 1.3.: Reflection axes of a regular 3-gon and a regular 4-gon.

Example 1.2.36 (Symmetry groups of regular polygons)**.** (i) A regular 3-gon π_3 is an equilateral triangle, and $|\Sigma(\pi_3)| = 6$, as we saw in Example 1.2.34.

(ii) Let π_4 be a square with vertices $\{v_0, v_1, v_2, v_3\}$, drawn with centre at the origin and sides parallel to the axes. Every $\varphi \in \Sigma(\pi_4)$ permutes the vertices, and φ is determined by $\{\varphi(v_i)\}$, so there are at most $24 = 4!$ symmetries. Not every permutation arises, however: if v_i and v_j are adjacent, then $\|v_i - v_j\| = 1$, whereas $\|v_0 - v_2\| = \sqrt{2} = \|v_1 - v_3\|$; since isometries preserve distance, φ must preserve adjacency. There are exactly eight symmetries (Theorem 1.2.39): the identity, the three rotations about O by $90°, 180°, 270°$, and four reflections with axes $L[v_0, v_2]$, $L[v_1, v_3]$, the x-axis and the y-axis. This group is the **dihedral group with 8 elements**, denoted D_8.

(iii) The symmetry group of a regular pentagon π_5 with vertices $v_0, \ldots, v_4$ and centre O has 10 elements: the rotations about the origin by $(72j)°$ for $0 \le j \le 4$, together with the reflections with axes $L[O, v_k]$ for $0 \le k \le 4$. It is denoted D_{10}.

Definition 1.2.37 (Dihedral group)**.** A group D_{2n} with exactly $2n$ elements is called a **dihedral group** if it contains an element a of order n and an element b of order 2 such that

$$bab = a^{-1}.$$

Note 1.2.38. The definition is purely algebraic: it does not depend on geometry. If $n = 2$, then D_4 is abelian; if $n \ge 3$, then D_{2n} is not abelian. One can show that there is essentially only one dihedral group with $2n$ elements (any two such are isomorphic). The name comes from F. Klein's *dihedron*, two congruent regular polygons of zero thickness pasted together.

Theorem 1.2.39

The symmetry group $\Sigma(\pi_n)$ of a regular n-gon π_n is a dihedral group with $2n$ elements.

Proof. Let π_n have vertices $v_0, \ldots, v_{n-1}$ and centre O. Define a to be the rotation about O by $(360/n)°$,

$$a(v_i) = \begin{cases} v_{i+1} & \text{if } 0 \le i < n-1, \\ v_0 & \text{if } i = n-1, \end{cases} \qquad \text{and} \qquad b(v_i) = \begin{cases} v_0 & \text{if } i = 0, \\ v_{n-i} & \text{if } 1 \le i \le n-1, \end{cases}$$

where b is the reflection with axis $L[O, v_0]$. Clearly a has order n and b has order 2. There are n distinct symmetries $1, a, a^2, \ldots, a^{n-1}$, and $b, ab, a^2b, \ldots, a^{n-1}b$ are also all distinct by cancellation. If $a^s = a^r b$ with $0 \le r \le n-1$ and $s = 0, 1$, then $a^s(v_i) = a^r b(v_i)$ for all i; now $a^s(v_0) = v_s$ while $a^r b(v_0) = v_{r-1}$, forcing $s = r - 1$; but with $i = 1$ we get $a^{r-1}(v_1) = v_{r+1}$ while $a^r b(v_1) = v_{r-1}$, a contradiction. So $a^s \ne a^r b$ for all r, s, and we have exhibited $2n$ distinct symmetries.

There are no others. We may assume the centre O of π_n is the origin, so every symmetry φ fixes O and hence is a linear transformation (Proposition 1.2.29). The vertices adjacent to v_0, namely v_1 and v_{n-1}, are the closest vertices to v_0; that is, $\|v_i - v_0\| > \|v_1 - v_0\|$ for $2 \le i \le n-2$. So if $\varphi(v_0) = v_j$, then $\varphi(v_1) = v_{j+1}$ or $\varphi(v_1) = v_{j-1}$. In the first case a^j agrees with φ on v_0 and v_1, so Corollary 1.2.32 gives $\varphi = a^j$; in the second case $\varphi = a^j b$ for the same reason. Therefore $|\Sigma(\pi_n)| = 2n$.

Finally, $bab = a^{-1}$: by Corollary 1.2.32 it suffices to evaluate both sides on v_0 and v_1, and $bab(v_0) = v_{n-1} = a^{-1}(v_0)$ while $bab(v_1) = v_0 = a^{-1}(v_1)$. ∎

Note 1.2.40 (Symmetry in calculus)**.** Symmetry arises in calculus when describing curves $f(x, y) = 0$:

(i) symmetry about the x-axis: the equation is unaltered when y is replaced by $-y$ — in our language, ρ_x;

(ii) symmetry about the y-axis: x replaced by $-x$ — ρ_y;

(iii) symmetry with respect to the origin: both replaced — R_{180};

(iv) symmetry about the line $y = x$: x and y interchanged — ρ_L with L the 45° line.

Thus $f(x, y) = f(x, -y)$ says the graph Γ of $f(x, y) = 0$ is symmetric about the x-axis, for $(a, b) \in \Gamma$ implies $(a, -b) \in \Gamma$.

§1.3. Subgroups and Lagrange's Theorem

A subgroup of a group G is a subset which is a group under the same operation as in G. The following definition makes this precise.

Definition 1.3.1 (Closed subset)**.** Let $*$ be an operation on a set G and let $S \subseteq G$. We say S is **closed** under $*$ if $x * y \in S$ for all $x, y \in S$.

The operation on G is a function $*\colon G \times G \to G$; if $S \subseteq G$, then $S \times S \subseteq G \times G$, and S being closed means $*(S \times S) \subseteq S$. For example, $\mathbb{Z}$ is a subset of the additive group $\mathbb{Q}$ closed under $+$. However $\mathbb{Q}^\times$ is closed under multiplication but not under $+$: both 2 and -2 lie in $\mathbb{Q}^\times$, yet $-2 + 2 = 0 \notin \mathbb{Q}^\times$.

Definition 1.3.2 (Subgroup)**.** A subset H of a group G is a **subgroup** if

(i) $1 \in H$;

(ii) if $x, y \in H$, then $xy \in H$ (that is, H is closed under $*$);

(iii) if $x \in H$, then $x^{-1} \in H$.

Notation . We write $H \leq G$ to denote that H is a subgroup of G. Both $\{1\}$ and G are always subgroups of G. A subgroup H is **proper** if $H \neq G$, written $H < G$, and **nontrivial** if $H \neq \{1\}$.

Proposition 1.3.3

Every subgroup $H \leq G$ of a group G is itself a group.

Proof. Axiom (ii) shows H is closed under the operation of G, so H has an operation (the restriction of $*$ to $H \times H$). This operation is associative, since $(xy)z = x(yz)$ holds for all $x, y, z \in G$, in particular for all $x, y, z \in H$. Axiom (i) gives the identity and axiom (iii) gives inverses. ∎

Intuition . It is *quicker* to check that a subset is a subgroup than to verify the group axioms from scratch, because associativity is inherited and need not be verified again.

Example 1.3.4 (Subgroups)**.** (i) $O_2(\mathbb{R})$, the isometries fixing the origin, is a subgroup of $\mathrm{Isom}(\mathbb{R}^2)$. If $\Omega \subseteq \mathbb{R}^2$, then $\Sigma(\Omega)$ is also a subgroup of $\mathrm{Isom}(\mathbb{R}^2)$; and if the centre of gravity of Ω is at the origin, then $\Sigma(\Omega) \leq O_2(\mathbb{R})$.

(ii) The four permutations

$$\mathbf{V} = \{(1),\ (1\ 2)(3\ 4),\ (1\ 3)(2\ 4),\ (1\ 4)(2\ 3)\}$$

form a group, because $\mathbf{V}$ is a subgroup of S_4: $(1) \in \mathbf{V}$; $\alpha^2 = (1)$ for each $\alpha \in \mathbf{V}$, so $\alpha^{-1} = \alpha \in \mathbf{V}$; and the product of any two distinct elements of $\mathbf{V} - \{(1)\}$ is the third one. One calls $\mathbf{V}$ the **four-group**, or **Klein group** ($\mathbf{V}$ abbreviates *Vierergruppe*).

Note how much work this saves: verifying $a(bc) = (ab)c$ directly would involve $4^3 = 64$ equations.

(iii) If $\mathbb{R}^2$ is the plane as an additive abelian group, then any line L through the origin is a subgroup: choose a point (a, b) on L and note that L consists of all scalar multiples (ra, rb).

One can shorten the list of items needed to verify that a subset is a subgroup.

Proposition 1.3.5

A subset H of a group G is a subgroup if and only if H is nonempty and, whenever $x, y \in H$, then $xy^{-1} \in H$.

Proof. If H is a subgroup, then $H \neq \varnothing$ since $1 \in H$; and if $x, y \in H$, then $y^{-1} \in H$ by (iii), so $xy^{-1} \in H$ by (ii).

Conversely, assume H satisfies the new condition. Since H is nonempty it contains some h; taking $x = h = y$ gives $1 = hh^{-1} \in H$, so (i) holds. If $y \in H$, set $x = 1$ (legitimate now that $1 \in H$) to get $y^{-1} = 1y^{-1} \in H$, so (iii) holds. Finally $(y^{-1})^{-1} = y$ by Lemma 1.2.10, so if $x, y \in H$, then $y^{-1} \in H$ and $xy = x(y^{-1})^{-1} \in H$, giving (ii). ■

Remark 1.3.6. Since every subgroup contains 1, one may replace "H is nonempty" by "$1 \in H$". If the operation in G is addition, the condition reads: H is a nonempty subset with $x, y \in H \Rightarrow x - y \in H$.

Note 1.3.7 (Historical)**.** For Galois a group was just a subset H of S_n closed under composition. A. Cayley, in 1854, was the first to define an abstract group, mentioning associativity, inverses and identity explicitly.

Proposition 1.3.8

A nonempty subset H of a *finite* group G is a subgroup if and only if H is closed under the operation of G. In particular, a nonempty subset of S_n is a subgroup if and only if it is closed under composition.

Proof. Every subgroup is nonempty and closed, by axioms (i) and (ii).

Conversely, assume H is a nonempty closed subset, so axiom (ii) holds. Then H contains all powers of its elements: there is some $a \in H$, and $a^n \in H$ for all $n \geq 1$. As in Example 1.2.19, every element of G has finite order, so there is m with $a^m = 1$; hence $1 \in H$ and axiom (i) holds. Finally, if $h \in H$ and $h^m = 1$, then $h^{-1} = h^{m-1} \in H$ (since $hh^{m-1} = 1 = h^{m-1}h$), so axiom (iii) holds. ■

Remark 1.3.9. This proposition can be false for infinite G: the subset $\mathbb{N}$ of the additive group $\mathbb{Z}$ is closed under $+$ but is not a subgroup.

Example 1.3.10 (The alternating group)**.** The subset of S_n consisting of all the *even* permutations is a subgroup, because it is closed under multiplication: even $\circ$ even = even (Corollary 1.1.40). This subgroup is called the **alternating group** on n letters and is denoted by A_n.

Note 1.3.11. The name comes from polynomials: if $f(x) = (x - u_1)\cdots(x - u_n)$, then $D = \prod_{i<j}(u_i - u_j)$ changes sign when one permutes the roots, since $\prod_{i<j}[\alpha(u_i) - \alpha(u_j)] = \pm D$. The sign *alternates* as various α are applied; it does not change for α in the alternating group.

§1.3.i. Cyclic Groups

Definition 1.3.12 (Cyclic subgroup, cyclic group)**.** If G is a group and $a \in G$, write

$$\langle a \rangle = \{a^n : n \in \mathbb{Z}\} = \{\text{all powers of } a\};$$

$\langle a \rangle$ is called the **cyclic subgroup** of G generated by a. A group G is **cyclic** if $G = \langle a \rangle$ for some $a \in G$, and then a is called a **generator** of G.

It is easy to see $\langle a \rangle$ is a subgroup: $1 = a^0 \in \langle a \rangle$; $a^n a^m = a^{n+m} \in \langle a \rangle$; $a^{-1} \in \langle a \rangle$. By Example 1.2.11(vi), for every $n \geq 1$ the multiplicative group Γ_n of all nth roots of unity is cyclic, with the primitive nth root of unity $\zeta = e^{2\pi i/n}$ as a generator. A cyclic group can have several different generators; for instance $\langle a \rangle = \langle a^{-1} \rangle$.

Proposition 1.3.13

If $G = \langle a \rangle$ is a cyclic group of order n, then a^k is a generator of G if and only if $\gcd(k, n) = 1$.

Proof. If a^k is a generator, then $a \in \langle a^k \rangle$, so $a = a^{ks}$ for some s. Hence $a^{ks-1} = 1$, and Lemma 1.2.21 gives $n \mid (ks - 1)$; that is, $ks - 1 = tn$ for some integer t, so $sk - tn = 1$ and $\gcd(k, n) = 1$.

Conversely, if $1 = sk + tn$, then $a = a^{sk+tn} = a^{sk}$ (because $a^{tn} = 1$), so $a \in \langle a^k \rangle$. Hence $G = \langle a \rangle \leq \langle a^k \rangle$, and so $G = \langle a^k \rangle$. ■

Corollary 1.3.14

The number of generators of a cyclic group of order n is $\phi(n)$, where ϕ is the Euler ϕ-function.

Proof. Immediate from Proposition 1.3.13. ■

Proposition 1.3.15

Every subgroup S of a cyclic group $G = \langle a \rangle$ is itself cyclic. In fact a^m is a generator of S, where m is the smallest positive integer with $a^m \in S$.

Proof. We may assume $S \neq \{1\}$, the statement being obvious otherwise. Let $I = \{m \in \mathbb{Z} : a^m \in S\}$. Then $0 \in I$, since $a^0 = 1 \in S$; if $m, n \in I$, then $a^m a^{-n} = a^{m-n} \in S$, so $m - n \in I$; and if $m \in I$, $i \in \mathbb{Z}$, then $(a^m)^i = a^{im} \in S$, so $im \in I$. Since $S \neq \{1\}$ there is $a^q \in S$ with $a^q \neq 1$, so $q \in I$ and $I \neq \{0\}$. Hence, if k is the smallest positive integer in I, then $k \mid m$ for every $m \in I$.

We claim $\langle a^k \rangle = S$. Clearly $\langle a^k \rangle \leq S$. Conversely, take $s \in S$; then $s = a^m$ for some $m \in I$, and $m = k\ell$ for some ℓ, so $s = a^{k\ell} \in \langle a^k \rangle$. ■

Proposition 1.3.16

Let G be a finite group and let $a \in G$. Then the order of a is the number of elements in $\langle a \rangle$.

Proof. Since G is finite, there is $k \geq 1$ such that $1, a, a^2, \ldots, a^{k-1}$ are k distinct elements while $1, a, \ldots, a^k$ has a repetition; hence $a^k = a^i$ for some $0 \leq i < k$. If $i \geq 1$, then $a^{k-i} = 1$, contradicting the original list having no repetitions. Therefore $a^k = a^0 = 1$, and k is the order of a, being the smallest such positive integer.

Put $H = \{1, a, a^2, \ldots, a^{k-1}\}$, so $|H| = k$; it suffices to show $H = \langle a \rangle$. Clearly $H \subseteq \langle a \rangle$. Conversely, given $a^i \in \langle a \rangle$, the division algorithm gives $i = qk + r$ with $0 \leq r < k$, and then $a^i = a^{qk+r} = (a^k)^q a^r = a^r \in H$. ∎

Definition 1.3.17 (Order of a group)**.** If G is a finite group, then the number of elements in G, denoted $|G|$, is called the **order** of G.

Abuse of Notation 1.3.18. The word "order" is used in two senses: the order of an element $a \in G$, and the order $|G|$ of a group. Proposition 1.3.16 reconciles them: the order of a equals $|\langle a \rangle|$.

Proposition 1.3.19

A group G of order n is cyclic if and only if, for each divisor d of n, there is at most one cyclic subgroup of order d.

Proof. Suppose $G = \langle a \rangle$ is cyclic of order n. We claim $\langle a^{n/d} \rangle$ has order d. Clearly $(a^{n/d})^d = a^n = 1$, and d is smallest with this property: if $(a^{n/d})^r = 1$, then $n \mid (n/d)r$ by Lemma 1.2.21, so $(n/d)r = ns$ for some integer s, giving $r = ds \geq d$.

For uniqueness, let $C \leq G$ have order d. By Proposition 1.3.15, C is cyclic, say $C = \langle x \rangle$ with $x = a^m$ of order d. Then $1 = x^d = a^{md}$, so $n \mid md$ by Lemma 1.2.21, say $md = nk$. Therefore $x = a^m = (a^{n/d})^k$, so $C = \langle x \rangle \subseteq \langle a^{n/d} \rangle$; since both subgroups have order d, $C = \langle a^{n/d} \rangle$.

Conversely, define a relation on G by $a \equiv b$ if $\langle a \rangle = \langle b \rangle$. This is an equivalence relation, and the class $[a]$ consists of all the generators of $C = \langle a \rangle$; denote it by $\mathrm{gen}(C)$. Then

$$G = \bigcup_{C \text{ cyclic}} \mathrm{gen}(C), \qquad \text{so} \qquad n = |G| = \sum_{C} |\mathrm{gen}(C)|,$$

the sum being over all cyclic subgroups of G. But $|\mathrm{gen}(C)| = \phi(|C|)$ by Corollary 1.3.14. By hypothesis G has at most one cyclic subgroup of any given order, so

$$n = \sum_{C} |\mathrm{gen}(C)| \leq \sum_{d \mid n} \phi(d) = n.$$

Therefore, for each divisor d of n, there must be a cyclic subgroup C of order d contributing $\phi(d)$ to the sum. In particular there is a cyclic subgroup of order n, and so G is cyclic. ∎

§1.3.ii. Subgroups Generated by a Subset

Proposition 1.3.20

The intersection $\bigcap_{i \in I} H_i$ of any family of subgroups of a group G is again a subgroup of G. In particular, if $H, K \leq G$, then $H \cap K \leq G$.

Proof. Let $D = \bigcap_{i\in I} H_i$. Then $D \neq \varnothing$ because $1 \in H_i$ for all i. If $x \in D$, then $x \in H_i$ for all i; each H_i is a subgroup, so $x^{-1} \in H_i$ for all i and $x^{-1} \in D$. Finally, if $x, y \in D$, then x, y lie in every H_i, hence so does xy, and $xy \in D$. ■

Corollary 1.3.21

If X is a subset of a group G, then there is a subgroup $\langle X\rangle$ of G containing X that is smallest in the sense that $\langle X\rangle \leq H$ for every subgroup $H \leq G$ with $X \subseteq H$.

Proof. There do exist subgroups of G containing X; for example G itself. Define $\langle X\rangle = \bigcap_{X\subseteq H} H$, the intersection of all subgroups H of G containing X. By Proposition 1.3.20 this is a subgroup, and it contains X because each H does. If H is any subgroup containing X, then H is one of the subgroups being intersected, so $\langle X\rangle \leq H$. ■

Note 1.3.22. There is no restriction on X; in particular $X = \varnothing$ is allowed. Since $\varnothing \subseteq H$ for every subgroup H, the subgroup $\langle\varnothing\rangle$ is the intersection of *all* subgroups of G, one of which is $\{1\}$; hence $\langle\varnothing\rangle = \{1\}$.

Definition 1.3.23 (Subgroup generated by a subset)**.** If X is a subset of a group G, then $\langle X\rangle$ is called the **subgroup generated by** X.

Example 1.3.24. (i) If $G = \langle a\rangle$ is cyclic with generator a, then G is generated by $X = \{a\}$.

(ii) The symmetry group $\Sigma(\pi_n)$ of a regular n-gon is generated by a, b, where a is the rotation about the origin by $(360/n)^\circ$ and b is a reflection (Theorem 1.2.39). These generators satisfy $a^n = 1$, $b^2 = 1$ and $bab = a^{-1}$, so $\Sigma(\pi_n)$ is a dihedral group D_{2n}.

Definition 1.3.25 (Word)**.** Let X be a nonempty subset of a group G. A **word** on X is either the identity element or an element of G of the form

$$w = x_1^{e_1} x_2^{e_2} \cdots x_n^{e_n},$$

where $n \geq 1$, $x_i \in X$ for all i, and $e_i = \pm 1$ for all i.

Proposition 1.3.26

If X is a nonempty subset of a group G, then $\langle X\rangle$ is the set of all words on X.

Proof. Let W be the set of all words on X; we first show $W \leq G$. By definition $1 \in W$. If $w = x_1^{e_1} \cdots x_n^{e_n}$ and $w' = y_1^{f_1} \cdots y_m^{f_m}$ lie in W, then $ww' = x_1^{e_1} \cdots x_n^{e_n} y_1^{f_1} \cdots y_m^{f_m}$ is again a word on X, so $ww' \in W$; and $w^{-1} = x_n^{-e_n} x_{n-1}^{-e_{n-1}} \cdots x_1^{-e_1} \in W$. Thus W is a subgroup containing every element of X, and so $\langle X\rangle \leq W$.

For the reverse inequality, let S be any subgroup containing X. Since S is a subgroup it contains x^e whenever $x \in X$ and $e = \pm 1$, and it contains all products of such elements; hence $W \leq S$. Therefore $W \leq \bigcap S = \langle X\rangle$. ■

Proposition 1.3.27 (gcd and lcm, group-theoretically)

Let a and b be integers and let $A = \langle a\rangle$ and $B = \langle b\rangle$ be the cyclic subgroups of $\mathbb{Z}$ they generate.

(i) If $A + B = \{\alpha + \beta : \alpha \in A,\ \beta \in B\}$, then $A + B = \langle d\rangle$, where $d = \gcd(a, b)$.

(ii) $A \cap B = \langle m \rangle$, where $m = \operatorname{lcm}(a, b)$.

Proof. (i) It is straightforward to check that $A+B$ is a subgroup of $\mathbb{Z}$; in fact $A+B$ is precisely the set of all linear combinations of a and b. By Proposition 1.3.15 the subgroup $A+B$ is cyclic: $A + B = \langle d \rangle$, where d can be chosen to be the smallest nonnegative number in $A + B$. Thus d is a common divisor of a and b, and it is the smallest such; that is, $d = \gcd(a, b)$.

(ii) If $c \in A \cap B$, then $a \mid c$ and $b \mid c$, so every element of $A \cap B$ is a common multiple of a and b; conversely every common multiple lies in the intersection. By Proposition 1.3.15, $A \cap B = \langle m \rangle$, where m may be taken to be the smallest nonnegative number in $A \cap B$; that is, $m = \operatorname{lcm}(a, b)$. ∎

§1.3.iii. Cosets and Lagrange's Theorem

Perhaps the most fundamental fact about subgroups H of a finite group G is that their orders are constrained. Certainly $|H| \leq |G|$, but it turns out that $|H|$ must *divide* $|G|$. To prove this we introduce cosets.

Definition 1.3.28 (Coset)**.** If $H \leq G$ and $a \in G$, then the **coset** aH is the subset

$$aH = \{ah : h \in H\}.$$

Of course $a = a1 \in aH$. Cosets are usually *not* subgroups: if $a \notin H$, then $1 \notin aH$, for $1 = ah$ would give the contradiction $a = h^{-1} \in H$. In $*$ notation we write $a * H = \{a * h : h \in H\}$; in additive notation, $a + H = \{a + h : h \in H\}$.

Note 1.3.29. The cosets just defined are often called *left* cosets; there are also *right* cosets $Ha = \{ha : h \in H\}$. We shall work almost exclusively with left cosets.

Example 1.3.30 (Cosets)**.** (i) Consider $\mathbb{R}^2$ as an additive abelian group and let ℓ be a line through the origin O; as in Example 1.3.4(iii), ℓ is a subgroup. If $\beta \in \mathbb{R}^2$, then the coset $\beta + \ell$ is the line ℓ' through β *parallel* to ℓ, for if $r\alpha \in \ell$, then the parallelogram law gives $\beta + r\alpha \in \ell'$ (see Figure 1.4).

(ii) If $G = S_3$ and $H = \langle (1\ 2) \rangle$, there are exactly three cosets of H, each of size 2:

$$\begin{aligned} H &= \{(1),\ (1\ 2)\} = (1\ 2)H, \\ (1\ 3)H &= \{(1\ 3),\ (1\ 2\ 3)\} = (1\ 2\ 3)H, \\ (2\ 3)H &= \{(2\ 3),\ (1\ 3\ 2)\} = (1\ 3\ 2)H. \end{aligned}$$

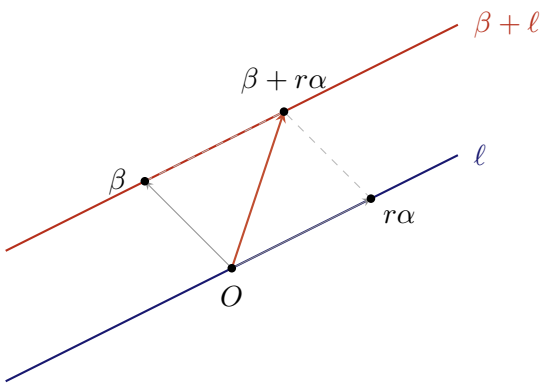


Figure 1.4.: The coset $\beta + \ell$ is the line through β parallel to ℓ.

Observation . In these examples different cosets of a given subgroup do not overlap. That is no accident: cosets are equivalence classes.

Proposition 1.3.31

If $H \leq G$, then the relation on G defined by

$$a \equiv b \quad \text{if} \quad a^{-1}b \in H$$

is an equivalence relation, and the equivalence class of $a \in G$ is the coset aH.

Proof. *Reflexive:* $a^{-1}a = 1 \in H$. *Symmetric:* if $a^{-1}b \in H$, then $(a^{-1}b)^{-1} = b^{-1}a \in H$, since subgroups are closed under inverses. *Transitive:* if $a^{-1}b, b^{-1}c \in H$, then $(a^{-1}b)(b^{-1}c) = a^{-1}c \in H$, since subgroups are closed under multiplication.

For the classes: if $x \equiv a$, then $x^{-1}a \in H$, and Lemma 1.3.32 gives $x \in xH = aH$; thus $[a] \subseteq aH$. Conversely, if $x = ah \in aH$, then $x^{-1}a = (ah)^{-1}a = h^{-1}a^{-1}a = h^{-1} \in H$, so $x \equiv a$ and $x \in [a]$. Hence $[a] = aH$. ■

Lemma 1.3.32

Let $H \leq G$ and let $a, b \in G$.

(i) $aH = bH$ if and only if $b^{-1}a \in H$. In particular, $aH = H$ if and only if $a \in H$.

(ii) If $aH \cap bH \neq \varnothing$, then $aH = bH$.

(iii) $|aH| = |H|$ for all $a \in G$.

Proof. (i) and (ii) are the general facts about equivalence classes applied to Proposition 1.3.31: distinct classes are disjoint, and two elements are equivalent if and only if their classes coincide. The second statement of (i) follows because $H = 1H$.

(iii) The function $f\colon H \to aH$ given by $f(h) = ah$ is a bijection, with inverse $ah \mapsto a^{-1}(ah) = h$. Therefore H and aH have the same number of elements. ■

Theorem 1.3.33 (Lagrange's theorem)

If H is a subgroup of a finite group G, then $|H|$ is a divisor of $|G|$.

Proof. Let $\{a_1H, a_2H, \ldots, a_tH\}$ be the family of all the distinct cosets of H in G. Then

$$G = a_1H \cup a_2H \cup \cdots \cup a_tH,$$

because each $g \in G$ lies in the coset gH, and $gH = a_iH$ for some i. Moreover distinct cosets a_iH and a_jH are disjoint, by Lemma 1.3.32(ii). It follows that

$$|G| = |a_1H| + |a_2H| + \cdots + |a_tH|.$$

But $|a_iH| = |H|$ for all i, by Lemma 1.3.32(iii), and so $|G| = t|H|$. ■

Definition 1.3.34 (Index)**.** The **index** of a subgroup H in G, denoted $[G : H]$, is the number of cosets of H in G.

When G is finite, the index $[G : H]$ is the number t in the formula $|G| = t|H|$ above, so that

$$|G| = [G : H]\,|H|.$$

In particular the index $[G : H]$ is also a divisor of $|G|$.

Corollary 1.3.35

If H is a subgroup of a finite group G, then $[G : H] = |G|/|H|$.

Proof. Immediate from Lagrange's theorem. ■

Example 1.3.36. By Theorem 1.2.39, the symmetry group $\Sigma(\pi_n)$ of a regular n-gon is a dihedral group of order $2n$. It contains the cyclic subgroup $\langle a\rangle$ of order n generated by a rotation a, and $[\Sigma(\pi_n) : \langle a\rangle] = 2$. Thus there are two cosets: $\langle a\rangle$ and $b\langle a\rangle$, where b is any symmetry outside $\langle a\rangle$.

Corollary 1.3.37

If G is a finite group and $a \in G$, then the order of a is a divisor of $|G|$.

Proof. By Proposition 1.3.16, the order of a equals the order of the subgroup $H = \langle a\rangle$; now apply Lagrange's theorem. ■

This explains why the orders of the elements of S_5 displayed in Table 1.2 are divisors of 120.

Corollary 1.3.38

If a finite group G has order m, then $a^m = 1$ for all $a \in G$.

Proof. By Corollary 1.3.37, a has order d with $d \mid m$, say $m = dk$. Thus $a^m = a^{dk} = (a^d)^k = 1$. ■

Corollary 1.3.39

If p is a prime, then every group G of order p is cyclic.

Proof. Choose $a \in G$ with $a \neq 1$ and let $H = \langle a\rangle$. By Lagrange's theorem $|H|$ divides $|G| = p$; since p is prime and $|H| > 1$, it follows that $|H| = p = |G|$, and so $H = G$. ■

Question 1.3.40 (Is the converse of Lagrange's theorem true?)**.** Lagrange's theorem says the order of a subgroup of a finite group G divides $|G|$. If d divides $|G|$, must there exist a subgroup of G of order d?

Remark 1.3.41. The answer is *no*: the alternating group A_4 has order 12 but has no subgroup of order 6.

Lecture 2: Residue Classes and Units Modulo n

§1.4. Residue classes

18 September 2026

Fix an integer $n \geq 2$. Under addition, the set

$$n\mathbb{Z} = \{nk : k \in \mathbb{Z}\}$$

is a subgroup of $\mathbb{Z}$. Its cosets are the basic objects of this lecture.

Definition 1.4.1 (Residue class)**.** For $a \in \mathbb{Z}$, the **residue class of a modulo n** is the coset

$$[a]_n = a + n\mathbb{Z} = \{a + kn : k \in \mathbb{Z}\}.$$

The set of all residue classes modulo n is denoted by

$$\mathbb{Z}_n = \{[a]_n : a \in \mathbb{Z}\}.$$

Proposition 1.4.2 (Equality of residue classes)
For $a, b \in \mathbb{Z}$, the following conditions are equivalent:

$$[a]_n = [b]_n, \qquad a - b \in n\mathbb{Z}, \qquad n \mid (a - b).$$

Proof. By the criterion for equality of cosets, $a + n\mathbb{Z} = b + n\mathbb{Z}$ if and only if $a - b \in n\mathbb{Z}$. The latter means precisely that $a - b = kn$ for some $k \in \mathbb{Z}$, which is equivalent to $n \mid (a - b)$. ∎

Thus $[a]_n = [b]_n$ exactly when $a \equiv b \pmod{n}$. In particular, changing an integer by a multiple of n does not change its residue class.

Proposition 1.4.3 (Division algorithm)
Given $a \in \mathbb{Z}$, there exist unique integers q, r such that

$$a = qn + r \qquad \text{and} \qquad 0 \leq r \leq n - 1.$$

Proof. For existence, consider

$$S = \{a - kn : k \in \mathbb{Z}\} \cap \mathbb{N}_0.$$

The set S is nonempty: for example, if $m \geq |a|$, then $a - (-m)n = a + mn \geq 0$. By the well-ordering principle, S has a least element r. Thus $r = a - qn$ for some $q \in \mathbb{Z}$, so $a = qn + r$. If $r \geq n$, then

$$r - n = a - (q + 1)n \geq 0,$$

so $r - n \in S$, contradicting the minimality of r. Hence $0 \leq r < n$, equivalently $0 \leq r \leq n - 1$.

For uniqueness, suppose $a = qn + r = q'n + r'$, where $0 \leq r, r' < n$. Then

$$r - r' = (q' - q)n.$$

But $|r - r'| < n$, and the only multiple of n with absolute value strictly less than n is 0. Therefore $r = r'$, and then $q = q'$. ∎

Corollary 1.4.4 (Standard representative)

Every residue class in $\mathbb{Z}_n$ contains exactly one integer r satisfying $0 \le r \le n-1$.

Proof. If $a = qn + r$ is the division-algorithm decomposition of a, then $a - r = qn$, so $[a]_n = [r]_n$ by Proposition 1.4.2. If both $r, r' \in \{0, \dots, n-1\}$ represent the same class, then $n \mid (r - r')$ and $|r - r'| < n$; hence $r = r'$. ∎

Corollary 1.4.5

The integers $0, 1, \dots, n-1$ form a complete set of coset representatives for $n\mathbb{Z}$ in $\mathbb{Z}$. Consequently,

$$\mathbb{Z}_n = \{[0]_n, [1]_n, \dots, [n-1]_n\}$$

and these n residue classes are pairwise distinct.

Proof. Existence and distinctness are exactly the two assertions of Corollary 1.4.4. ∎

§1.5. The additive group $\mathbb{Z}_n$

We would like to add residue classes by adding representatives. Since a class has many representatives, we must first check that the answer does not depend on the choices.

Lemma 1.5.1 (Addition is well-defined)

The assignment

$$([a]_n, [b]_n) \longmapsto [a+b]_n$$

is well-defined.

Proof. Suppose $[a_1]_n = [a_2]_n$ and $[b_1]_n = [b_2]_n$. Then $n \mid (a_1 - a_2)$ and $n \mid (b_1 - b_2)$. Therefore

$$n \mid (a_1 - a_2) + (b_1 - b_2) = (a_1 + b_1) - (a_2 + b_2),$$

which gives $[a_1 + b_1]_n = [a_2 + b_2]_n$. Thus different choices of representatives produce the same residue class. ∎

Define addition on $\mathbb{Z}_n$ by

$$[a]_n + [b]_n = [a+b]_n.$$

Theorem 1.5.2

The set $\mathbb{Z}_n$ is an abelian group of order n under addition.

Proof. The operation is well-defined by Lemma 1.5.1, and its value is again an element of $\mathbb{Z}_n$. For $[a]_n, [b]_n, [c]_n \in \mathbb{Z}_n$, associativity follows from associativity in $\mathbb{Z}$:

$$([a]_n + [b]_n) + [c]_n = [(a+b)+c]_n = [a+(b+c)]_n = [a]_n + ([b]_n + [c]_n).$$

The identity is $[0]_n$, because $[0]_n + [a]_n = [a]_n = [a]_n + [0]_n$. The inverse of $[a]_n$ is $[-a]_n$, since $[a]_n + [-a]_n = [0]_n$. Finally,

$$[a]_n + [b]_n = [a+b]_n = [b+a]_n = [b]_n + [a]_n,$$

so the group is abelian. By Corollary 1.4.5, it has exactly n elements. ∎

Example 1.5.3 (Subgroups of $\mathbb{Z}_{12}$)**.** For $[a]_{12} \in \mathbb{Z}_{12}$, write

$$H_a = \langle [a]_{12} \rangle = \{[ka]_{12} : k \in \mathbb{Z}\}.$$

The distinct possibilities are

$$\begin{aligned} H_0 &= \{[0]_{12}\}, \\ H_6 &= \{[0]_{12}, [6]_{12}\}, \\ H_4 = H_8 &= \{[0]_{12}, [4]_{12}, [8]_{12}\}, \\ H_3 = H_9 &= \{[0]_{12}, [3]_{12}, [6]_{12}, [9]_{12}\}, \\ H_2 = H_{10} &= \{[0]_{12}, [2]_{12}, [4]_{12}, [6]_{12}, [8]_{12}, [10]_{12}\}, \\ H_1 = H_5 = H_7 = H_{11} &= \mathbb{Z}_{12}. \end{aligned}$$

For example, repeatedly adding $[5]_{12}$ gives $[0]_{12}, [5]_{12}, [10]_{12}, [3]_{12}, \ldots$, and eventually every residue class occurs. Notice that $1, 5, 7, 11$ are precisely the standard representatives that are relatively prime to 12.

The subgroup sizes in this example are governed by a general formula, proved in Proposition 1.6.6 after we establish Bézout's identity.

§1.6. Multiplication and the group of units

Lemma 1.6.1 (Multiplication is well-defined)
The assignment

$$([a]_n, [b]_n) \longmapsto [ab]_n$$

is well-defined.

Proof. Suppose $[a_1]_n = [a_2]_n$ and $[b_1]_n = [b_2]_n$. Write $a_1 - a_2 = kn$ and $b_1 - b_2 = \ell n$. Then

$$a_1 b_1 - a_2 b_2 = a_1(b_1 - b_2) + b_2(a_1 - a_2) = n(a_1 \ell + b_2 k).$$

Thus $n \mid (a_1 b_1 - a_2 b_2)$, so $[a_1 b_1]_n = [a_2 b_2]_n$. ∎

We may therefore define multiplication on $\mathbb{Z}_n$ by

$$[a]_n [b]_n = [ab]_n.$$

Not every nonzero residue class has a multiplicative inverse. The invertible classes are characterized by the greatest common divisor.

Lemma 1.6.2 (The gcd is constant on residue classes)
If $[a]_n = [b]_n$, then $\gcd(a, n) = \gcd(b, n)$.

Proof. We have $a = b + kn$ for some $k \in \mathbb{Z}$. An integer divides both a and n if and only if it divides both $a - kn = b$ and n. Hence the pairs (a, n) and (b, n) have exactly the same common divisors, and therefore the same greatest common divisor. ∎

Definition 1.6.3 (Units modulo n)**.** The set of **units modulo** n is

$$\mathbb{Z}_n^* = \{[a]_n \in \mathbb{Z}_n : \gcd(a, n) = 1\}.$$

This definition does not depend on the representative a, by Lemma 1.6.2.

The following result supplies the inverse needed to make $\mathbb{Z}_n^*$ a group.

Proposition 1.6.4 (Bézout's identity)

Let $a, b \in \mathbb{Z}$, not both zero, and let d be the least positive integer in

$$S = \{ax + by : x, y \in \mathbb{Z}\}.$$

Then $d = \gcd(a, b)$. In particular, there exist $x, y \in \mathbb{Z}$ such that

$$\gcd(a, b) = ax + by.$$

Proof. First, S contains a positive integer: one of a, b is nonzero, and either that integer or its negative is positive. Hence the least positive element d exists by the well-ordering principle. Write $d = ax_0 + by_0$.

We claim that $d \mid a$. By the division algorithm, write $a = qd + r$ with $0 \leq r < d$. Then

$$r = a - qd = a - q(ax_0 + by_0) = a(1 - qx_0) + b(-qy_0),$$

so $r \in S$. If $r > 0$, this contradicts the minimality of d; hence $r = 0$ and $d \mid a$. The same argument gives $d \mid b$, so d is a common divisor of a and b.

Conversely, we show that d is the *greatest* common divisor, and for this we use the other half of the information we have about d: that it is a $\mathbb{Z}$-linear combination of a and b. Let c be any common divisor of a and b, say $a = ca'$ and $b = cb'$ with $a', b' \in \mathbb{Z}$. Then

$$d = ax_0 + by_0 = ca'x_0 + cb'y_0 = c\,(a'x_0 + b'y_0),$$

and $a'x_0 + b'y_0 \in \mathbb{Z}$, so $c \mid d$. Thus every common divisor of a and b divides d.

Finally, $c \mid d$ together with $d > 0$ forces $|c| \leq d$, in particular $c \leq d$. So d is a common divisor of a and b that is at least as large as every other one, which is exactly to say $d = \gcd(a, b)$. ∎

Corollary 1.6.5 (Bézout criterion)

For integers a, b, one has $\gcd(a, b) = 1$ if and only if there exist $x, y \in \mathbb{Z}$ such that $ax + by = 1$.

Proof. If $\gcd(a, b) = 1$, the required integers exist by Proposition 1.6.4. Conversely, if $ax + by = 1$, every common divisor of a and b divides 1; hence their greatest common divisor is 1. ∎

Proposition 1.6.6 (Order of a residue class)

In the additive group $\mathbb{Z}_n$,

$$|\langle [a]_n \rangle| = \frac{n}{\gcd(a, n)}.$$

Consequently, $[a]_n$ generates $\mathbb{Z}_n$ if and only if $\gcd(a, n) = 1$.

Proof. Let $d = \gcd(a, n)$, and write $a = da'$ and $n = dn'$. Then $\gcd(a', n') = 1$. For a positive integer k,

$$k[a]_n = [0]_n \Leftrightarrow n \mid ka \Leftrightarrow dn' \mid kda' \Leftrightarrow n' \mid ka'.$$

Choose $x, y \in \mathbb{Z}$ with $a'x + n'y = 1$. If $n' \mid ka'$, then

$$k = k(a'x + n'y) = (ka')x + n'(ky),$$

so $n' \mid k$. Conversely, $n' \mid k$ plainly implies $n' \mid ka'$. Therefore the least positive k for which $k[a]_n = [0]_n$ is $n' = n/d$. This is the order of $[a]_n$, and hence the size of the cyclic subgroup it generates. The final assertion follows because this size is n exactly when $d = 1$. ∎

Corollary 1.6.7 (Criterion for invertibility)

A residue class $[a]_n \in \mathbb{Z}_n$ has a multiplicative inverse if and only if $\gcd(a, n) = 1$.

Proof. If $\gcd(a, n) = 1$, choose $x, y \in \mathbb{Z}$ with $ax + ny = 1$. Reducing this equality modulo n gives $[a]_n[x]_n = [ax]_n = [1]_n$, so $[x]_n$ is an inverse of $[a]_n$.

Conversely, suppose $[a]_n[x]_n = [1]_n$ for some $x \in \mathbb{Z}$. Then $ax \equiv 1 \pmod{n}$, so $ax - 1 = kn$ for some $k \in \mathbb{Z}$. Therefore $ax + n(-k) = 1$, and Corollary 1.6.5 gives $\gcd(a, n) = 1$. ■

Theorem 1.6.8

The set $\mathbb{Z}_n^*$ is an abelian group under multiplication.

Proof. Multiplication is well-defined by Lemma 1.6.1. If $[a]_n, [b]_n \in \mathbb{Z}_n^*$, then $\gcd(a, n) = \gcd(b, n) = 1$. Choose Bézout relations $ax + ny = 1$ and $bu + nv = 1$. Multiplying them gives

$$1 = (ax + ny)(bu + nv) = ab(xu) + n(axv + buy + nyv),$$

so $\gcd(ab, n) = 1$ by Corollary 1.6.5. Hence $[ab]_n$ is again in $\mathbb{Z}_n^*$, proving closure.

Associativity and commutativity are inherited from integer multiplication, and $[1]_n$ is the identity. Every element of $\mathbb{Z}_n^*$ has an inverse by Corollary 1.6.7; moreover that inverse also belongs to $\mathbb{Z}_n^*$, since it is itself invertible. Thus $\mathbb{Z}_n^*$ is an abelian group. ■

Example 1.6.9 (The units modulo 12)**.** We have

$$\mathbb{Z}_{12}^* = \{[1]_{12}, [5]_{12}, [7]_{12}, [11]_{12}\}.$$

Each nonidentity element is its own inverse: $5^2 \equiv 7^2 \equiv 11^2 \equiv 1 \pmod{12}$. Thus $\mathbb{Z}_{12}^*$ is a group of order 4, but it is not cyclic.

Lecture 3: Group Actions

§1.7. Group Actions

Rotman §2.7

Groups of permutations led us to abstract groups. In this section we go back the other way: every group can be realized as a group of permutations, and, more usefully, a group can be studied by letting it *act* on a set. The payoff includes a classification of the groups of order at most 7, Cauchy's theorem, the structure of p-groups, and the simplicity of the alternating groups A_n for $n \geq 5$.

§1.7.i. What We Need from Homomorphisms and Quotient Groups

This section uses some vocabulary from Rotman ğ2.5–ğ2.6. We collect it here.

Recall (Homomorphisms, normal subgroups, center)**.** Let G and H be groups.

- A function $f\colon G \to H$ is a **homomorphism** if $f(ab) = f(a)f(b)$ for all $a, b \in G$. A bijective homomorphism is an **isomorphism**; if one exists we write $G \cong H$.
- The **kernel** of a homomorphism f is $\ker f = \{a \in G : f(a) = 1\}$, and its **image** is $\operatorname{Im} f = \{f(a) : a \in G\}$.
- A subgroup $K \leq G$ is **normal**, written $K \trianglelefteq G$, if $gkg^{-1} \in K$ for all $g \in G$ and

$k \in K$.

- The **center** of G is $Z(G) = \{z \in G : zg = gz \text{ for all } g \in G\}$.

Proposition 1.7.1 (Facts from Rotman §2.5–§2.6)
Let G be a group.

(i) If $f\colon G \to H$ is a homomorphism, then $f(1) = 1$, $f(a^{-1}) = f(a)^{-1}$, and $\ker f \trianglelefteq G$. Moreover f is injective if and only if $\ker f = \{1\}$. If f is an isomorphism, then a and $f(a)$ have the same order for every $a \in G$.

(ii) Every subgroup of $Z(G)$ is a normal subgroup of G; in particular $Z(G) \trianglelefteq G$, and every subgroup of an abelian group is normal.

(iii) *(Quotient groups)* If $K \trianglelefteq G$, then the family G/K of all cosets of K is a group under $(aK)(bK) = abK$, with identity K; it has order $|G/K| = [G : K]$.

(iv) *(Correspondence theorem)* If $K \trianglelefteq G$, then $S \mapsto S/K$ is a bijection from the subgroups S with $K \leq S \leq G$ onto the subgroups of G/K; moreover $S \trianglelefteq G$ if and only if $S/K \trianglelefteq G/K$.

(v) Two cyclic groups of the same order are isomorphic. Together with Corollary 1.3.39: every group of prime order p is isomorphic to $\mathbb{Z}_p$.

(vi) *(Rotman, Proposition 2.122)* If G is a finite *abelian* group, then G has a subgroup of order d for every divisor d of $|G|$. In particular, if a prime p divides $|G|$, then G contains an element of order p.

The next few facts appear as exercises in Rotman; we shall use them repeatedly, so we prove them here.

Lemma 1.7.2 (Rotman, Exercise 2.56)
If $f\colon X \to Y$ is a bijection, then $\varphi\colon S_X \to S_Y$, $\varphi(\sigma) = f\sigma f^{-1}$, is an isomorphism. In particular, if $|X| = n$, then $S_X \cong S_n$. If X and Y are finite, then φ carries each cycle $(x_1 \ \dots \ x_r)$ to the cycle $(f(x_1) \ \dots \ f(x_r))$, so it preserves cycle structure and parity.

Proof. $f\sigma f^{-1}$ is a composite of bijections $Y \to X \to X \to Y$, so it lies in S_Y. We have $\varphi(\sigma)\varphi(\tau) = f\sigma f^{-1} f\tau f^{-1} = f\sigma\tau f^{-1} = \varphi(\sigma\tau)$, and $\pi \mapsto f^{-1}\pi f$ is an inverse of φ. The statement about cycles is checked exactly as in Lemma 1.1.26: if $\sigma(x) = x'$, then $f\sigma f^{-1}(f(x)) = f(x')$. ∎

Lemma 1.7.3 (Rotman, Exercise 2.38)
If $x^2 = 1$ for every x in a group G, then G is abelian.

Proof. The hypothesis says $x = x^{-1}$ for all x. Hence, for $x, y \in G$, $xy = (xy)^{-1} = y^{-1}x^{-1} = yx$. ∎

Lemma 1.7.4 (Rotman, Exercise 2.40)
A finite group G of even order contains an element of order 2.

Proof. Pair each $x \in G$ with x^{-1}. The elements with $x \neq x^{-1}$ fall into pairs $\{x, x^{-1}\}$, so there is an even number of them. Hence the set $\{x \in G : x = x^{-1}\}$ has even size too. It contains 1, so it contains some $x \neq 1$, and such an x satisfies $x^2 = 1$, that is, x has order 2. ∎

Lemma 1.7.5 (Rotman, Exercise 2.87)

If $G/Z(G)$ is cyclic, then G is abelian.

Proof. Write $Z = Z(G)$ and $G/Z = \langle aZ \rangle$. Every coset has the form $(aZ)^i = a^i Z$, so every $g \in G$ can be written $g = a^i z$ with $z \in Z$. If also $h = a^j z'$, then, since z, z' commute with everything,

$$gh = a^i z a^j z' = a^{i+j} z z' = a^j z' a^i z = hg.$$ ∎

Lemma 1.7.6 (Rotman, Proposition 2.127)

Let a, b be commuting elements of a group, of orders m and n. If $\gcd(m, n) = 1$, then ab has order mn.

Proof. Since $ab = ba$, we have $(ab)^r = a^r b^r$ for all r, so $(ab)^{mn} = (a^m)^n (b^n)^m = 1$. Conversely suppose $(ab)^k = 1$. Then $a^k = b^{-k}$ lies in $\langle a \rangle \cap \langle b \rangle$. The order of this subgroup divides both $m = |\langle a \rangle|$ and $n = |\langle b \rangle|$ by Lagrange's theorem, hence equals 1; so $a^k = 1 = b^k$, and Lemma 1.2.21 gives $m \mid k$ and $n \mid k$. Write $k = mq$ and choose x, y with $mx + ny = 1$ (Corollary 1.6.5); then $q = mqx + nqy = kx + nqy$ is divisible by n, so $mn \mid k$. Therefore the order of ab is mn. ∎

§1.7.ii. Cayley's Theorem

The next result, due to A. Cayley (1821–1895), shows that abstract groups are not so far removed from permutations.

Theorem 1.7.7 (Cayley)

Every group G is (isomorphic to) a subgroup of the symmetric group S_G. In particular, if $|G| = n$, then G is isomorphic to a subgroup of S_n.

Proof. For each $a \in G$, define the **translation** $\tau_a \colon G \to G$ by $\tau_a(x) = ax$. (If $a \neq 1$, then τ_a is *not* a homomorphism, since $\tau_a(1) = a \neq 1$.) For $a, b \in G$ and every $x \in G$,

$$(\tau_a \circ \tau_b)(x) = \tau_a(bx) = a(bx) = (ab)x = \tau_{ab}(x),$$

by associativity, so that

$$\tau_a \tau_b = \tau_{ab}.$$

It follows that each τ_a is a bijection, with inverse $\tau_{a^{-1}}$: $\tau_a \tau_{a^{-1}} = \tau_1 = 1_G = \tau_{a^{-1}} \tau_a$. Thus $\tau_a \in S_G$.

Define $\varphi \colon G \to S_G$ by $\varphi(a) = \tau_a$. The displayed equation says $\varphi(a)\varphi(b) = \varphi(ab)$, so φ is a homomorphism. It is injective: if $\varphi(a) = \varphi(b)$, then $\tau_a(x) = \tau_b(x)$ for all x, and $x = 1$ gives $a = b$. Hence $G \cong \operatorname{Im} \varphi \leq S_G$.

The last statement follows from Lemma 1.7.2: if $|G| = n$, then $S_G \cong S_n$. ∎

Remark 1.7.8. In the proof, the permutation τ_a is just the row labelled a of the multiplication table of G.

Example 1.7.9 (Cayley's theorem for $\mathbb{Z}_3$)**.** Take $G = \mathbb{Z}_3 = \{[0], [1], [2]\}$ (written additively, so $\tau_a(x) = a + x$). Labelling $[0], [1], [2]$ as $1, 2, 3$, we get $\tau_{[0]} = (1)$, $\tau_{[1]} = (1\ 2\ 3)$, $\tau_{[2]} = (1\ 3\ 2)$. Thus $\mathbb{Z}_3$ is isomorphic to the subgroup $A_3 = \langle (1\ 2\ 3) \rangle$ of S_3.

To tell the truth, Cayley's theorem by itself is only mildly interesting. However, the identical proof works in a larger setting that is more useful. Even when H is not a normal subgroup, we still write G/H for the family of all (left) cosets of H in G.

Theorem 1.7.10 (Representation on cosets)

Let G be a group and let H be a subgroup of G having finite index n. Then there is a homomorphism $\varphi\colon G \to S_n$ with $\ker \varphi \leq H$.

Proof. For each $a \in G$ define $\tau_a\colon G/H \to G/H$ by $\tau_a(xH) = axH$; this is well defined, because $xH = yH$ means $y^{-1}x \in H$, and then $(ay)^{-1}(ax) = y^{-1}x \in H$, so $axH = ayH$. For $a, b \in G$,

$$(\tau_a \circ \tau_b)(xH) = \tau_a(bxH) = a(bx)H = (ab)xH = \tau_{ab}(xH),$$

so again $\tau_a\tau_b = \tau_{ab}$. As before each τ_a is a bijection with inverse $\tau_{a^{-1}}$, for $\tau_a\tau_{a^{-1}} = \tau_1 = 1_{G/H} = \tau_{a^{-1}}\tau_a$; so $\tau_a \in S_{G/H}$, and $\varphi\colon G \to S_{G/H}$, $\varphi(a) = \tau_a$, is a homomorphism.

If $a \in \ker \varphi$, then $\tau_a = 1_{G/H}$, so $axH = xH$ for all $x \in G$; taking $x = 1$ gives $aH = H$, and hence $a \in H$ by Lemma 1.3.32(i). Thus $\ker \varphi \leq H$.

Finally $|G/H| = n$, so $S_{G/H} \cong S_n$ by Lemma 1.7.2; composing φ with this isomorphism does not change the kernel. ∎

When $H = \{1\}$, this is Cayley's theorem.

§1.7.iii. Groups of Small Order

We now classify all groups of order at most 7. By Proposition 1.7.1(v), every group of prime order p is isomorphic to $\mathbb{Z}_p$, so up to isomorphism there is just one group of order p. Of the orders $2, \ldots, 7$, four are primes $(2, 3, 5, 7)$, so we need only look at orders 4 and 6.

Proposition 1.7.11

Every group G of order 4 is isomorphic to either $\mathbb{Z}_4$ or the four-group $\mathbf{V}$. Moreover, $\mathbb{Z}_4$ and $\mathbf{V}$ are not isomorphic.

Proof. By Lagrange's theorem, every element of G other than 1 has order 2 or 4. If there is an element of order 4, then G is cyclic, and $G \cong \mathbb{Z}_4$. Otherwise $x^2 = 1$ for all $x \in G$, so G is abelian by Lemma 1.7.3.

Choose distinct $x, y \in G$, neither equal to 1. Then $xy \notin \{1, x, y\}$: $xy = 1$ would give $y = x^{-1} = x$, $xy = x$ would give $y = 1$, and $xy = y$ would give $x = 1$. Hence

$$G = \{1, x, y, xy\}.$$

The product of any two distinct elements of order 2 is the third one (e.g. $x \cdot xy = x^2y = y$), and exactly the same is true in $\mathbf{V}$. Therefore the bijection $f\colon G \to \mathbf{V}$ defined by

$$f(1) = (1), \quad f(x) = (1\ 2)(3\ 4), \quad f(y) = (1\ 3)(2\ 4), \quad f(xy) = (1\ 4)(2\ 3)$$

is an isomorphism.

Finally, $\mathbb{Z}_4 \ncong \mathbf{V}$: an isomorphism preserves orders of elements (Proposition 1.7.1(i)), but $\mathbb{Z}_4$ has an element of order 4, while every element of $\mathbf{V}$ has order at most 2. ∎

Proposition 1.7.12

If G is a group of order 6, then G is isomorphic to either $\mathbb{Z}_6$ or S_3. Moreover, $\mathbb{Z}_6$ and S_3 are not isomorphic.

Proof. By Lagrange's theorem, the only possible orders of nonidentity elements are 2, 3, and 6. Of course $G \cong \mathbb{Z}_6$ if G has an element of order 6. By Lemma 1.7.4, G contains an element t of order 2; let $T = \langle t \rangle = \{1, t\}$.

Since $[G : T] = 3$, the representation on the cosets of T (Theorem 1.7.10) is a homomorphism $\rho\colon G \to S_{G/T} \cong S_3$ with $\ker \rho \leq T$. Thus $\ker \rho = \{1\}$ or $\ker \rho = T$.

Case $\ker \rho = \{1\}$. Then ρ is injective, and hence bijective, since $|G| = 6 = |S_3|$. So $G \cong S_3$.

Case $\ker \rho = T$. Then $T \trianglelefteq G$, and the quotient group G/T is defined. It has order 3, so it is cyclic: there is $a \in G$ with $G/T = \{T, aT, a^2T\}$. Moreover ρ_t is the permutation

$$\rho_t = \begin{pmatrix} T & aT & a^2T \\ tT & taT & ta^2T \end{pmatrix}.$$

Since $t \in T = \ker \rho$, ρ_t is the identity; in particular $aT = \rho_t(aT) = taT$, so that $a^{-1}ta \in T = \{1, t\}$ by Lemma 1.3.32(i). But $a^{-1}ta \neq 1$ because $t \neq 1$; hence $a^{-1}ta = t$, that is, $ta = at$.

Now $aT \neq T$ and $a^2T \neq T$, so $a \neq 1$ and $a^2 \neq 1$; thus a has order 3 or 6. In either case G has an element of order 6: if a has order 3, then at has order 6 by Lemma 1.7.6. Therefore G is cyclic of order 6, and $G \cong \mathbb{Z}_6$.

Finally, $\mathbb{Z}_6 \ncong S_3$, for one is abelian and the other is not. ∎

Remark 1.7.13. Cayley stated this proposition in 1854. In 1878, however, he wrote that for $n = 6$ "there are three groups": the cyclic one, and two more generated by α, β with $\alpha^2 = 1$, $\beta^3 = 1$, one with $\alpha\beta = \beta\alpha$ and one with $\alpha\beta = \beta^2\alpha$. His list is $\mathbb{Z}_6$, $\mathbb{Z}_2 \times \mathbb{Z}_3$, and S_3; but $\mathbb{Z}_2 \times \mathbb{Z}_3 \cong \mathbb{Z}_6$. Even Homer nods.

Classifying groups of order 8 is more difficult, for we have not yet developed enough theory. It turns out that there are exactly 5 nonisomorphic groups of order 8: three are abelian,

$$\mathbb{Z}_8, \qquad \mathbb{Z}_4 \times \mathbb{Z}_2, \qquad \mathbb{Z}_2 \times \mathbb{Z}_2 \times \mathbb{Z}_2,$$

and two are nonabelian: the dihedral group D_8 and the quaternions $\mathbf{Q}$.

Order of group	Number of groups
2	1
4	2
8	5
16	14
32	51
64	267
128	2,328
256	56,092
512	10,494,213
1024	49,487,365,422

Table 1.3.: The number of nonisomorphic groups of order 2^k.

One can continue this discussion for larger orders, but things soon get out of hand, as Table 1.3 shows. A. McIver and P. M. Neumann proved that the number of nonisomorphic groups of order n is at most $n^{\mu^2+\mu+2}$, where μ is the largest exponent occurring in the prime factorization of n. Obviously, making a telephone directory of groups is not the way to study them.

§1.7.iv. Actions, Orbits, and Stabilizers

Groups arose by abstracting the fundamental properties enjoyed by permutations. But there is an important feature of permutations that the axioms do not mention: permutations are *functions*. We shall see that there are interesting consequences when this feature is restored.

> **Notation .** A function of two variables $\alpha\colon X \times Y \to Z$ can be regarded as a one-parameter family of functions of one variable: each $x \in X$ gives a function $\alpha_x\colon Y \to Z$, namely $\alpha_x(y) = \alpha(x, y)$.

Definition 1.7.14 (Group action)**.** If X is a set and G is a group, then G **acts on** X if there is a function $\alpha\colon G \times X \to X$, called an **action**, such that

(i) $\alpha_g \circ \alpha_h = \alpha_{gh}$ for all $g, h \in G$;

(ii) $\alpha_1 = 1_X$, the identity function.

If G acts on X, one often calls X a **G-set**.

> **Notation .** If G acts on X, we usually write gx instead of $\alpha_g(x)$. In this notation, axiom (i) reads $g(hx) = (gh)x$, and axiom (ii) reads $1x = x$.

Of course, every subgroup $G \leq S_X$ acts on X by evaluation, $\alpha_g(x) = g(x)$. More generally, actions of a group G on a set X are the same thing as homomorphisms $G \to S_X$.

> **Proposition 1.7.15**
>
> If $\alpha\colon G \times X \to X$ is an action of a group G on a set X, then $g \mapsto \alpha_g$ defines a homomorphism $G \to S_X$. Conversely, if $B\colon G \to S_X$ is a homomorphism, then $\beta\colon G \times X \to X$, defined by $\beta(g, x) = B(g)(x)$, is an action.

Proof. Let α be an action. Each α_g is a permutation of X, with inverse $\alpha_{g^{-1}}$, because $\alpha_g\alpha_{g^{-1}} = \alpha_{gg^{-1}} = \alpha_1 = 1_X$ and likewise $\alpha_{g^{-1}}\alpha_g = 1_X$. Hence $A\colon G \to S_X$, $A(g) = \alpha_g$, is a function with the stated target, and axiom (i) says precisely that it is a homomorphism:

$$A(gh) = \alpha_{gh} = \alpha_g \circ \alpha_h = A(g) \circ A(h).$$

Conversely, let $B\colon G \to S_X$ be a homomorphism and $\beta(g, x) = B(g)(x)$, so that $\beta_g = B(g)$. Axiom (i) says $B(g) \circ B(h) = B(gh)$, which holds because B is a homomorphism; axiom (ii), $B(1) = 1_X$, holds because every homomorphism takes the identity to the identity (Proposition 1.7.1(i)). ∎

Cayley's theorem says that a group G acts on itself by (left) translation, and its generalization Theorem 1.7.10 says that G also acts on the family G/H of cosets of a subgroup H by (left) translation.

Example 1.7.16 (Conjugation)**.** A group G acts on itself by **conjugation**: for each $g \in G$, define $\alpha_g\colon G \to G$ by

$$\alpha_g(x) = gxg^{-1}.$$

To verify axiom (i), note that for each $x \in G$,

$$(\alpha_g \circ \alpha_h)(x) = \alpha_g(hxh^{-1}) = g(hxh^{-1})g^{-1} = (gh)x(gh)^{-1} = \alpha_{gh}(x).$$

Therefore $\alpha_g \circ \alpha_h = \alpha_{gh}$. For axiom (ii), $\alpha_1(x) = 1x1^{-1} = x$ for each x, so $\alpha_1 = 1_G$.

The following two definitions are fundamental.

Definition 1.7.17 (Orbit, stabilizer)**.** Let G act on X and let $x \in X$. The **orbit** of x is the subset

$$\mathcal{O}(x) = \{gx : g \in G\} \subseteq X,$$

and the **stabilizer** of x is

$$G_x = \{g \in G : gx = x\}.$$

We say G acts **transitively** on X if there is some $x \in X$ with $\mathcal{O}(x) = X$.

Lemma 1.7.18

The stabilizer G_x of a point $x \in X$ is a subgroup of G.

Proof. $1 \in G_x$ since $1x = x$. If $g, h \in G_x$, then $(gh)x = g(hx) = gx = x$, so $gh \in G_x$; and $g^{-1}x = g^{-1}(gx) = (g^{-1}g)x = 1x = x$, so $g^{-1} \in G_x$. ∎

Let us find orbits and stabilizers in the examples above.

Example 1.7.19 (Translations)**.** (i) Let G act on itself by translation, $\tau_a \colon x \mapsto ax$. If $x \in G$, then $\mathcal{O}(x) = G$, for if $g \in G$, then $g = (gx^{-1})x$. The stabilizer G_x is $\{1\}$, for if $x = \tau_a(x) = ax$, then $a = 1$. So G acts transitively on itself.

(ii) Let G act on G/H by translation, $\tau_a \colon xH \mapsto axH$. Then $\mathcal{O}(xH) = G/H$, for if $g \in G$ and $a = gx^{-1}$, then $\tau_a \colon xH \mapsto gH$. Thus G acts transitively on G/H. The stabilizer of xH is $G_{xH} = xHx^{-1}$, because, by Lemma 1.3.32(i),

$$axH = xH \Leftrightarrow x^{-1}ax \in H \Leftrightarrow a \in xHx^{-1}.$$

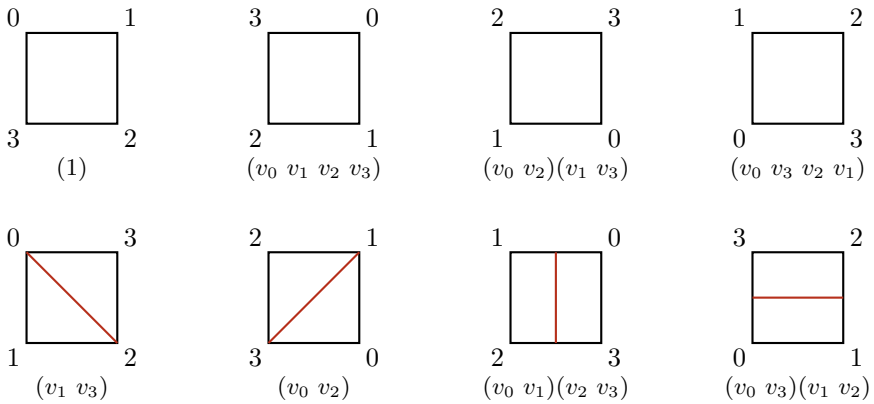


Figure 1.5.: The dihedral group D_8 acting on the vertices of a square. Label i marks where the vertex v_i is sent; the top row shows the rotations, the bottom row the reflections (with their axes).

Example 1.7.20 (D_8 acting on a square)**.** Let $X = \{v_0, v_1, v_2, v_3\}$ be the vertices of a square (labelled as in Figure 1.3), and let $G = D_8$ act on X as in Figure 1.5:

$$\begin{aligned} G = \{&\text{rotations: } (1),\ (v_0\ v_1\ v_2\ v_3),\ (v_0\ v_2)(v_1\ v_3),\ (v_0\ v_3\ v_2\ v_1);\\ &\text{reflections: } (v_1\ v_3),\ (v_0\ v_2),\ (v_0\ v_1)(v_2\ v_3),\ (v_0\ v_3)(v_1\ v_2)\}. \end{aligned}$$

For each vertex v_i there is some $g \in G$ with $gv_0 = v_i$ (a rotation will do); therefore $\mathcal{O}(v_0) = X$, and D_8 acts transitively.

What is the stabilizer G_{v_0}? Aside from the identity, there is only one $g \in D_8$ fixing v_0, namely $g = (v_1\ v_3)$; therefore G_{v_0} is a subgroup of order 2. (This generalizes to D_{2n} acting on the vertices of a regular n-gon: the action is transitive, and the stabilizer of v_0 is $\{1, b\}$, where b is the reflection with axis $L[O, v_0]$.)

Example 1.7.21 (Conjugacy classes and centralizers)**.** Let a group G act on itself by conjugation. If $x \in G$, then

$$\mathcal{O}(x) = \{y \in G : y = axa^{-1} \text{ for some } a \in G\};$$

$\mathcal{O}(x)$ is called the **conjugacy class** of x, and it is often denoted by x^G. For example, Proposition 1.1.29 shows that if $\alpha \in S_n$, then the conjugacy class of α consists of all the permutations in S_n having the same cycle structure as α.

If $x \in G$, then the stabilizer G_x of x is

$$C_G(x) = \{g \in G : gxg^{-1} = x\}.$$

This subgroup of G, consisting of all $g \in G$ that commute with x, is called the **centralizer** of x in G.

Example 1.7.22 (A cyclic subgroup of S_n)**.** Let $X = \{1, 2, \ldots, n\}$, let $\sigma \in S_n$, and let the cyclic group $G = \langle\sigma\rangle$ act on X. If $i \in X$, then

$$\mathcal{O}(i) = \{\sigma^k(i) : k \in \mathbb{Z}\}.$$

Let $\sigma = \beta_1 \cdots \beta_t$ be the complete factorization of σ, and suppose the cycle containing $i = i_1$ is $\beta_j = (i_1\ i_2\ \ldots\ i_r)$. As in the proof of Proposition 1.1.16, $\sigma^k(i_1) = i_{k+1}$ for $0 \le k \le r-1$, while $\sigma^r(i_1) = i_1$; so the powers $\sigma^k(i_1)$ run cyclically through $i_1, \ldots, i_r$. Therefore

$$\mathcal{O}(i) = \{i_1, i_2, \ldots, i_r\}, \qquad |\mathcal{O}(i)| = r.$$

The stabilizer G_ℓ of a symbol ℓ is all of G if σ fixes ℓ, and it is a proper subgroup of G if σ moves ℓ (for then $\sigma \notin G_\ell$).

A group G acting on a set X gives an equivalence relation on X: define

$$x \equiv y \quad \text{if there exists } g \in G \text{ with } y = gx.$$

Reflexive: $x = 1x$. *Symmetric:* if $y = gx$, then $g^{-1}y = g^{-1}(gx) = (g^{-1}g)x = x$, so $y \equiv x$. *Transitive:* if $y = gx$ and $z = hy$, then $z = h(gx) = (hg)x$, so $x \equiv z$. The equivalence class of $x \in X$ is exactly its orbit:

$$[x] = \{y \in X : y \equiv x\} = \{gx : g \in G\} = \mathcal{O}(x).$$

Proposition 1.7.23

If G acts on a set X, then X is the disjoint union of the orbits. If X is finite, then

$$|X| = \sum_i |\mathcal{O}(x_i)|,$$

where one x_i is chosen from each orbit.

Proof. The orbits are the equivalence classes of $\equiv$, and the equivalence classes of an equivalence relation form a partition of X. The count is correct because the orbits are disjoint, so no element of X is counted twice. ∎

Here is the connection between orbits and stabilizers.

Theorem 1.7.24 (Orbit–stabilizer)

If G acts on a set X and $x \in X$, then

$$|\mathcal{O}(x)| = [G : G_x],$$

the index of the stabilizer G_x in G.

Proof. Let G/G_x denote the family of all cosets of G_x in G. We exhibit a bijection $\varphi\colon \mathcal{O}(x) \to G/G_x$; this gives the result, since $|G/G_x| = [G : G_x]$ by definition of the index.

If $y \in \mathcal{O}(x)$, then $y = gx$ for some $g \in G$; define $\varphi(y) = gG_x$.

- *Well defined:* if also $y = hx$, then $h^{-1}gx = h^{-1}y = x$, so $h^{-1}g \in G_x$, and $hG_x = gG_x$ by Lemma 1.3.32(i).
- *Injective:* suppose $\varphi(y) = \varphi(z)$, where $y = gx$, $z = hx$, and $gG_x = hG_x$. Then $h^{-1}g \in G_x$, so $h^{-1}gx = x$; applying h gives $y = gx = hx = z$.
- *Surjective:* given $gG_x \in G/G_x$, let $y = gx \in \mathcal{O}(x)$; then $\varphi(y) = gG_x$. ■

Example 1.7.25. In Example 1.7.20, D_8 acting on the four corners of a square, we saw $|\mathcal{O}(v_0)| = 4$, $|G_{v_0}| = 2$, and indeed $[G : G_{v_0}] = 8/2 = 4$.

In Example 1.7.22, $G = \langle\sigma\rangle \leq S_n$ acting on $X = \{1, \ldots, n\}$, if an r-cycle β_j in the complete factorization of σ moves ℓ, then $r = |\mathcal{O}(\ell)|$. Theorem 1.7.24 says that r divides $|G|$, which is the order k of σ. (Proposition 1.2.22 tells us more: k is the lcm of the lengths of the cycles in the factorization.)

Corollary 1.7.26

If a finite group G acts on a set X, then the number of elements in any orbit is a divisor of $|G|$.

Proof. $|\mathcal{O}(x)| = [G : G_x] = |G|/|G_x|$, by Theorem 1.7.24 and Corollary 1.3.35. ■

Corollary 1.7.27

If x lies in a finite group G, then the number of conjugates of x is the index of its centralizer:

$$\left|x^G\right| = [G : C_G(x)],$$

and hence it is a divisor of $|G|$.

Proof. As in Example 1.7.21, for the conjugation action the orbit of x is its conjugacy class x^G, and the stabilizer G_x is the centralizer $C_G(x)$. ■

In Table 1.1 we counted the permutations in S_4 of each cycle structure: $1, 6, 8, 6, 3$. Each of these numbers divides $|S_4| = 24$. The corresponding numbers for S_5 are $1, 10, 20, 30, 24, 20, 15$, and these all divide $|S_5| = 120$. We now recognize these subsets as conjugacy classes, and the next corollary explains why these numbers divide the group order.

Corollary 1.7.28

If $\alpha \in S_n$, then the number of permutations in S_n having the same cycle structure as α is a divisor of $n!$.

Proof. By Proposition 1.1.29, two permutations in S_n are conjugate if and only if they have the same cycle structure. So the permutations with the cycle structure of α form the conjugacy class α^{S_n}, whose size divides $|S_n| = n!$ by Corollary 1.7.27. ∎

§1.7.v. Cauchy's Theorem and p-Groups

When we classified groups of order 6, it would have been helpful to know that any such group has an element of order 3 (we could only use Lemma 1.7.4 to find an element of order 2). We now prove that every finite group G contains an element of prime order p for every prime $p \mid |G|$.

If the conjugacy class x^G of an element x consists of x alone, then $gxg^{-1} = x$, that is, $gx = xg$, for every $g \in G$; so $x \in Z(G)$. Conversely, if $x \in Z(G)$, then $x^G = \{x\}$. Thus

$$Z(G) = \{x \in G : |x^G| = 1\}.$$

Theorem 1.7.29 (Cauchy)

If G is a finite group whose order is divisible by a prime p, then G contains an element of order p.

Proof. We prove the theorem by induction on $|G|$; the base step $|G| = 1$ is vacuously true, for there are no prime divisors of 1.

If $x \in G$, then the number of conjugates of x is $|x^G| = [G : C_G(x)]$ by Corollary 1.7.27. If $x \notin Z(G)$, then x^G has more than one element, and so $|C_G(x)| < |G|$. If $p \mid |C_G(x)|$ for some noncentral x, then the inductive hypothesis gives an element of order p in $C_G(x) \leq G$, and we are done. Therefore we may assume that $p \nmid |C_G(x)|$ for all noncentral $x \in G$. Better, since $|G| = [G : C_G(x)]\,|C_G(x)|$ is divisible by p, Euclid's lemma gives

$$p \mid [G : C_G(x)] \quad \text{for every noncentral } x.$$

Now apply Proposition 1.7.23 to the conjugation action. The orbits of size 1 are exactly the singletons $\{z\}$ with $z \in Z(G)$, and there are $|Z(G)|$ of them; so

$$|G| = |Z(G)| + \sum_i [G : C_G(x_i)],$$

where one x_i is selected from each conjugacy class having more than one element. Since $|G|$ and all $[G : C_G(x_i)]$ are divisible by p, it follows that $|Z(G)|$ is divisible by p. But $Z(G)$ is abelian, so Proposition 1.7.1(vi) says that $Z(G)$, and hence G, contains an element of order p. ∎

Remark 1.7.30. The proof uses the abelian case of Cauchy's theorem, which was proved in Rotman ğ2.6 with quotient groups. Example 1.7.37 below gives a completely independent proof.

Definition 1.7.31 (Class equation)**.** The **class equation** of a finite group G is

$$|G| = |Z(G)| + \sum_i [G : C_G(x_i)],$$

where one x_i is selected from each conjugacy class having more than one element.

Example 1.7.32. In S_3 the center is trivial, and the other conjugacy classes are the transpositions and the 3-cycles, so the class equation is $6 = 1 + 3 + 2$. In S_4, Table 1.1 gives $24 = 1 + 6 + 8 + 6 + 3$.

Definition 1.7.33 (p-group)**.** If p is a prime, then a **p-group** is a group of order p^n for some $n \geq 0$.

There are groups whose center is trivial; for example, $Z(S_3) = \{1\}$. For p-groups with more than one element, however, this never happens.

Theorem 1.7.34

If p is a prime and G is a p-group with more than one element, then $Z(G) \neq \{1\}$.

Proof. Consider the class equation

$$|G| = |Z(G)| + \sum_i [G : C_G(x_i)].$$

Each $C_G(x_i)$ is a proper subgroup of G, for $x_i \notin Z(G)$. Since $[G : C_G(x_i)]$ is a divisor of $|G|$, a power of p, it is itself a power of p, and it is not $p^0 = 1$. Thus p divides $|G|$ and each term of the sum, and so $p \mid |Z(G)|$ as well. Therefore $|Z(G)| \geq p$, and $Z(G) \neq \{1\}$. ∎

Corollary 1.7.35

If p is a prime, then every group G of order p^2 is abelian.

Proof. If G is not abelian, then $Z(G)$ is a proper subgroup, so $|Z(G)| = 1$ or p by Lagrange's theorem. But $Z(G) \neq \{1\}$ by Theorem 1.7.34, so $|Z(G)| = p$. The center is a normal subgroup, so the quotient $G/Z(G)$ is defined; it has order p, hence is cyclic. Then Lemma 1.7.5 says G is abelian, a contradiction. ∎

Example 1.7.36 (Nonabelian groups of order p^3)**.** For every prime p, there exist nonabelian groups of order p^3. Let $UT(3, p)$ be the set of all upper triangular 3×3 matrices over $\mathbb{Z}_p$ with 1's on the diagonal:

$$UT(3,p) = \left\{ \begin{pmatrix} 1 & a & b \\ 0 & 1 & c \\ 0 & 0 & 1 \end{pmatrix} : a, b, c \in \mathbb{Z}_p \right\}.$$

It is a subgroup of $GL(3, \mathbb{Z}_p)$: it contains I, and

$$\begin{pmatrix} 1 & a & b \\ 0 & 1 & c \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & a' & b' \\ 0 & 1 & c' \\ 0 & 0 & 1 \end{pmatrix} = \begin{pmatrix} 1 & a+a' & b+ac'+b' \\ 0 & 1 & c+c' \\ 0 & 0 & 1 \end{pmatrix},$$

$$\begin{pmatrix} 1 & a & b \\ 0 & 1 & c \\ 0 & 0 & 1 \end{pmatrix}^{-1} = \begin{pmatrix} 1 & -a & ac-b \\ 0 & 1 & -c \\ 0 & 0 & 1 \end{pmatrix}.$$

It has order p^3, because there are p choices for each of a, b, c. It is not abelian: by the product formula, the matrices with $(a, b, c) = (1, 0, 0)$ and $(0, 0, 1)$ multiply to $(1, 1, 1)$ in one order and to $(1, 0, 1)$ in the other. (Rotman, Exercise 2.111: $UT(3, 2) \cong D_8$.)

Example 1.7.37 (McKay's proof of Cauchy's and Fermat's theorems)**.** Who would have guessed that Cauchy's theorem and Fermat's theorem are special cases of a common theorem? The following elementary yet ingenious proof is due to J. H. McKay.

Let G be a finite group and p a prime, and let

$$X = \{(a_1, a_2, \ldots, a_p) \in G^p : a_1 a_2 \cdots a_p = 1\},$$

where G^p is the cartesian product of p copies of G. Then $|X| = |G|^{p-1}$: having chosen the first $p-1$ entries arbitrarily, the pth entry must equal $(a_1 a_2 \cdots a_{p-1})^{-1}$.

Make X into a $\mathbb{Z}_p$-set by cyclically shifting: for $0 \le i \le p-1$,

$$[i](a_1, a_2, \ldots, a_p) = (a_{i+1}, a_{i+2}, \ldots, a_p, a_1, a_2, \ldots, a_i).$$

(Shifting by j and then by i is shifting by $i+j$, so this is an action.) The shifted tuple lies in X, because the product of its entries is a conjugate of $a_1 a_2 \cdots a_p = 1$:

$$a_{i+1} \cdots a_p a_1 \cdots a_i = (a_1 \cdots a_i)^{-1}(a_1 a_2 \cdots a_p)(a_1 \cdots a_i) = 1.$$

By Corollary 1.7.26, every orbit has size dividing $|\mathbb{Z}_p| = p$, so its size is 1 or p. An orbit has size 1 exactly when the tuple is unchanged by the shift [1], that is, when all its entries are equal: $(a, a, \ldots, a)$, which lies in X if and only if $a^p = 1$. So if $s = |\{a \in G : a^p = 1\}|$, then Proposition 1.7.23 gives

$$|G|^{p-1} = |X| = s + kp \quad \text{for some } k \ge 0.$$

- *Cauchy.* If $p \mid |G|$, then $p \mid s$. Since $s \ge 1$ (take $a = 1$), we get $s \ge p \ge 2$, so some $a \ne 1$ satisfies $a^p = 1$; such an a has order p.
- *Fermat.* Let $G = \mathbb{Z}_n$ with $n \ge 1$ (written additively, so "$a^p = 1$" reads $pa = 0$). If $p \nmid n$, then an element with $pa = 0$ has order dividing both p and n (Lemma 1.2.21 and Corollary 1.3.37), so $a = 0$ and $s = 1$. Hence $n^{p-1} = 1 + kp$, i.e. $n^{p-1} \equiv 1 \pmod p$, and multiplying by n gives $n^p \equiv n \pmod p$. This congruence also holds when $p \mid n$, both sides being $\equiv 0$. Since both sides depend only on $n \bmod p$, it holds for every integer n: this is **Fermat's theorem**.

Remark 1.7.38. The argument proves more: if $|G| = n$, then the number of solutions of $x^p = 1$ in G is $s \equiv n^{p-1} \pmod p$.

We have seen that A_4 is a group of order 12 having no subgroup of order 6. So the converse of Lagrange's theorem is false. It is true, however, when G is a p-group; indeed, more is true.

Proposition 1.7.39

If G is a group of order p^e, then G has a normal subgroup of order p^k for every $k \le e$.

Proof. We induct on $e \ge 0$. The base step $e = 0$ is obvious (take $\{1\}$), so assume $e \ge 1$. By Theorem 1.7.34, $Z(G) \ne \{1\}$.

If $Z(G) = G$, then G is abelian; Proposition 1.7.1(vi) gives a subgroup of order p^k, and it is normal because G is abelian.

Assume that $Z(G)$ is a proper subgroup, say $|Z(G)| = p^c$ with $1 \le c < e$. If $k \le c$, then the abelian group $Z(G)$ contains a subgroup of order p^k by Proposition 1.7.1(vi), and this subgroup is normal in G since it lies in the center (Proposition 1.7.1(ii)). If $k > c$, consider $G/Z(G)$, a p-group of order $p^{e-c} < p^e$. By induction it contains a normal subgroup S^* of order p^{k-c}. The correspondence theorem gives a normal subgroup S of G with

$$Z(G) \le S \trianglelefteq G \qquad \text{and} \qquad S/Z(G) = S^*.$$

Then $|S| = [S : Z(G)]\,|Z(G)| = |S^*|\,|Z(G)| = p^{k-c} \cdot p^c = p^k$. ∎

§1.7.vi. Simple Groups

Abelian groups (and the quaternions) have the property that every subgroup is normal. At the opposite pole are the groups having no normal subgroups other than the two obvious ones, $\{1\}$ and G.

Definition 1.7.40 (Simple group)**.** A group G is **simple** if $G \neq \{1\}$ and G has no normal subgroups other than $\{1\}$ and G itself.

Proposition 1.7.41

An abelian group G is simple if and only if it is finite of prime order.

Proof. If G is finite of prime order p, then G has no subgroups H other than $\{1\}$ and G, since $|H|$ divides p by Lagrange's theorem. Therefore G is simple.

Conversely, assume that G is simple. Since G is abelian, every subgroup is normal, so G has no subgroups other than $\{1\}$ and G. Choose $x \in G$ with $x \neq 1$; since $\langle x\rangle \neq \{1\}$, we have $\langle x\rangle = G$. If x had infinite order, then all powers of x would be distinct, and $\langle x^2\rangle$ would be a subgroup with $\{1\} \neq \langle x^2\rangle \neq \langle x\rangle$ (as $x \notin \langle x^2\rangle$), a contradiction. So x has finite order m, and $|G| = m$. If m were composite, say $m = k\ell$ with $1 < k < m$, then $\langle x^k\rangle$ would be a subgroup of order ℓ, neither $\{1\}$ nor G, a contradiction. Therefore $G = \langle x\rangle$ has prime order. ∎

We are now going to show that A_5 is a nonabelian simple group; indeed, it is the smallest one, for there is no nonabelian simple group of order less than 60.

Suppose that $x \in G$ has k conjugates: $\left|x^G\right| = k$. If H is a subgroup with $x \in H \leq G$, how many conjugates does x have in H? Since

$$x^H = \{hxh^{-1} : h \in H\} \subseteq \{gxg^{-1} : g \in G\} = x^G,$$

we have $\left|x^H\right| \leq \left|x^G\right|$, and the inequality can be strict. For example, take $G = S_3$, $x = (1\ 2)$, and $H = \langle x\rangle$: then $\left|x^G\right| = 3$ (all transpositions are conjugate), whereas $\left|x^H\right| = 1$ (because H is abelian). Let us consider this question for $G = S_5$, $x = (1\ 2\ 3)$, and $H = A_5$.

Lemma 1.7.42

All 3-cycles are conjugate in A_5.

Proof. Let $\alpha = (1\ 2\ 3)$. There are 20 3-cycles in S_5 (Table 1.1), and they are all conjugate in S_5, so $\left|\alpha^{S_5}\right| = 20$. By Corollary 1.7.27, $20 = |S_5|/|C_{S_5}(\alpha)| = 120/|C_{S_5}(\alpha)|$, so $|C_{S_5}(\alpha)| = 6$: there are exactly six permutations in S_5 commuting with α. Here they are:

$$(1),\quad (1\ 2\ 3),\quad (1\ 3\ 2),\quad (4\ 5),\quad (4\ 5)(1\ 2\ 3),\quad (4\ 5)(1\ 3\ 2).$$

The last three are odd, so $C_{A_5}(\alpha) = C_{S_5}(\alpha) \cap A_5$ has order 3. We conclude that

$$\left|\alpha^{A_5}\right| = |A_5|/|C_{A_5}(\alpha)| = 60/3 = 20;$$

that is, all twenty 3-cycles are conjugate to α in A_5, hence to each other. ∎

Lemma 1.7.43

If $n \geq 3$, then every element of A_n is a 3-cycle or a product of 3-cycles.

Proof. The identity is $(1\ 2\ 3)^3$. If $\alpha \in A_n$, then α is a product of an even number of transpositions, $\alpha = \tau_1\tau_2\cdots\tau_{2m-1}\tau_{2m}$ (Theorem 1.1.39). Grouping the transpositions in pairs $\tau_{2i-1}\tau_{2i}$, it suffices to consider a product $\tau\tau'$ of two transpositions. If $\tau = \tau'$, then $\tau\tau' = (1)$, which can be dropped. If τ and τ' share exactly one symbol, then $\tau = (i\ j)$, $\tau' = (i\ k)$, and $\tau\tau' = (i\ k\ j)$. If they are disjoint, then

$$\tau\tau' = (i\ j)(k\ \ell) = (i\ j)(j\ k)(j\ k)(k\ \ell) = (i\ j\ k)(j\ k\ \ell).$$ ■

Theorem 1.7.44

A_5 is a simple group.

Proof. We show that if $H \trianglelefteq A_5$ and $H \neq \{(1)\}$, then $H = A_5$. If H contains a 3-cycle, then normality forces H to contain all its conjugates in A_5, so H contains every 3-cycle by Lemma 1.7.42, and $H = A_5$ by Lemma 1.7.43. Therefore it suffices to prove that H contains a 3-cycle.

As $H \neq \{(1)\}$, it contains some $\sigma \neq (1)$. The even cycle structures in S_5 are those of (1), $(1\ 2\ 3)$, $(1\ 2)(3\ 4)$, and $(1\ 2\ 3\ 4\ 5)$, so σ is a 3-cycle (and we are done), or $\sigma = (a\ b)(c\ d)$, or $\sigma = (a\ b\ c\ d\ e)$; in either of the last two cases write $\{a,b,c,d,e\} = \{1,2,3,4,5\}$.

- If $\sigma = (a\ b)(c\ d) \in H$, conjugate by $\beta = (c\ d\ e) \in A_5$: by Proposition 1.1.27, $\sigma' = \beta\sigma\beta^{-1} = (a\ b)(d\ e) \in H$, because $H \trianglelefteq A_5$. Hence $\sigma\sigma' = (c\ d)(d\ e) = (c\ d\ e) \in H$.
- If $\sigma = (a\ b\ c\ d\ e) \in H$, conjugate by $\gamma = (a\ b\ c) \in A_5$: $\sigma'' = \gamma\sigma\gamma^{-1} = (b\ c\ a\ d\ e) \in H$. Hence
$$\sigma''\sigma^{-1} = (b\ c\ a\ d\ e)(e\ d\ c\ b\ a) = (a\ b\ d) \in H.$$

In all cases H contains a 3-cycle. Therefore the only normal subgroups of A_5 are $\{(1)\}$ and A_5 itself, and A_5 is simple. ■

Remark 1.7.45. Where did the last computation come from? If $\sigma \in H$ and γ is a 3-cycle, then $\gamma\sigma\gamma^{-1} \in H$, so $(\gamma\sigma\gamma^{-1})\sigma^{-1} = \gamma(\sigma\gamma^{-1}\sigma^{-1}) \in H$. But $\sigma\gamma^{-1}\sigma^{-1}$ is a 3-cycle, so H contains a product of two 3-cycles. We chose γ carefully to force this product to be a 3-cycle. The same trick drives the proof of Theorem 1.7.50.

Remark 1.7.46. As we shall see in Chapter 5, Theorem 1.7.44 turns out to be the basic reason why the quadratic formula has no generalization giving the roots of polynomials of degree 5 or higher.

Observe that A_4 is *not* simple: the four-group $\mathbf{V}$ consists of (1) and the three permutations in S_4 of cycle structure $(a\ b)(c\ d)$, all of which are even; since conjugation preserves cycle structure (Proposition 1.1.27), $g\mathbf{V}g^{-1} = \mathbf{V}$ for all $g \in S_4$. Thus $\{(1)\} \neq \mathbf{V} \trianglelefteq A_4$, and $\mathbf{V} \neq A_4$. Without much more effort, however, we can prove that A_n is simple for every $n \geq 5$. We first generalize the two lemmas used for A_5.

Lemma 1.7.47 (Rotman, Exercise 2.116)

Let $n \geq 5$.

(i) All 3-cycles are conjugate in A_n.

(ii) If $H \trianglelefteq A_n$ contains a 3-cycle, then $H = A_n$.

Proof. (i) Let γ, γ' be 3-cycles. By Proposition 1.1.29 there is $\alpha \in S_n$ with $\alpha\gamma\alpha^{-1} = \gamma'$. If α is even we are done. If α is odd, choose two symbols k, ℓ fixed by γ (possible since $n \geq 5$) and let $\tau = (k\ \ell)$. Then τ and γ are disjoint, so they commute (Lemma 1.1.13), and $\alpha\tau$ is even with

$$(\alpha\tau)\gamma(\alpha\tau)^{-1} = \alpha(\tau\gamma\tau^{-1})\alpha^{-1} = \alpha\gamma\alpha^{-1} = \gamma'.$$

(ii) By normality and (i), H contains every 3-cycle, so $H = A_n$ by Lemma 1.7.43. ■

Lemma 1.7.48 (Rotman, Exercise 2.118)
Let $Y = \{i_1, \ldots, i_r\} \subseteq \{1, 2, \ldots, n\}$, and let

$$F = \{\sigma \in A_n : \sigma \text{ fixes every } i \notin Y\}.$$

Then F is a subgroup of A_n and $F \cong A_r$.

Proof. Let $E = \{\sigma \in S_n : \sigma$ fixes every $i \notin Y\}$. If $\sigma \in E$, then σ maps the complement of Y onto itself, hence (being a bijection) maps Y onto Y; so the restriction $\sigma|_Y$ lies in S_Y. The map $E \to S_Y$, $\sigma \mapsto \sigma|_Y$, is a homomorphism (restricting a composite gives the composite of restrictions, as $\sigma(Y) = Y$); it is injective, since σ is known off Y; and it is surjective, since every $\pi \in S_Y$ extends by the identity to an element of E. So $E \cong S_Y$.

This isomorphism preserves parity: if $\pi \in S_Y$ has a complete factorization with t cycles, then its extension has the same cycles plus $n - r$ extra 1-cycles, so its signum is $(-1)^{n-(t+n-r)} = (-1)^{r-t} = \operatorname{sgn}(\pi)$. Hence $F = E \cap A_n$ corresponds to the even permutations of Y, and $F \cong A_Y \cong A_r$, the last isomorphism by Lemma 1.7.2 (which preserves parity). In particular F, being the image of a subgroup under an isomorphism, is a subgroup. ■

Lemma 1.7.49
A_6 is a simple group.

Proof. Let $H \neq \{(1)\}$ be a normal subgroup of A_6; we must show $H = A_6$.

Assume first that some $\alpha \in H$ with $\alpha \neq (1)$ fixes some i, $1 \leq i \leq 6$. Define

$$F = \{\sigma \in A_6 : \sigma(i) = i\}.$$

Then $\alpha \in H \cap F$, so $H \cap F \neq \{(1)\}$. Moreover $H \cap F \trianglelefteq F$: if $\sigma \in F$ and $h \in H \cap F$, then $\sigma h \sigma^{-1} \in H$ (as $H \trianglelefteq A_6$) and $\sigma h \sigma^{-1} \in F$. But $F \cong A_5$ by Lemma 1.7.48, so F is simple by Theorem 1.7.44, and hence $H \cap F = F$; that is, $F \leq H$. Now F contains 3-cycles (any 3-cycle not involving i), so H contains a 3-cycle, and $H = A_6$ by Lemma 1.7.47.

We may now assume that no $\alpha \in H$ with $\alpha \neq (1)$ fixes any i. An even permutation of six symbols fixing none of them has cycle structure $(1\ 2)(3\ 4\ 5\ 6)$ or $(1\ 2\ 3)(4\ 5\ 6)$ (the structures $(1\ 2\ \ldots\ 6)$ and $(1\ 2)(3\ 4)(5\ 6)$ are odd). In the first case, say $\alpha = (a\ b)(c\ d\ e\ f)$, the element $\alpha^2 = (c\ e)(d\ f) \in H$ is nontrivial and fixes a, a contradiction. In the second case, say $\alpha = (1\ 2\ 3)(4\ 5\ 6)$ after renaming, let $\beta = (2\ 3\ 4)$. Then $\beta\alpha^{-1}\beta^{-1} \in H$ by normality, and $\beta\alpha^{-1}\beta^{-1} = (1\ 4\ 3)(2\ 6\ 5)$ by Proposition 1.1.27, so

$$\alpha(\beta\alpha^{-1}\beta^{-1}) = (1\ 2\ 3)(4\ 5\ 6)(1\ 4\ 3)(2\ 6\ 5) = (1\ 5\ 3\ 2\ 4) \in H$$

is nontrivial and fixes 6, another contradiction. Therefore no such H exists in this case, and A_6 is simple. ■

Theorem 1.7.50

A_n is a simple group for all $n \geq 5$.

Proof. The case $n = 5$ is Theorem 1.7.44, so let $n \geq 6$. Let H be a nontrivial normal subgroup of A_n; by Lemma 1.7.47 it suffices to prove that H contains a 3-cycle.

Choose $\beta \in H$ with $\beta \neq (1)$. Then β moves some i, say $\beta(i) = j \neq i$. Choose a 3-cycle α which fixes i and moves j. Then α and β do not commute: $\beta\alpha(i) = \beta(i) = j$, while $\alpha\beta(i) = \alpha(j) \neq j$. It follows that

$$\gamma = (\alpha\beta\alpha^{-1})\beta^{-1}$$

is a nontrivial element of H: it lies in H because $\alpha\beta\alpha^{-1} \in H$ by normality, and it is nontrivial because $\alpha\beta\alpha^{-1} \neq \beta$. But $\beta\alpha^{-1}\beta^{-1}$ is a 3-cycle by Proposition 1.1.27, so $\gamma = \alpha(\beta\alpha^{-1}\beta^{-1})$ is a product of two 3-cycles. Hence γ moves at most 6 symbols, say $i_1, \ldots, i_6$ (if γ moves fewer than 6 symbols, adjoin others so that we have a list of 6). Define

$$F = \{\sigma \in A_n : \sigma \text{ fixes all } i \neq i_1, \ldots, i_6\}.$$

Then $F \cong A_6$ by Lemma 1.7.48, and $\gamma \in H \cap F$. As in the proof of Lemma 1.7.49, $H \cap F$ is a nontrivial normal subgroup of F. But F is simple, being isomorphic to A_6, so $H \cap F = F$; that is, $F \leq H$. Therefore H contains a 3-cycle, and $H = A_n$. ■

Lecture 4: Homomorphisms

§1.8. Homomorphisms

Rotman §2.5

An important problem is deciding whether two given groups G and H are somehow the same. For example, S_3 is the group of all permutations of $X = \{1, 2, 3\}$, and the group S_Y of all permutations of $Y = \{a, b, c\}$ is a different group, because permutations of $\{1, 2, 3\}$ are not permutations of $\{a, b, c\}$. But even though S_3 and S_Y are different, they surely bear a strong resemblance to each other (Example 1.8.6). The notions of homomorphism and isomorphism allow one to compare different groups.

§1.8.i. Homomorphisms and Isomorphisms

Definition 1.8.1 (Homomorphism, isomorphism)**.** If $(G, *)$ and $(H, \circ)$ are groups (we have displayed the operation in each), then a function $f\colon G \to H$ is a **homomorphism** if

$$f(x * y) = f(x) \circ f(y)$$

for all $x, y \in G$. If f is also a bijection, then f is called an **isomorphism**. We say that G and H are **isomorphic**, denoted by $G \cong H$, if there exists an isomorphism $f\colon G \to H$.

Note 1.8.2 (Etymology)**.** *Homomorphism* comes from the Greek *homo* ("same") and *morph* ("shape" or "form"): a homomorphism carries a group to another group (its image) of similar form. *Isomorphism* involves the Greek *iso* ("equal"), and isomorphic groups have identical form.

Lemma 1.8.3 (Rotman, Exercise 2.57) (i) The composite of homomorphisms is a homomorphism.

(ii) The inverse of an isomorphism is an isomorphism.

(iii) Isomorphism is an equivalence relation on any family of groups. In particular, if $G \cong H$, then $H \cong G$.

Proof. (i) If $f\colon G \to H$ and $g\colon H \to L$ are homomorphisms, then $gf(xy) = g(f(x)f(y)) = gf(x)\,gf(y)$.

(ii) Let $f\colon G \to H$ be an isomorphism with inverse function $f^{-1}\colon H \to G$. Given $u, v \in H$, write $u = f(x)$ and $v = f(y)$; then $f^{-1}(uv) = f^{-1}(f(x)f(y)) = f^{-1}(f(xy)) = xy = f^{-1}(u)f^{-1}(v)$.

(iii) $1_G\colon G \to G$ is an isomorphism (reflexivity); (ii) gives symmetry; and (i), together with the fact that a composite of bijections is a bijection, gives transitivity. ∎

Example 1.8.4 (First examples)**.** (i) The identity $1_G\colon G \to G$ is an isomorphism, and the **trivial homomorphism** $f\colon G \to H$, $f(a) = 1$ for all $a \in G$, is a homomorphism.

(ii) Let $\mathbb{R}$ be the additive group of real numbers and $\mathbb{R}^>$ the multiplicative group of positive reals. Then $f\colon \mathbb{R} \to \mathbb{R}^>$, $f(x) = e^x$, is a homomorphism, for

$$f(x+y) = e^{x+y} = e^x e^y = f(x)f(y).$$

It is an isomorphism, since its inverse function is $g(x) = \log x$. Therefore $\mathbb{R} \cong \mathbb{R}^>$. Note that g is also an isomorphism, as Lemma 1.8.3(ii) predicts: $g(xy) = \log(xy) = \log x + \log y = g(x) + g(y)$.

(iii) The additive group $\mathbb{C}$ is isomorphic to the plane $\mathbb{R}^2$ (Example 1.2.11(vii)). Define $f\colon \mathbb{C} \to \mathbb{R}^2$ by $f(a+ib) = (a,b)$. Clearly f is a bijection, and it is a homomorphism:

$$\begin{aligned} f\big((a+ib)+(a'+ib')\big) &= f\big((a+a')+i(b+b')\big) = (a+a', b+b') \\ &= (a,b)+(a',b') = f(a+ib)+f(a'+ib'). \end{aligned}$$

Definition 1.8.5 (Multiplication table)**.** Let $a_1, a_2, \dots, a_n$ be a list, with no repetitions, of all the elements of a finite group G of order n. A **multiplication table** for G is the $n \times n$ array whose ij entry is $a_i a_j$:

G	a_1	$\cdots$	a_j	$\cdots$	a_n
a_1	a_1a_1	$\cdots$	a_1a_j	$\cdots$	a_1a_n
$\vdots$	$\vdots$		$\vdots$		$\vdots$
a_i	a_ia_1	$\cdots$	a_ia_j	$\cdots$	a_ia_n
$\vdots$	$\vdots$		$\vdots$		$\vdots$
a_n	a_na_1	$\cdots$	a_na_j	$\cdots$	a_na_n

We agree that the identity is listed first, $a_1 = 1$; then the first row and first column merely repeat the listing, and we usually omit them.

Consider two almost trivial groups: the multiplicative group $\Gamma_2 = \{1, -1\}$ and the parity group $\mathbb{P}$ (Example 1.2.11(viii)). Their multiplication tables are

Γ_2:

1	−1
−1	1

; $\mathbb{P}$:

even	odd
odd	even

.

Clearly Γ_2 and $\mathbb{P}$ are distinct groups; equally clearly, there is no significant difference between them. Isomorphism formalizes this: the function $f\colon \Gamma_2 \to \mathbb{P}$ with $f(1) = \text{even}$ and $f(-1) = \text{odd}$ is an isomorphism.

There are many multiplication tables for a group of order n, one for each of the $n!$ lists of its elements. If $a_1, \dots, a_n$ lists the elements of G and $f\colon G \to H$ is a bijection, then $f(a_1), \dots, f(a_n)$ lists the elements of H and determines a multiplication table for H. That f is an isomorphism says precisely that the two tables match: if a_ia_j is the ij entry of the table of G, then

$f(a_i)f(a_j) = f(a_ia_j)$ is the ij entry of the table of H. In this sense isomorphic groups have the same multiplication table; they are essentially the same, differing only in the notation for the elements and the operations.

Example 1.8.6 (S_3 and S_Y)**.** Here is an algorithm to check whether a bijection $f\colon G \to H$ between finite groups is an isomorphism: list the elements $a_1, \dots, a_n$ of G, form the multiplication table of G from this list and that of H from $f(a_1), \dots, f(a_n)$, and compare the n^2 entries.

Take $G = S_3$ and $H = S_Y$, $Y = \{a, b, c\}$. List S_3 as

$$(1),\ (1\ 2),\ (1\ 3),\ (2\ 3),\ (1\ 2\ 3),\ (1\ 3\ 2),$$

and let $\varphi\colon S_3 \to S_Y$ be the obvious function replacing numbers by letters:

$$(1),\ (a\ b),\ (a\ c),\ (b\ c),\ (a\ b\ c),\ (a\ c\ b).$$

Superimposing the two tables, they match. For instance, the $4,5$ entry for S_3 is $(2\ 3)(1\ 2\ 3) = (1\ 3)$, while the $4,5$ entry for S_Y is $(b\ c)(a\ b\ c) = (a\ c)$. This is the special case $X = \{1, 2, 3\}$ of Lemma 1.7.2.

We now turn from isomorphisms to general homomorphisms.

Lemma 1.8.7

Let $f\colon G \to H$ be a homomorphism.

(i) $f(1) = 1$;

(ii) $f(x^{-1}) = f(x)^{-1}$;

(iii) $f(x^n) = f(x)^n$ for all $n \in \mathbb{Z}$.

Proof. (i) Applying f to $1 \cdot 1 = 1$ gives $f(1)f(1) = f(1)$, and multiplying both sides by $f(1)^{-1}$ gives $f(1) = 1$.

(ii) Applying f to $x^{-1}x = 1$ gives $f(x^{-1})f(x) = 1$, and uniqueness of inverses (Proposition 1.2.8(iv)) gives $f(x^{-1}) = f(x)^{-1}$.

(iii) An easy induction gives $f(x^n) = f(x)^n$ for all $n \geq 0$. For negative exponents, $(y^{-1})^n = y^{-n}$ in any group, so

$$f(x^{-n}) = f\big((x^{-1})^n\big) = f(x^{-1})^n = \big(f(x)^{-1}\big)^n = f(x)^{-n}. \qquad \blacksquare$$

Example 1.8.8 (Cyclic groups of the same order)**.** Any two finite cyclic groups G and H of the same order m are isomorphic. Together with Corollary 1.3.39, it follows that any two groups of prime order p are isomorphic.

Suppose $G = \langle x \rangle$ and $H = \langle y \rangle$, and define $f\colon G \to H$ by $f(x^i) = y^i$ for $0 \leq i < m$. Since $G = \{1, x, \dots, x^{m-1}\}$ and $H = \{1, y, \dots, y^{m-1}\}$ (Proposition 1.3.16), f is a bijection. To see that f is a homomorphism, we must show $f(x^ix^j) = f(x^i)f(x^j)$ for $0 \leq i, j < m$. If $i + j < m$, then

$$f(x^ix^j) = f(x^{i+j}) = y^{i+j} = y^iy^j = f(x^i)f(x^j).$$

If $i + j \geq m$, then $i + j = m + r$ with $0 \leq r < m$, so $x^{i+j} = x^mx^r = x^r$ (because $x^m = 1$); similarly $y^{i+j} = y^r$. Hence

$$f(x^ix^j) = f(x^{i+j}) = f(x^r) = y^r = y^{i+j} = y^iy^j = f(x^i)f(x^j).$$

Therefore f is an isomorphism and $G \cong H$. (Example 1.9.20 gives a nicer proof.)

Definition 1.8.9 (Invariant)**.** A property of a group G that is shared by every group isomorphic to G is called an **invariant** of G.

For example, the order $|G|$ is an invariant, for isomorphic groups have the same order. Being abelian is an invariant: if $ab = ba$, then

$$f(a)f(b) = f(ab) = f(ba) = f(b)f(a),$$

so $f(a)$ and $f(b)$ commute. Thus $\mathbb{R}$ and $GL(2,\mathbb{R})$ are not isomorphic, for $\mathbb{R}$ is abelian and $GL(2,\mathbb{R})$ is not. Here are some further invariants.

Lemma 1.8.10 (Rotman, Exercise 2.59)

Let $f\colon G \to H$ be an isomorphism.

(i) For every $a \in G$, the elements a and $f(a)$ have the same order (both infinite, or both finite and equal). Consequently, if G has an element of some order n and H does not, then $G \ncong H$.

(ii) For every k, G and H have the same number of elements of order k.

Proof. (i) By Lemma 1.8.7, $f(a)^n = f(a^n)$, and since f is injective with $f(1) = 1$, we have $f(a^n) = 1$ if and only if $a^n = 1$. Thus $\{n : a^n = 1\} = \{n : f(a)^n = 1\}$, and the orders agree.
(ii) By (i), f restricts to a bijection from the elements of order k in G to those in H. ■

Whether or not a group is cyclic is also an invariant. In general, however, it is a challenge to decide whether two given groups are isomorphic.

Example 1.8.11 ($\mathbf{V} \ncong \Gamma_4$)**.** Here are two nonisomorphic groups of the same order. Let $\mathbf{V} = \{(1), (1\ 2)(3\ 4), (1\ 3)(2\ 4), (1\ 4)(2\ 3)\}$ be the four-group (Example 1.3.4), and let $\Gamma_4 = \langle i\rangle = \{1, i, -1, -i\}$ be the multiplicative cyclic group of fourth roots of unity. If there were an isomorphism $f\colon \mathbf{V} \to \Gamma_4$, then surjectivity would provide $x \in \mathbf{V}$ with $i = f(x)$. But $x^2 = (1)$ for all $x \in \mathbf{V}$, so $i^2 = f(x)^2 = f(x^2) = f((1)) = 1$, contradicting $i^2 = -1$. Therefore $\mathbf{V} \ncong \Gamma_4$.
There are other proofs: Γ_4 is cyclic and $\mathbf{V}$ is not; Γ_4 has an element of order 4 and $\mathbf{V}$ does not; Γ_4 has a unique element of order 2, while $\mathbf{V}$ has three.

§1.8.ii. Kernels, Images, and Normal Subgroups

Definition 1.8.12 (Kernel, image)**.** If $f\colon G \to H$ is a homomorphism, define its **kernel** and **image** by

$$\ker f = \{x \in G : f(x) = 1\} \qquad \text{and} \qquad \operatorname{Im} f = \{h \in H : h = f(x) \text{ for some } x \in G\}.$$

Note 1.8.13. *Kernel* comes from the German word for "grain" or "seed" (*corn* comes from the same word): it indicates an important ingredient of a homomorphism.

Example 1.8.14 (Kernels and images)**.** (i) Let $\Gamma_n = \langle\zeta\rangle$, where $\zeta = e^{2\pi i/n}$ is a primitive nth root of unity. Then $f\colon \mathbb{Z} \to \Gamma_n$, $f(m) = \zeta^m$, is a surjective homomorphism, and $\ker f$ consists of all the multiples of n, because ζ has order n (Lemma 1.2.21).

(ii) $\operatorname{sgn}\colon S_n \to \Gamma_2 = \{\pm 1\}$ is a homomorphism by Theorem 1.1.37. For $n \geq 2$ it is surjective, since $\operatorname{sgn}(\tau) = -1$ for a transposition τ; its kernel is the alternating group A_n of all even permutations.

(iii) Since $\det(AB) = \det(A)\det(B)$, the determinant $\det\colon GL(2,\mathbb{R}) \to \mathbb{R}^\times$ is a homomorphism. It is surjective, since each $r \in \mathbb{R}^\times$ is the determinant of $\left(\begin{smallmatrix} r & 0 \\ 0 & 1 \end{smallmatrix}\right)$. Its kernel is the **special linear group**

$$SL(2,\mathbb{R}) = \{A \in GL(2,\mathbb{R}) : \det(A) = 1\}.$$

(This extends to $\det\colon GL(n,\mathbb{R}) \to \mathbb{R}^\times$.)

(iv) We generalize the construction of $\ker f$. Recall the **inverse image** of a subset $B \subseteq Y$ under a function $f\colon X \to Y$:

$$f^{-1}(B) = \{x \in X : f(x) \in B\}.$$

If $f\colon G \to H$ is a homomorphism and $B \leq H$, then $f^{-1}(B)$ is a subgroup of G containing $\ker f$. Indeed, $1 \in f^{-1}(B)$, for $f(1) = 1 \in B$. If $x, y \in f^{-1}(B)$, then $f(xy) = f(x)f(y) \in B$, so $xy \in f^{-1}(B)$. If $x \in f^{-1}(B)$, then $f(x^{-1}) = f(x)^{-1} \in B$, so $x^{-1} \in f^{-1}(B)$. Finally $\ker f = f^{-1}(\{1\}) \subseteq f^{-1}(B)$.

Proposition 1.8.15

Let $f\colon G \to H$ be a homomorphism.

(i) $\ker f$ is a subgroup of G, and $\operatorname{Im} f$ is a subgroup of H.

(ii) If $x \in \ker f$ and $a \in G$, then $axa^{-1} \in \ker f$.

(iii) f is an injection if and only if $\ker f = \{1\}$.

Proof. (i) $\ker f = f^{-1}(\{1\})$ is a subgroup by Example 1.8.14(iv). As for the image: $1 = f(1) \in \operatorname{Im} f$; if $h = f(x) \in \operatorname{Im} f$, then $h^{-1} = f(x)^{-1} = f(x^{-1}) \in \operatorname{Im} f$; and if also $k = f(y) \in \operatorname{Im} f$, then $hk = f(x)f(y) = f(xy) \in \operatorname{Im} f$.

(ii) If $f(x) = 1$, then $f(axa^{-1}) = f(a)f(x)f(a)^{-1} = f(a)f(a)^{-1} = 1$.

(iii) If f is an injection, then $x \neq 1$ implies $f(x) \neq f(1) = 1$, so $x \notin \ker f$. Conversely, assume $\ker f = \{1\}$ and $f(x) = f(y)$. Then $1 = f(x)f(y)^{-1} = f(xy^{-1})$, so $xy^{-1} \in \ker f = \{1\}$; hence $x = y$. ∎

Definition 1.8.16 (Normal subgroup)**.** A subgroup K of a group G is a **normal subgroup** if $k \in K$ and $g \in G$ imply $gkg^{-1} \in K$. We then write $K \trianglelefteq G$.

Proposition 1.8.15 thus says that *the kernel of a homomorphism is always a normal subgroup.* We record some first examples.

- If G is abelian, then every subgroup K is normal: for $k \in K$ and $g \in G$, $gkg^{-1} = kgg^{-1} = k \in K$.
- The cyclic subgroup $H = \langle(1\ 2)\rangle = \{(1), (1\ 2)\}$ of S_3 is *not* normal: if $\alpha = (1\ 2\ 3)$, then $\alpha^{-1} = (3\ 2\ 1)$ and

$$\alpha(1\ 2)\alpha^{-1} = (1\ 2\ 3)(1\ 2)(3\ 2\ 1) = (2\ 3) \notin H.$$

- The cyclic subgroup $K = \langle(1\ 2\ 3)\rangle = \{(1), (1\ 2\ 3), (1\ 3\ 2)\}$ of S_3 is normal: by Proposition 1.1.27, a conjugate of a 3-cycle is a 3-cycle, and K contains both 3-cycles of S_3.
- By Example 1.8.14(ii) and (iii), $A_n \trianglelefteq S_n$ and $SL(2,\mathbb{R}) \trianglelefteq GL(2,\mathbb{R})$ (both facts are also easy to prove directly).

§1.8.iii. Conjugation and the Center

Definition 1.8.17 (Conjugate)**.** If G is a group and $a \in G$, then a **conjugate** of a is any element of the form gag^{-1} with $g \in G$.

Clearly, a subgroup $K \leq G$ is normal if and only if K contains all the conjugates of its elements. By Proposition 1.1.29, $\alpha, \beta \in S_n$ are conjugate in S_n if and only if they have the same cycle structure.

Remark 1.8.18. If $H \leq S_n$, then $\alpha, \beta \in H$ being conjugate in S_n does *not* imply that they are conjugate in H. For example, (1 2)(3 4) and (1 3)(2 4) are conjugate in S_4, but not in $\mathbf{V}$: since $\mathbf{V}$ is abelian, the only conjugate of an element of $\mathbf{V}$ in $\mathbf{V}$ is the element itself.

Remark 1.8.19 (Similarity)**.** In linear algebra, a linear transformation $T\colon V \to V$ of an n-dimensional real vector space determines an $n \times n$ matrix A once a basis of V is chosen; another basis gives another matrix B, and A and B are *similar*: $B = PAP^{-1}$ for some nonsingular P. Thus conjugacy in $GL(n, \mathbb{R})$ is similarity.

Definition 1.8.20 (Conjugation)**.** If G is a group and $g \in G$, define **conjugation** $\gamma_g\colon G \to G$ by

$$\gamma_g(a) = gag^{-1} \qquad \text{for all } a \in G.$$

Proposition 1.8.21 (i) If G is a group and $g \in G$, then conjugation $\gamma_g\colon G \to G$ is an isomorphism.

(ii) Conjugate elements have the same order.

Proof. (i) If $g, h \in G$, then

$$(\gamma_g \circ \gamma_h)(a) = \gamma_g(hah^{-1}) = g(hah^{-1})g^{-1} = (gh)a(gh)^{-1} = \gamma_{gh}(a);$$

that is, $\gamma_g \circ \gamma_h = \gamma_{gh}$. It follows that each γ_g is a bijection, for $\gamma_g \circ \gamma_{g^{-1}} = \gamma_1 = 1_G = \gamma_{g^{-1}} \circ \gamma_g$. And γ_g is a homomorphism:

$$\gamma_g(ab) = g(ab)g^{-1} = (gag^{-1})(gbg^{-1}) = \gamma_g(a)\gamma_g(b).$$

(ii) If a and b are conjugate, then $b = gag^{-1} = \gamma_g(a)$ for some $g \in G$. Since γ_g is an isomorphism, a and b have the same order by Lemma 1.8.10(i). ■

Example 1.8.22 (The center)**.** The **center** of a group G is

$$Z(G) = \{z \in G : zg = gz \text{ for all } g \in G\};$$

it consists of all elements commuting with everything in G. (The equation $zg = gz$ can be rewritten as $z = gzg^{-1}$, so no other element of G is conjugate to z.)

$Z(G)$ is a subgroup: $1 \in Z(G)$. If $y, z \in Z(G)$, then for all g, $(yz)g = y(zg) = y(gz) = (yg)z = g(yz)$, so $yz \in Z(G)$. If $z \in Z(G)$, then in particular $zg^{-1} = g^{-1}z$ for all g, and taking inverses (Lemma 1.2.10(iii)) gives $gz^{-1} = z^{-1}g$, so $z^{-1} \in Z(G)$.

$Z(G)$ is a normal subgroup: if $z \in Z(G)$ and $g \in G$, then $gzg^{-1} = zgg^{-1} = z \in Z(G)$. The same computation shows that *every* subgroup of $Z(G)$ is normal in G.

A group G is abelian if and only if $Z(G) = G$. At the other extreme are groups with $Z(G) = \{1\}$; these are called **centerless**. For example, S_n is centerless for every $n \geq 3$ (Rotman, Exercise 2.31): if $\alpha \neq (1)$, then α moves some i, say $\alpha(i) = j \neq i$; choose $k \notin \{i, j\}$ and let $\beta = (j\ k)$. By Lemma 1.1.26, $\beta\alpha\beta^{-1}$ sends $\beta(i) = i$ to $\beta(j) = k \neq j$, so $\beta\alpha\beta^{-1} \neq \alpha$, and $\alpha \notin Z(S_n)$.

Example 1.8.23 ($\mathbf{V} \trianglelefteq S_4$)**.** The four-group $\mathbf{V} = \{(1), (1\ 2)(3\ 4), (1\ 3)(2\ 4), (1\ 4)(2\ 3)\}$ is a normal subgroup of S_4. By Proposition 1.1.27, every conjugate of a product of two disjoint transpositions is another such; but there are only 3 permutations in S_4 with this cycle structure (Table 1.1), and all of them lie in $\mathbf{V}$.

Proposition 1.8.24

Let H be a subgroup of index 2 in a group G.

(i) $g^2 \in H$ for every $g \in G$.

(ii) H is a normal subgroup of G.

Proof. Since H has index 2, there are exactly two cosets, H and aH with $a \notin H$, and G is the disjoint union $G = H \cup aH$.

(i) If $g \in H$, then $g^2 \in H$. Otherwise $g \in aH$, say $g = ah$ with $h \in H$. If $g^2 \notin H$, then $g^2 = ah'$ with $h' \in H$, and

$$g = g^{-1}g^2 = (ah)^{-1}ah' = h^{-1}a^{-1}ah' = h^{-1}h' \in H,$$

a contradiction.

(ii) Let $h \in H$ and $g \in G$; we show $ghg^{-1} \in H$. If $g \in H$, this holds because H is a subgroup. Otherwise $g \in aH$, say $g = ax$ with $x \in H$. Then $ghg^{-1} = a(xhx^{-1})a^{-1} = ah'a^{-1}$, where $h' = xhx^{-1} \in H$. If $ghg^{-1} \notin H$, then $ah'a^{-1} \in aH$, say $ah'a^{-1} = ay$ with $y \in H$. Canceling a gives $h'a^{-1} = y$, and then $a = y^{-1}h' \in H$, a contradiction. Hence $H \trianglelefteq G$. ∎

§1.8.iv. The Quaternions

Definition 1.8.25 (Quaternions)**.** The **group of quaternions** is the set $\mathbf{Q}$ of the following eight matrices in $GL(2, \mathbb{C})$:

$$\mathbf{Q} = \{I, A, A^2, A^3, B, BA, BA^2, BA^3\}, \qquad \text{where} \quad A = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}, \quad B = \begin{pmatrix} 0 & i \\ i & 0 \end{pmatrix}.$$

Proposition 1.8.26 (Rotman, Exercise 2.76)

$\mathbf{Q}$ is a nonabelian group of order 8. Its only element of order 2 is $-I$, every $M \in \mathbf{Q}$ with $M \neq \pm I$ satisfies $M^2 = -I$, and $Z(\mathbf{Q}) = \{I, -I\}$.

Proof. Put $C = BA$. Direct computation gives

$$A^2 = B^2 = -I, \qquad C = BA = \begin{pmatrix} -i & 0 \\ 0 & i \end{pmatrix}, \qquad AB = \begin{pmatrix} i & 0 \\ 0 & -i \end{pmatrix} = -C, \qquad C^2 = -I.$$

Hence $A^3 = -A$, $BA^2 = -B$, $BA^3 = -C$, and $\mathbf{Q} = \{\pm I, \pm A, \pm B, \pm C\}$, eight distinct matrices. From the relations above,

$$\begin{aligned} &AB = -C, && BC = B^2A = -A, && CA = BA^2 = -B, \\ &BA = C, && CB = BAB = -B^2A = A, && AC = ABA = -CA = B, \end{aligned}$$

Thus the product of any two elements of $\mathbf{Q}$ lies in $\mathbf{Q}$, and $\mathbf{Q}$ is a subgroup of $GL(2, \mathbb{C})$ by Proposition 1.3.8. It is not abelian, since $AB = -BA \neq BA$.

Since $(-M)^2 = M^2$, every element other than $\pm I$ squares to $-I$ and hence has order 4, while $-I$ has order 2. Finally $\pm I$ are central (they are scalar matrices), whereas $\pm A$ does not commute with B, and $\pm B$ and $\pm C$ do not commute with A; so $Z(\mathbf{Q}) = \{\pm I\}$. ∎

Note 1.8.27 (Hamilton's quaternions)**.** Addition, subtraction, multiplication, and division can be extended from $\mathbb{R}$ to the plane so that all the usual laws hold: this is $\mathbb{C}$. W. R. Hamilton found a way of extending these operations to four-dimensional space so that all the usual laws still hold except commutativity of multiplication; he called the new "numbers" *quaternions* (Latin for "four"). The multiplication is determined by the basis vectors $1, \mathbf{i}, \mathbf{j}, \mathbf{k}$ with

$$\mathbf{i}^2 = \mathbf{j}^2 = \mathbf{k}^2 = -1, \quad \mathbf{ij} = \mathbf{k} = -\mathbf{ji}, \quad \mathbf{jk} = \mathbf{i} = -\mathbf{kj}, \quad \mathbf{ki} = \mathbf{j} = -\mathbf{ik}.$$

The nonzero quaternions form a multiplicative group, and $\mathbf{Q}$ is isomorphic to its subgroup $\{\pm 1, \pm\mathbf{i}, \pm\mathbf{j}, \pm\mathbf{k}\}$.

In $\mathbf{Q}$, the element A has order 4, so $\langle A\rangle$ has index 2; its other coset is

$$B\langle A\rangle = \{B, BA, BA^2, BA^3\}.$$

Example 1.8.28 (Every subgroup of $\mathbf{Q}$ is normal)**.** By Lagrange's theorem the possible orders of subgroups of $\mathbf{Q}$ are $1, 2, 4, 8$. The subgroups $\{I\}$ and $\mathbf{Q}$ are clearly normal. A subgroup of order 4 has index 2, hence is normal by Proposition 1.8.24. Finally, $-I$ is the only element of order 2, so $\langle -I\rangle$ is the only subgroup of order 2; it is normal because it equals the center $Z(\mathbf{Q})$ (Example 1.8.22).

Thus $\mathbf{Q}$ is a nonabelian group which is like an abelian group in the sense that every subgroup is normal. This is essentially the only such example: every finite group all of whose subgroups are normal has the form $\mathbf{Q} \times A$, where $A = B \times C$, every nonidentity element of B has order 2, and every element of C has odd order (direct products are introduced in the next section).

Lagrange's theorem says that the order of a subgroup of a finite group G divides $|G|$. As remarked after Corollary 1.3.39, the converse is false; we can now prove it.

Proposition 1.8.29

The alternating group A_4 is a group of order 12 having no subgroup of order 6.

Proof. By Table 1.1, the even permutations in S_4 are the identity, the 8 three-cycles, and the 3 products of two disjoint transpositions; so $|A_4| = 12$. If A_4 had a subgroup H of order 6, then H would have index 2, and so $\alpha^2 \in H$ for every $\alpha \in A_4$, by Proposition 1.8.24(i). If α is a 3-cycle, then α has order 3, so $\alpha = \alpha^4 = (\alpha^2)^2 \in H$. Thus H contains all 8 three-cycles, which is impossible since $|H| = 6$. ∎

Proposition 1.9.28 will show, however, that an *abelian* group of order n has a subgroup of order d for every divisor d of n.

Lecture 5: Quotient Groups

§1.9. Quotient Groups

Rotman §2.6

The group $\mathbb{Z}_m$ of residue classes was built from $\mathbb{Z}$ and its subgroup $m\mathbb{Z}$. This construction is the prototype of a general way of building new groups from given ones, called *quotient groups*. Along the way, group theory gives new proofs of the theorems of Fermat, Euler, and Wilson.

§1.9.i. The Integers mod m, Revisited

Fix $m \geq 2$. We constructed the residue classes $[a]_m = a + m\mathbb{Z}$ and the set $\mathbb{Z}_m$ in Lecture 2. Rotman's Propositions 2.98–2.103 are results we have already proved:

- $[a]_m = [b]_m$ if and only if $a \equiv b \pmod m$ (Proposition 1.4.2); in particular $[a]_m = [0]_m$ if and only if $m \mid a$;
- $\mathbb{Z}_m = \{[0]_m, [1]_m, \dots, [m-1]_m\}$ has exactly m elements (Corollary 1.4.5);
- $[a]_m + [b]_m = [a+b]_m$ is well defined (Lemma 1.5.1), and $\mathbb{Z}_m$ is an abelian group of order m under it (Theorem 1.5.2);
- $[a]_m[b]_m = [ab]_m$ is well defined (Lemma 1.6.1); it is associative and commutative with identity $[1]_m$, because these laws are inherited from $\mathbb{Z}$ exactly as for addition.

The group axioms in $\mathbb{Z}_m$ are "inherited" from those of $\mathbb{Z}$. When the modulus is clear from context, we sometimes drop the subscript and write $[a]$ for $[a]_m$.

Proposition 1.9.1

$\mathbb{Z}_m$ is a cyclic group of order m with generator $[1]_m$. Consequently, every cyclic group of order m is isomorphic to $\mathbb{Z}_m$.

Proof. If $0 \le r < m$, then $[r]_m = [1]_m + \cdots + [1]_m$ (r summands), so $\mathbb{Z}_m = \langle [1]_m \rangle$. The second statement follows from Example 1.8.8. ∎

Remark 1.9.2. Here is an alternative construction. Let $G_m = \{0, 1, \dots, m-1\}$, with

$$a \oplus b = \begin{cases} a+b & \text{if } a+b \le m-1, \\ a+b-m & \text{if } a+b > m-1. \end{cases}$$

Although this definition looks simpler, proving associativity is now very tedious, and proofs with it usually require case analyses (compare Example 1.8.8).

$\mathbb{Z}_m$ is *not* a group under multiplication, since some elements, e.g. $[0]_m$, have no inverse. We shall repeatedly need two consequences of Bézout's identity.

Lemma 1.9.3

Let a, b, m, n be integers.

(i) *(Euclid's lemma)* If $\gcd(m, a) = 1$ and $m \mid ab$, then $m \mid b$. In particular, if p is prime and $p \mid ab$, then $p \mid a$ or $p \mid b$.

(ii) If $\gcd(m, n) = 1$, $m \mid a$, and $n \mid a$, then $mn \mid a$.

Proof. (i) By Corollary 1.6.5 there are x, y with $mx + ay = 1$. Then $b = mbx + (ab)y$, and m divides both terms. If p is prime and $p \nmid a$, then $\gcd(p, a) = 1$, since the only positive divisors of p are 1 and p.

(ii) Write $a = mq$. Then $n \mid mq$ with $\gcd(n, m) = 1$, so $n \mid q$ by (i), and $mn \mid mq = a$. ∎

Proposition 1.9.4 (i) If $\gcd(a, m) = 1$, then $[a]_m[x]_m = [b]_m$ can be solved for $[x]_m \in \mathbb{Z}_m$.

(ii) If p is prime, then $\mathbb{Z}_p^\times$, the set of nonzero elements of $\mathbb{Z}_p$, is a multiplicative abelian group of order $p-1$.

Proof. (i) Choose s, t with $sa + tm = 1$. Then $[s]_m[a]_m = [1]_m$, so $[x]_m = [sb]_m$ is a solution: $[a]_m[sb]_m = [s]_m[a]_m[b]_m = [b]_m$.

(ii) If $0 < a < p$, then $\gcd(a, p) = 1$. Hence $\mathbb{Z}_p^\times$ coincides with the group $\mathbb{Z}_p^*$ of units of Theorem 1.6.8, and it has $p - 1$ elements, being $\mathbb{Z}_p$ with $[0]_p$ removed. ■

We will prove later (Rotman, Theorem 3.122) that $\mathbb{Z}_p^\times$ is a *cyclic* group of order $p - 1$. Here is a new proof of Fermat's theorem, entirely different from the one in Example 1.7.37.

Corollary 1.9.5 (Fermat)

If p is a prime and $a \in \mathbb{Z}$, then $a^p \equiv a \pmod{p}$.

Proof. It suffices to show that $[a^p]_p = [a]_p$. If $[a]_p = [0]_p$, then $[a^p]_p = [a]_p^p = [0]_p = [a]_p$. If $[a]_p \neq [0]_p$, then $[a]_p \in \mathbb{Z}_p^\times$, a group of order $p - 1$, so $[a]_p^{p-1} = [1]_p$ by Corollary 1.3.38. Multiplying by $[a]_p$ gives $[a^p]_p = [a]_p^p = [a]_p$. ■

If m is not prime, then $\mathbb{Z}_m^\times = \mathbb{Z}_m - \{[0]_m\}$ is *not* a group: if $m = ab$ with $1 < a, b < m$, then $[a]_m, [b]_m \neq [0]_m$, but $[a]_m[b]_m = [m]_m = [0]_m$. The right analogue of $\mathbb{Z}_p^\times$ is the group $\mathbb{Z}_m^*$ of units from Lecture 2, which Rotman denotes by $U(\mathbb{Z}_m)$.

Definition 1.9.6 (Euler ϕ-function)**.** For $m \geq 1$, $\phi(m)$ is the number of integers k with $1 \leq k \leq m$ and $\gcd(k, m) = 1$.

Lemma 1.9.7

$\mathbb{Z}_m^*$ is exactly the set of classes in $\mathbb{Z}_m$ having a multiplicative inverse, and it is a multiplicative abelian group of order $\phi(m)$.

Proof. The first statement is Corollary 1.6.7, and $\mathbb{Z}_m^*$ is an abelian group by Theorem 1.6.8. By Corollary 1.4.5, $\mathbb{Z}_m^*$ consists of the $[r]_m$ with $0 \leq r \leq m - 1$ and $\gcd(r, m) = 1$, and these are distinct. Since $\gcd(0, m) = m = \gcd(m, m) \neq 1$, replacing the range $0 \leq r \leq m - 1$ by $1 \leq r \leq m$ does not change the count, which is therefore $\phi(m)$. ■

If p is prime, then $\phi(p) = p - 1$ and $\mathbb{Z}_p^* = \mathbb{Z}_p^\times$.

Theorem 1.9.8 (Euler)

If $\gcd(r, m) = 1$, then $r^{\phi(m)} \equiv 1 \pmod{m}$.

Proof. If G is a finite group of order n, then $x^n = 1$ for all $x \in G$ (Corollary 1.3.38). Here $[r]_m \in \mathbb{Z}_m^*$, a group of order $\phi(m)$ by Lemma 1.9.7, so $[r]_m^{\phi(m)} = [1]_m$. In congruence notation, $r^{\phi(m)} \equiv 1 \pmod{m}$. ■

Example 1.9.9 ($\mathbb{Z}_8^*$ and $\mathbb{Z}_{10}^*$)**.** We have

$$\mathbb{Z}_8^* = \{[1], [3], [5], [7]\} \cong \mathbf{V},$$

for $[3]^2 = [9] = [1]$, $[5]^2 = [25] = [1]$, and $[7]^2 = [49] = [1]$, so every nonidentity element has order 2 (see Proposition 1.7.11). Moreover,

$$\mathbb{Z}_{10}^* = \{[1], [3], [7], [9]\} \cong \mathbb{Z}_4,$$

for $[3]^4 = [81] = [1]$ while $[3]^2 = [9] = [-1] \neq [1]$; so $[3]$ has order 4, and $\mathbb{Z}_{10}^*$ is cyclic.

Theorem 1.9.10 (Wilson)
An integer $p \geq 2$ is a prime if and only if

$$(p-1)! \equiv -1 \pmod{p}.$$

Proof. Assume p is prime. If $a_1, \ldots, a_n$ is a list of all the elements of a finite abelian group G, then the product $a_1 a_2 \cdots a_n$ equals the product of all elements a with $a^2 = 1$, for every other element cancels against its inverse ($a \neq a^{-1}$ pairs off with a^{-1}). In $\mathbb{Z}_p^\times$, if $[a]^2 = [1]$, then $p \mid a^2 - 1 = (a-1)(a+1)$, so $p \mid a-1$ or $p \mid a+1$ by Lemma 1.9.3(i); that is, $[a] = [\pm 1]$. Hence the product of all elements of $\mathbb{Z}_p^\times$, namely $[1][2]\cdots[p-1] = [(p-1)!]$, is $[1][-1] = [-1]$ if p is odd, and $[1] = [-1]$ if $p = 2$. Therefore $(p-1)! \equiv -1 \pmod{p}$.

Conversely, suppose $(m-1)! \equiv -1 \pmod{m}$, say $(m-1)! + 1 = km$. Then every common divisor of m and $(m-1)!$ divides 1, so $\gcd(m, (m-1)!) = 1$. If m were composite, there would be $a \mid m$ with $1 < a \leq m-1$; then $a \mid (m-1)!$, and $a > 1$ would be a common divisor of m and $(m-1)!$, a contradiction. Therefore m is prime. ∎

Remark 1.9.11. Wilson's theorem generalizes the way Euler's theorem generalizes Fermat's: replace $\mathbb{Z}_p^\times$ by $\mathbb{Z}_n^*$. For example, one can show for all $m \geq 3$ that $\mathbb{Z}_{2^m}^*$ has exactly 3 elements of order 2, namely $[-1]$, $[1 + 2^{m-1}]$, and $[-(1 + 2^{m-1})]$. It follows that the product of all odd numbers r with $1 \leq r < 2^m$ is congruent to 1 modulo 2^m, because

$$(-1)(1 + 2^{m-1})(-1 - 2^{m-1}) = (1 + 2^{m-1})^2 = 1 + 2^m + 2^{2m-2} \equiv 1 \pmod{2^m}.$$

§1.9.ii. Quotient Groups

The map $\pi\colon \mathbb{Z} \to \mathbb{Z}_m$, $\pi(a) = [a]_m$, is a surjective homomorphism, so $\mathbb{Z}_m = \operatorname{Im}\pi$: every element of $\mathbb{Z}_m$ has the form $\pi(a)$, and $\pi(a) + \pi(b) = \pi(a+b)$. This description of $\mathbb{Z}_m$ in terms of $\mathbb{Z}$ generalizes to arbitrary groups. Suppose $f\colon G \to H$ is a surjective homomorphism. Every element of H has the form $f(a)$, and the operation in H is given by $f(a)f(b) = f(ab)$. Now $K = \ker f$ is a normal subgroup of G, and we are going to reconstruct $H = \operatorname{Im} f$ (and a surjective homomorphism $\pi\colon G \to H$) from G and K alone.

Definition 1.9.12 (Product of subsets)**.** Let $\mathcal{P}(G)$ be the family of all nonempty subsets of a group G. For $X, Y \in \mathcal{P}(G)$, define

$$XY = \{xy : x \in X \text{ and } y \in Y\}.$$

This multiplication is associative: $X(YZ)$ is the set of all $x(yz)$ and $(XY)Z$ is the set of all $(xy)z$ with $x \in X$, $y \in Y$, $z \in Z$, and these are the same elements. Some instances:

- The product of a one-point subset $\{a\}$ and a subgroup H is the coset aH.
- If $H \leq G$, then $HH = H$: $HH \subseteq H$ because H is closed under multiplication, and $h = h1 \in HH$ for every $h \in H$.

Two subsets may commute even though their elements do not: $HH = HH$ for a nonabelian subgroup H. More interestingly, let $G = S_3$ and $K = \langle (1\ 2\ 3) \rangle$. Then $(1\ 2)$ does not commute with $(1\ 2\ 3) \in K$, but $(1\ 2)K = K(1\ 2)$. In fact, normal subgroups are characterized by this property.

Lemma 1.9.13
A subgroup K of a group G is normal if and only if $bK = Kb$ for every $b \in G$.

Proof. Assume $K \trianglelefteq G$, and let $bk \in bK$. Then $bkb^{-1} = k' \in K$, so $bk = (bkb^{-1})b = k'b \in Kb$; hence $bK \subseteq Kb$. For the reverse inclusion, let $kb \in Kb$. Since K is normal, $b^{-1}kb = b^{-1}k(b^{-1})^{-1} = k'' \in K$, so $kb = b(b^{-1}kb) = bk'' \in bK$. Therefore $bK = Kb$.

Conversely (Rotman, Exercise 2.69), assume $bK = Kb$ for all b. If $k \in K$ and $b \in G$, then $bk \in bK = Kb$, say $bk = k'b$; hence $bkb^{-1} = k' \in K$. ■

Here is a fundamental construction of a new group from a given group.

Theorem 1.9.14

Let G/K denote the family of all cosets of a subgroup K of G. If K is a normal subgroup, then

$$aK\,bK = abK$$

for all $a, b \in G$, and G/K is a group under this operation.

Proof. The product $(aK)(bK)$ is a product of four elements of $\mathcal{P}(G)$, so associativity in $\mathcal{P}(G)$ gives

$$(aK)(bK) = a(Kb)K = a(bK)K = ab(KK) = abK,$$

using $Kb = bK$ (Lemma 1.9.13) and $KK = K$. Thus the product of two cosets of K is again a coset of K, and an operation on G/K has been defined. It is associative, because multiplication in $\mathcal{P}(G)$ is. The identity is $K = 1K$, for $(1K)(bK) = bK$, and the inverse of aK is $a^{-1}K$, for $(a^{-1}K)(aK) = a^{-1}aK = K$. Therefore G/K is a group. ■

Definition 1.9.15 (Quotient group)**.** The group G/K of Theorem 1.9.14 is called the **quotient group** G mod K. When G is finite, its order is

$$|G/K| = [G : K] = |G|/|K|$$

(presumably the reason for the name).

Example 1.9.16 ($\mathbb{Z}/\langle m\rangle = \mathbb{Z}_m$)**.** Let $\langle m\rangle = m\mathbb{Z}$ be the subgroup of all multiples of m. Since $\mathbb{Z}$ is abelian, $\langle m\rangle$ is normal. The sets $\mathbb{Z}/\langle m\rangle$ and $\mathbb{Z}_m$ coincide, because they have the same elements: the coset $a + \langle m\rangle = \{a + km : k \in \mathbb{Z}\}$ is the class $[a]_m$. The operations also coincide: addition in $\mathbb{Z}/\langle m\rangle$ is $(a + \langle m\rangle) + (b + \langle m\rangle) = (a + b) + \langle m\rangle$, which is just $[a]_m + [b]_m = [a + b]_m$. Therefore $\mathbb{Z}_m = \mathbb{Z}/\langle m\rangle$.

Recall Lemma 1.3.32(i): if $K \leq G$, then $aK = bK$ if and only if $b^{-1}a \in K$; in particular $aK = K$ if and only if $a \in K$. We can now prove the converse of Proposition 1.8.15(ii).

Corollary 1.9.17

Every normal subgroup $K \trianglelefteq G$ is the kernel of some homomorphism.

Proof. Define the **natural map** $\pi\colon G \to G/K$ by $\pi(a) = aK$. The formula $aK\,bK = abK$ says $\pi(a)\pi(b) = \pi(ab)$, so π is a (surjective) homomorphism. Since K is the identity of G/K,

$$\ker \pi = \{a \in G : aK = K\} = K.$$ ■

The next theorem shows that every homomorphism gives rise to an isomorphism, and that quotient groups are merely constructions of homomorphic images. E. Noether (1882–1935) emphasized the fundamental importance of this fact.

Theorem 1.9.18 (First isomorphism theorem)
If $f\colon G \to H$ is a homomorphism, then

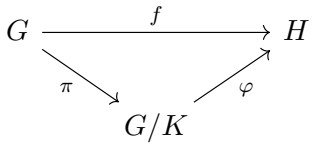
$$\ker f \trianglelefteq G \qquad \text{and} \qquad G/\ker f \cong \operatorname{Im} f.$$

In more detail, if $\ker f = K$, then $\varphi\colon G/K \to \operatorname{Im} f \leq H$, $\varphi(aK) = f(a)$, is an isomorphism.

Proof. $K = \ker f$ is normal by Proposition 1.8.15(ii).

- *Well defined:* if $aK = bK$, then $a = bk$ for some $k \in K$, and $f(a) = f(b)f(k) = f(b)$, because $f(k) = 1$.
- *Homomorphism:* $\varphi(aK\,bK) = \varphi(abK) = f(ab) = f(a)f(b) = \varphi(aK)\varphi(bK)$.
- *Surjective onto* $\operatorname{Im} f$*:* clearly $\operatorname{Im}\varphi \subseteq \operatorname{Im} f$; and if $y = f(a) \in \operatorname{Im} f$, then $y = \varphi(aK)$.
- *Injective:* if $\varphi(aK) = \varphi(bK)$, then $f(a) = f(b)$, so $1 = f(b)^{-1}f(a) = f(b^{-1}a)$ and $b^{-1}a \in K$; thus $aK = bK$ by Lemma 1.3.32(i). ∎

Remark 1.9.19. The following commutative diagram describes the proof, where $\pi\colon G \to G/K$ is the natural map; the theorem says $f = \varphi\pi$ with φ an isomorphism onto $\operatorname{Im} f$.



For example, the identity $1_G\colon G \to G$ is a surjective homomorphism with kernel $\{1\}$, so $G/\{1\} \cong G$. Given any homomorphism $f\colon G \to H$, one should immediately ask for its kernel and image; the first isomorphism theorem then provides an isomorphism $G/\ker f \cong \operatorname{Im} f$. Since there is no significant difference between isomorphic groups, it also says that there is no significant difference between quotient groups and homomorphic images.

Example 1.9.20 (Cyclic groups, again)**.** We revisit Example 1.8.8. If $G = \langle a\rangle$ is cyclic of order m, define $f\colon \mathbb{Z} \to G$ by $f(n) = a^n$. Then f is a homomorphism (laws of exponents, Proposition 1.2.18), it is surjective because a generates G, and $\ker f = \{n \in \mathbb{Z} : a^n = 1\} = \langle m\rangle$ by Lemma 1.2.21. The first isomorphism theorem gives $\mathbb{Z}/\langle m\rangle \cong G$. So every cyclic group of order m is isomorphic to $\mathbb{Z}/\langle m\rangle = \mathbb{Z}_m$, and hence any two cyclic groups of order m are isomorphic.

Example 1.9.21 ($\mathbb{R}/\mathbb{Z} \cong S^1$)**.** What is the quotient group $\mathbb{R}/\mathbb{Z}$? Define $f\colon \mathbb{R} \to S^1$, where S^1 is the circle group, by $f(x) = e^{2\pi ix}$. Then f is a homomorphism, $f(x+y) = f(x)f(y)$, by the addition formulas for sine and cosine, and f is surjective. Its kernel consists of all $x \in \mathbb{R}$ with $e^{2\pi ix} = \cos 2\pi x + i \sin 2\pi x = 1$; but $\cos 2\pi x = 1$ forces $x \in \mathbb{Z}$, and every integer lies in the kernel, so $\ker f = \mathbb{Z}$. The first isomorphism theorem gives

$$\mathbb{R}/\mathbb{Z} \cong S^1.$$

§1.9.iii. Products of Subgroups and the Isomorphism Theorems

A natural question is whether HK is a subgroup when H and K are subgroups. In general it is not: in $G = S_3$, with $H = \langle(1\ 2)\rangle$ and $K = \langle(1\ 3)\rangle$,

$$HK = \{(1), (1\ 2), (1\ 3), (1\ 2)(1\ 3)\} = \{(1), (1\ 2), (1\ 3), (1\ 3\ 2)\}$$

has 4 elements, so it is not a subgroup lest we contradict Lagrange's theorem.

Proposition 1.9.22 (i) If H and K are subgroups of a group G, and one of them is normal, then HK is a subgroup of G; moreover $HK = KH$ in this case.

(ii) If both H and K are normal subgroups, then HK is a normal subgroup.

Proof. (i) Assume first that $K \trianglelefteq G$. We claim $HK = KH$. If $hk \in HK$, then $k' = hkh^{-1} \in K$, and $hk = (hkh^{-1})h = k'h \in KH$; so $HK \subseteq KH$. Similarly $kh = h(h^{-1}kh) = hk'' \in HK$ with $k'' = h^{-1}kh \in K$; so $KH \subseteq HK$.

Now HK is a subgroup: $1 = 1 \cdot 1 \in HK$; if $hk \in HK$, then $(hk)^{-1} = k^{-1}h^{-1} \in KH = HK$; and if $hk, h_1k_1 \in HK$, then $h_1^{-1}kh_1 = k' \in K$, so

$$hk\,h_1k_1 = hh_1(h_1^{-1}kh_1)k_1 = (hh_1)(k'k_1) \in HK.$$

If instead $H \trianglelefteq G$, the same argument with the roles of H and K exchanged shows that $KH = HK$ and that KH is a subgroup.

(ii) If $g \in G$, then $ghkg^{-1} = (ghg^{-1})(gkg^{-1}) \in HK$. ∎

Here is a useful counting result.

Proposition 1.9.23 (Product formula)

If H and K are subgroups of a finite group G, then

$$|HK|\,|H \cap K| = |H|\,|K|,$$

where $HK = \{hk : h \in H,\ k \in K\}$ (not necessarily a subgroup).

Proof. Define $f\colon H \times K \to HK$ by $f(h,k) = hk$; clearly f is surjective. Since $H \times K$ is the disjoint union of the fibres $f^{-1}(x) = \{(h,k) \in H \times K : hk = x\}$ for $x \in HK$, it suffices to show $\left|f^{-1}(x)\right| = |H \cap K|$ for every $x \in HK$.

We claim that if $x = hk$, then

$$f^{-1}(x) = \{(hd, d^{-1}k) : d \in H \cap K\}.$$

Each $(hd, d^{-1}k)$ lies in $f^{-1}(x)$, for $hdd^{-1}k = hk = x$. Conversely, if $(h', k') \in f^{-1}(x)$, then $h'k' = hk$, so $h^{-1}h' = kk'^{-1}$ lies in $H \cap K$; call it d. Then $h' = hd$ and $k' = d^{-1}k$. Finally, $d \mapsto (hd, d^{-1}k)$ is a bijection from $H \cap K$ onto $f^{-1}(x)$ (it is injective by cancellation), so $\left|f^{-1}(x)\right| = |H \cap K|$. ∎

The next two results are variants of the first isomorphism theorem.

Theorem 1.9.24 (Second isomorphism theorem)

If H and K are subgroups of a group G with $H \trianglelefteq G$, then HK is a subgroup, $H \cap K \trianglelefteq K$, and

$$K/(H \cap K) \cong HK/H.$$

Proof. HK is a subgroup by Proposition 1.9.22. Moreover $H \trianglelefteq HK$: in general, if $H \leq S \leq G$ and H is normal in G, then H is normal in S (if $ghg^{-1} \in H$ for all $g \in G$, then in particular for all $g \in S$). So HK/H makes sense.

Each coset $xH \in HK/H$ has the form kH with $k \in K$: write $x = hk$; then $hk = k(k^{-1}hk) = kh'$ with $h' \in H$, so $xH = kh'H = kH$. Hence the function $f\colon K \to HK/H$, $f(k) = kH$, is

surjective. It is a homomorphism, being the restriction to K of the natural map $HK \to HK/H$, and

$$\ker f = \{k \in K : kH = H\} = K \cap H.$$

Therefore $H \cap K \trianglelefteq K$, and the first isomorphism theorem gives $K/(H \cap K) \cong HK/H$. ■

The second isomorphism theorem gives the product formula when one of the subgroups is normal: $|K|/|H \cap K| = |HK|/|H|$, that is, $|HK|\,|H \cap K| = |H|\,|K|$.

Theorem 1.9.25 (Third isomorphism theorem)

If H and K are normal subgroups of a group G with $K \leq H$, then $H/K \trianglelefteq G/K$ and

$$(G/K)/(H/K) \cong G/H.$$

Proof. Define $f\colon G/K \to G/H$ by $f(aK) = aH$. It is well defined: if $a'K = aK$, then $a^{-1}a' \in K \leq H$, so $a'H = aH$. It is a homomorphism, $f(aK\,bK) = f(abK) = abH = aH\,bH$, and it is surjective. Moreover

$$\ker f = \{aK : aH = H\} = \{aK : a \in H\} = H/K,$$

so H/K is a normal subgroup of G/K. Since f is surjective, the first isomorphism theorem gives $(G/K)/(H/K) \cong G/H$. ■

The third isomorphism theorem is easy to remember: the K's in the fraction $(G/K)/(H/K)$ can be canceled. One better appreciates the first isomorphism theorem after proving the third: $(G/K)/(H/K)$ consists of cosets (of H/K) whose representatives are themselves cosets (of K), and a direct proof could be nasty.

The next result, which describes the subgroups of a quotient group G/K, can be regarded as a fourth isomorphism theorem. Recall that a function $f\colon X \to Y$ sets up a correspondence between subsets of X and subsets of Y via direct and inverse images; we shall use the elementary facts

$$S \subseteq T \Rightarrow f(S) \subseteq f(T), \qquad S \subseteq f^{-1}(f(S)), \qquad f(f^{-1}(U)) = U \text{ if } f \text{ is surjective.}$$

If $K \trianglelefteq G$, let $\mathrm{Sub}(G;K)$ be the family of all subgroups S of G containing K, and $\mathrm{Sub}(G/K)$ the family of all subgroups of G/K.

Proposition 1.9.26 (Correspondence theorem)

Let G be a group and $K \trianglelefteq G$. Then $S \mapsto S/K$ is a bijection $\mathrm{Sub}(G;K) \to \mathrm{Sub}(G/K)$. Writing $S^* = S/K$:

(i) $T \leq S$ in $\mathrm{Sub}(G;K)$ if and only if $T^* \leq S^*$, in which case $[S:T] = [S^*:T^*]$;

(ii) $T \trianglelefteq S$ in $\mathrm{Sub}(G;K)$ if and only if $T^* \trianglelefteq S^*$, in which case $S/T \cong S^*/T^*$.

Proof. Let $\pi\colon G \to G/K$ be the natural map, and note that $S/K = \{sK : s \in S\} = \pi(S)$. This is a subgroup of G/K: the image of a subgroup under a homomorphism is a subgroup (apply Proposition 1.8.15(i) to π restricted to S). Let $\Phi(S) = S/K$.

Key fact: if $K \leq S \leq G$, then $\pi^{-1}(\pi(S)) = S$. Certainly $S \subseteq \pi^{-1}(\pi(S))$. Conversely, if $a \in \pi^{-1}(\pi(S))$, then $\pi(a) = \pi(s)$ for some $s \in S$; so $s^{-1}a \in \ker\pi = K$, that is, $a = sk$ with $k \in K \leq S$, and $a \in S$.

Φ *is injective:* if $\pi(S) = \pi(S')$ with $S, S' \in \mathrm{Sub}(G;K)$, then $S = \pi^{-1}\pi(S) = \pi^{-1}\pi(S') = S'$.

Φ *is surjective:* if $U \leq G/K$, then $\pi^{-1}(U)$ is a subgroup of G containing $\pi^{-1}(\{1\}) = K$ (Example 1.8.14(iv)), and $\pi(\pi^{-1}(U)) = U$ because π is surjective.

(i) If $T \leq S$, then $T^* = \pi(T) \subseteq \pi(S) = S^*$. Conversely, if $T^* \leq S^*$, then $T = \pi^{-1}(T^*) \subseteq \pi^{-1}(S^*) = S$ by the key fact. For the indices, we show that $sT \mapsto \pi(s)T^*$ $(s \in S)$ is a bijection from the cosets of T in S to the cosets of T^* in S^*. For $s, s' \in S$,

$$sT = s'T \Leftrightarrow s'^{-1}s \in T = \pi^{-1}(T^*) \Leftrightarrow \pi(s')^{-1}\pi(s) \in T^* \Leftrightarrow \pi(s)T^* = \pi(s')T^*,$$

which shows that the map is well defined and injective; it is surjective because every element of S^* is $\pi(s)$ for some $s \in S$. Hence $[S : T] = [S^* : T^*]$. (When G is finite this is simply $[S^* : T^*] = \frac{|S|/|K|}{|T|/|K|} = |S|/|T| = [S : T]$.)

(ii) If $T \trianglelefteq S$, then $K \leq T$ are normal subgroups of the group S, and the third isomorphism theorem (applied to S) gives $T/K \trianglelefteq S/K$ and $(S/K)/(T/K) \cong S/T$, that is, $S^*/T^* \cong S/T$. Conversely, assume $T^* \trianglelefteq S^*$, and let $t \in T$, $s \in S$. Then

$$\pi(sts^{-1}) = \pi(s)\pi(t)\pi(s)^{-1} \in \pi(s)T^*\pi(s)^{-1} = T^*,$$

so $sts^{-1} \in \pi^{-1}(T^*) = T$. Hence $T \trianglelefteq S$. ■

When dealing with quotient groups, one usually says, without explicitly mentioning the correspondence theorem, that every subgroup of G/K has the form S/K for a unique subgroup S with $K \leq S \leq G$.

Example 1.9.27 (The second center)**.** If G is a finite p-group and $G \neq \{1\}$, then $Z(G) \neq \{1\}$ by Theorem 1.7.34. Hence, if G is not abelian, then $G/Z(G)$ is a nontrivial p-group and $Z(G/Z(G)) \neq \{1\}$. An important role in the study of finite p-groups is played by $Z_2(G)$, defined as the inverse image of $Z(G/Z(G))$ under the natural map $G \to G/Z(G)$. Since the center $Z(G/Z(G))$ is normal in $G/Z(G)$, the correspondence theorem gives $Z_2(G) \trianglelefteq G$.

Proposition 1.9.28

If G is a finite abelian group, then G has a subgroup of order d for every divisor d of $|G|$. In particular, if p is a prime divisor of $|G|$, then G contains an element of order p. (Compare Proposition 1.8.29.)

Proof. First we prove, by induction on $n = |G|$, that for every prime $p \mid n$ there is an element of order p in G. The base step $n = 1$ is vacuous. For the inductive step, $n > 1$, so we may choose $a \in G$ of order $k > 1$.

If $p \mid k$, say $k = p\ell$, then a^ℓ has order p: indeed $(a^\ell)^p = a^k = 1$, while $(a^\ell)^j = a^{\ell j} \neq 1$ for $0 < j < p$, since $0 < \ell j < k$.

If $p \nmid k$, let $H = \langle a \rangle$. Since G is abelian, $H \trianglelefteq G$, and $|G/H| = n/k$ is divisible by p (Euclid's lemma). By induction, G/H has an element bH of order p. If b has order m, then $(bH)^m = b^m H = H$, so $p \mid m$ by Lemma 1.2.21. We are back in the first case: $b^{m/p}$ has order p.

Now we prove the general statement by induction on $d \geq 1$, for all finite abelian groups at once. The base step $d = 1$ is obvious. If $d > 1$, let p be a prime divisor of d. By the first part, G has a subgroup H of order p; it is normal because G is abelian, and $|G/H| = |G|/p$ is divisible by d/p. By induction, G/H has a subgroup S^* of order d/p. By the correspondence theorem, $S^* = S/H$ for a subgroup S with $H \leq S \leq G$, and $|S| = |S^*|\,|H| = (d/p) \cdot p = d$. ■

Cauchy's theorem (Theorem 1.7.29) says that the second statement holds for every finite group, abelian or not.

§1.9.iv. Direct Products

Here is another construction of a new group from two given groups.

Definition 1.9.29 (Direct product)**.** If H and K are groups, their **direct product** $H \times K$ is the set of all ordered pairs (h, k) with $h \in H$ and $k \in K$, with the operation

$$(h, k)(h', k') = (hh', kk').$$

It is routine to check that $H \times K$ is a group: associativity holds coordinatewise, the identity is $(1, 1)$, and $(h, k)^{-1} = (h^{-1}, k^{-1})$. Note that $H \times K$ is abelian if and only if both H and K are abelian.

Example 1.9.30 ($\mathbf{V} \cong \mathbb{Z}_2 \times \mathbb{Z}_2$)**.** The function $f\colon \mathbf{V} \to \mathbb{Z}_2 \times \mathbb{Z}_2$ defined by

$$\begin{aligned} f\colon (1) &\mapsto ([0],[0]), & (1\ 2)(3\ 4) &\mapsto ([1],[0]), \\ (1\ 3)(2\ 4) &\mapsto ([0],[1]), & (1\ 4)(2\ 3) &\mapsto ([1],[1]) \end{aligned}$$

is an isomorphism: in both groups, every nonidentity element has order 2 and the product of any two distinct nonidentity elements is the third one.

We now apply the first isomorphism theorem to direct products.

Proposition 1.9.31

Let G and G' be groups, and let $K \trianglelefteq G$ and $K' \trianglelefteq G'$. Then $K \times K'$ is a normal subgroup of $G \times G'$, and

$$(G \times G')/(K \times K') \cong (G/K) \times (G'/K').$$

Proof. Let $\pi\colon G \to G/K$ and $\pi'\colon G' \to G'/K'$ be the natural maps, and define $f\colon G \times G' \to (G/K) \times (G'/K')$ by

$$f(g, g') = (\pi(g), \pi'(g')) = (gK, g'K').$$

Since π and π' are surjective homomorphisms and the operations are coordinatewise, f is a surjective homomorphism. Its kernel is $\{(g, g') : gK = K,\ g'K' = K'\} = K \times K'$. The first isomorphism theorem now gives both assertions. ∎

Here is a characterization of direct products.

Proposition 1.9.32

If a group G contains normal subgroups H and K with $H \cap K = \{1\}$ and $HK = G$, then $G \cong H \times K$.

Proof. First, each $g \in G$ has a *unique* factorization $g = hk$ with $h \in H$ and $k \in K$: if $hk = h'k'$, then $h'^{-1}h = k'k^{-1} \in H \cap K = \{1\}$, so $h' = h$ and $k' = k$. We may therefore define $\varphi\colon G \to H \times K$ by $\varphi(g) = (h, k)$, where $g = hk$.

Next, elements of H commute with elements of K. Let $h \in H$ and $k \in K$. Since K is normal, $(hkh^{-1})k^{-1} \in K$; since H is normal, $h(kh^{-1}k^{-1}) \in H$. But $H \cap K = \{1\}$, so $hkh^{-1}k^{-1} = 1$, and $hk = kh$.

Now φ is a homomorphism: if $g = hk$ and $g' = h'k'$, then $gg' = hkh'k' = (hh')(kk')$, so

$$\varphi(gg') = (hh', kk') = (h, k)(h', k') = \varphi(g)\varphi(g').$$

It is surjective, since $\varphi(hk) = (h, k)$, and injective, since $\varphi(g) = (1, 1)$ forces $g = 1$. Therefore φ is an isomorphism. ∎

All the hypotheses are needed. For example, let $G = S_3$, $H = \langle(1\ 2\ 3)\rangle$, and $K = \langle(1\ 2)\rangle$. Then $S_3 = HK$ (by the product formula, $|HK| = 3 \cdot 2/1 = 6$), $H \cap K = \{1\}$, and $H \trianglelefteq S_3$, but K is not normal. And $S_3 \ncong H \times K$, for S_3 is not abelian, while a direct product of abelian groups is abelian.

Theorem 1.9.33

If m and n are relatively prime, then

$$\mathbb{Z}_{mn} \cong \mathbb{Z}_m \times \mathbb{Z}_n.$$

Proof. Define $f\colon \mathbb{Z} \to \mathbb{Z}_m \times \mathbb{Z}_n$ by $f(a) = ([a]_m, [a]_n)$; it is a homomorphism because each coordinate is. We claim $\ker f = \langle mn\rangle$. Clearly $\langle mn\rangle \leq \ker f$. Conversely, if $a \in \ker f$, then $[a]_m = [0]_m$ and $[a]_n = [0]_n$, so $m \mid a$ and $n \mid a$; since $\gcd(m, n) = 1$, Lemma 1.9.3(ii) gives $mn \mid a$, so $a \in \langle mn\rangle$.

By the first isomorphism theorem and Example 1.9.16, $\mathbb{Z}_{mn} = \mathbb{Z}/\langle mn\rangle \cong \operatorname{Im} f$. Thus $\operatorname{Im} f$ is a subgroup of $\mathbb{Z}_m \times \mathbb{Z}_n$ with mn elements; since $\mathbb{Z}_m \times \mathbb{Z}_n$ also has mn elements, $\operatorname{Im} f$ is all of it, and $\mathbb{Z}_{mn} \cong \mathbb{Z}_m \times \mathbb{Z}_n$. ■

Remark 1.9.34. Surjectivity of f says: for any $a, b \in \mathbb{Z}$ there is $x \in \mathbb{Z}$ with $x \equiv a \pmod m$ and $x \equiv b \pmod n$. This is the **Chinese remainder theorem**; Rotman uses it to prove surjectivity, while the counting argument above proves it.

For example, $\mathbb{Z}_6 \cong \mathbb{Z}_2 \times \mathbb{Z}_3$. There is no such isomorphism if m and n are not relatively prime: Example 1.9.30 shows $\mathbf{V} \cong \mathbb{Z}_2 \times \mathbb{Z}_2$, which is not isomorphic to $\mathbb{Z}_4$, because $\mathbf{V}$ has no element of order 4.

Since $\langle a\rangle \cong \mathbb{Z}_n$ exactly when a has order n (Proposition 1.3.16 and Proposition 1.9.1), Theorem 1.9.33 can be read as saying that if a and b are commuting elements of relatively prime orders m and n, then ab has order mn. Here is a direct proof (this is the statement we already used as Lemma 1.7.6).

Proposition 1.9.35

Let G be a group, and let $a, b \in G$ be commuting elements of orders m and n. If $\gcd(m, n) = 1$, then ab has order mn.

Proof. Since a and b commute, $(ab)^r = a^r b^r$ for all r, so $(ab)^{mn} = a^{mn}b^{mn} = 1$. It suffices to prove that $(ab)^k = 1$ implies $mn \mid k$. If $1 = (ab)^k = a^k b^k$, then $a^k = b^{-k}$. Since a has order m, $1 = a^{mk} = b^{-mk}$; since b has order n, Lemma 1.2.21 gives $n \mid mk$, and so $n \mid k$ by Euclid's lemma. Similarly $m \mid k$. Finally $mn \mid k$ by Lemma 1.9.3(ii). ■

Here is a number-theoretic application of direct products.

Corollary 1.9.36

If $\gcd(m, n) = 1$, then $\phi(mn) = \phi(m)\phi(n)$.

Proof. The isomorphism of Theorem 1.9.33 is $f\colon \mathbb{Z}_{mn} \to \mathbb{Z}_m \times \mathbb{Z}_n$, $f([a]_{mn}) = ([a]_m, [a]_n)$ (this is the φ of the first isomorphism theorem). It also preserves multiplication:

$$f([a]_{mn}[b]_{mn}) = f([ab]_{mn}) = ([a]_m[b]_m, [a]_n[b]_n) = f([a]_{mn})f([b]_{mn}),$$

where pairs are multiplied coordinatewise, and $f([1]_{mn}) = ([1]_m, [1]_n)$.

We claim $f(\mathbb{Z}_{mn}^*) = \mathbb{Z}_m^* \times \mathbb{Z}_n^*$. If $[a]_{mn}[b]_{mn} = [1]_{mn}$, then applying f gives $[a]_m[b]_m = [1]_m$ and $[a]_n[b]_n = [1]_n$, so $f([a]_{mn}) \in \mathbb{Z}_m^* \times \mathbb{Z}_n^*$. Conversely, suppose $f([c]_{mn}) = ([c]_m, [c]_n)$ with $[c]_m[d]_m = [1]_m$ and $[c]_n[e]_n = [1]_n$. Since f is surjective, there is b with $([b]_m, [b]_n) = ([d]_m, [e]_n)$; then $f([c]_{mn}[b]_{mn}) = ([1]_m, [1]_n) = f([1]_{mn})$, and injectivity gives $[c]_{mn}[b]_{mn} = [1]_{mn}$, so $[c]_{mn} \in \mathbb{Z}_{mn}^*$.

By Lemma 1.9.7, $\phi(mn) = |\mathbb{Z}_{mn}^*| = |\mathbb{Z}_m^* \times \mathbb{Z}_n^*| = \phi(m)\phi(n)$. ■

Definition 1.9.37 (Direct product of n groups)**.** If $H_1, \ldots, H_n$ are groups, their **direct product** $H_1 \times \cdots \times H_n$ is the set of all n-tuples $(h_1, \ldots, h_n)$ with $h_i \in H_i$, with coordinatewise multiplication

$$(h_1, \ldots, h_n)(h_1', \ldots, h_n') = (h_1h_1', \ldots, h_nh_n').$$

The *basis theorem* (Rotman, Theorem 6.11) says that every finite abelian group is a direct product of cyclic groups. We end with a variant of Proposition 1.3.19.

Proposition 1.9.38

If G is a finite abelian group having a unique subgroup of order p for every prime divisor p of $|G|$, then G is cyclic.

Proof. Choose $a \in G$ of largest order, say n. We first show: $\langle a \rangle$ *contains every element* $x \in G$ *with* $x^p = 1$ *for some prime* p. If $x \neq 1$, then x has order p, so $p \mid |G|$ and $C = \langle x \rangle$ is the unique subgroup of order p. If $\gcd(p, n) = 1$, then xa has order $pn > n$ by Proposition 1.9.35, contradicting the choice of a. So $p \mid n$, say $n = pq$; then a^q has order p, so $\langle a^q \rangle = C$ by uniqueness, and $x \in C \leq \langle a \rangle$.

If $\langle a \rangle = G$, we are done; so assume there is $b \in G$ with $b \notin \langle a \rangle$. Since $b^{|G|} = 1 \in \langle a \rangle$, there is a smallest positive integer k with $b^k \in \langle a \rangle$, say

$$b^k = a^r.$$

Now k is the order of $b\langle a \rangle$ in $G/\langle a \rangle$, so $k \mid |G|$; and $k \neq 1$. Write $k = p\ell$ with p prime; then $p \mid |G|$. By Proposition 1.9.28, G has an element of order p, and the argument of the first paragraph shows that $p \mid n$. There are two possibilities.

- If $p \mid r$, say $r = pu$, then $b^{p\ell} = b^k = a^{pu}$, so $(b^\ell a^{-u})^p = 1$ (as G is abelian). By the first paragraph, $b^\ell a^{-u} \in \langle a \rangle$, so $b^\ell \in \langle a \rangle$ with $0 < \ell < k$, contradicting the minimality of k.
- If $p \nmid r$, then $\gcd(p, r) = 1$, so $1 = sp + tr$ for some integers s, t, and

  $$a = a^{sp+tr} = a^{sp}(a^r)^t = a^{sp}b^{p\ell t} = (a^s b^{\ell t})^p.$$

  So $a = x^p$ with $x = a^s b^{\ell t}$. Let x have order N. Then $a^N = (x^N)^p = 1$, so $n \mid N$. If $N = n$, then, since $p \mid n$, the element x^p would have order n/p (as $(x^p)^{n/p} = 1$ and $(x^p)^j = x^{pj} \neq 1$ for $0 < j < n/p$), contradicting $x^p = a$ of order n. Hence $N > n$, contradicting the choice of a.

Both cases are impossible, so $G = \langle a \rangle$. ■

The proposition is false for nonabelian groups: the quaternions $\mathbf{Q}$ form a noncyclic group of order 8 having a unique subgroup of order 2 (Example 1.8.28).